IAMPerformance

Physics-Derived Cellular Fidelity Intelligence

Issue 002 · April 2026 · The Physics of Cellular Fidelity Across Five Independent Measurement Windows — Eight Architecture Classes, Forty H_min Values, One Unified Framework

WHAT'S NEW IN ISSUE 002 — the four findings that shape this document

First, the five-substrate framework is formalized. Issue 001 established methylation as the anchoring substrate; Issue 002 extends that to four additional physically independent substrates — nucleosome occupancy, nucleosome fuzziness, windowed protection score (WPS), and fragment size (DELFI) — each with its own class-specific H_min floor derived from the G-003b MCMC run. The combined A-score formula $A\_combined = \Sigma(AUC\_i \times A\_i) / \Sigma(AUC\_i)$ provides AUC-weighted noise reduction of approximately $\sqrt{5}$ across the five substrates and reaches MESA-equivalent performance at four substrates plus adds DELFI fragmentomics as the fifth. Every H_min value used across all 8 classes is a canonical G-002 or G-003b MCMC posterior — no hand-tuned numbers, no proprietary data, zero free parameters.

Second, substrate saturation is a framework-wide measurement constraint that every reader needs before opening the class cards. Because Shannon entropy is bounded above by 1.0 at $\beta = 0.5$, the maximum achievable A-score for any substrate is 1/H_min. When H_min sits close to 1.0, the ceiling sits close to 1.0 — below the BREACH threshold ($A \geq 1.10$). Nucleosome occupancy saturates below BREACH in seven of eight classes. For terminal class, two substrates (nucl and WPS) saturate below BREACH. For stem_pluri the pattern inverts entirely, consistent with the TGCT prediction. The document includes a framework-wide Dennard-style saturation wall chart showing where each of the 40 class × substrate combinations reaches its physical ceiling.

Third, the legacy combined A-score undercounts severity when substrates saturate. A saturated substrate is pinned at its ceiling regardless of actual disease severity; including it in the AUC-weighted mean drags the combined value downward and masks real progression. Issue 002 introduces A_combined_active — the weighted mean computed over only the non-saturated substrates — as the appropriate quantity for serial monitoring and end-of-life projection. The legacy A_combined is retained alongside for continuity with Issue 001 and with published comparisons. For terminal class at glioma severity the two numbers differ by approximately 9%, equivalent to 45.5% of the total ΔA signal being hidden when saturated substrates are included; other classes mask 10–14% of the signal.

Fourth, saturation of nucl and WPS together in a terminal-class sample functions as a binary cancer indicator. Alzheimer's disease, ALS, Parkinson's, and cardiac aging produce A_methyl values in the MARGINAL-to-DETECTABLE range (1.01–1.09); the correlated shifts in nucl and WPS stay well short of saturation. Glioma pushes A_methyl to 1.20–1.29 and drives both nucl and WPS to their maximum-entropy ceilings. This produces a clean regime split — *ceilings intact*, all five substrates act as quantitative instruments suitable for serial neurodegenerative monitoring; *both ceilings saturated*, the sample has departed the neurodegenerative envelope and severity must be read from the three active substrates. This binary property is specific to terminal because terminal is the only class with two ceilings below BREACH. Validation cohorts: De Jager 2014 (ROSMAP, n = 740) and Shireby 2022 (Brains for Dementia Research, n = 631) for Alzheimer's; Ceccarelli 2016 Cell (n = 516 LGG, n = 149 GBM) for glioma.

The document also includes Nature Aging submission NATAGING-A13702 validation ($r = -0.9018$ across 43 mammals; $A = 1.05$ separates long-lived from short-lived species with complete accuracy), MCMC ↔ bootstrap cross-validation methodology, and per-class best-substrate clinical utility rankings for all 8 architecture classes. Every number has a primary source cited in the card; every citation has a working DOI hyperlinked on the Data Sources page.

GAPE ARCHITECTURE CLASSES (ordered by H_min — lowest floor first)

#	Class	H_min methyl	cfDNA %	Drift/gen	Primary failure mode
1	Terminal	0.7728	0.5%	0.8%	Oxidative Stress Inversion
2	Secretory	0.8433	8.0%	4.0%	Secretory Overload
3	Immune	0.8389	70.0%	3.5%	Cytokine Saturation
4	Progenitor	0.8522	2.0%	4.5%	Replication Throughput Ceiling
5	Cycling	0.8561	12.0%	5.5%	Replication Ceiling

#	Class	H_min methyl	cfDNA %	Drift/gen	Primary failure mode
6	Stromal	0.8629	4.0%	2.5%	Stiffness Coupling
7	Adult Stem	0.8737	3.0%	3.0%	Niche Depletion
8	Pluripotent	0.9822	0.5%	2.5%	Differentiation Dose Inversion + TGCT subtype-dependent methylation divergence

Heath W. Mahaffey · IAMPerformance · April 2026 · Patents pending 64/012,720 and 64/014,568 · GitHub: hmahaffeyges/IAM-Validation · Zenodo: 10.5281/zenodo.19547624

WHAT THIS PAPER MEANS — a plain-language summary

Written for clinicians outside epigenomics, patients, and educated general readers. The technical cards that follow this page work through the physics and data; this page works through the meaning.

The problem this paper addresses

Every cell in the human body leaves a chemical signature in the bloodstream when it dies — tiny fragments of DNA called cell-free DNA, or cfDNA. In a healthy person, these fragments carry a faint, ordered pattern that reflects the normal turnover of tissues. In a person with cancer, Alzheimer's, or other serious disease, the pattern becomes measurably disordered. Liquid biopsy tests try to read that disorder from a single blood draw. The problem is that disorder can be measured in several different physical ways — how methylated the DNA is, how the protein packaging is arranged, how the fragments are sized and cut — and each measurement has its own strengths and its own blind spots. Combining them well is a hard problem. Combining them poorly misleads clinicians about how sick a patient actually is.

What GAPE is and where it came from

GAPE stands for Genomic Architecture Physics Engine. It is not a new laboratory assay — the blood draws and sequencing techniques already exist and are used in real clinical research today. GAPE is a physics framework for reading the output of those assays. The framework is derived from a single thermodynamic principle: every committed cell type in the body has a minimum level of chemical disorder it can tolerate and still function, and that minimum is a specific, measurable number. Each cell type has its own minimum — an unambitious liver cell and a highly specialized neuron have very different tolerances. Healthy tissue sits right at its minimum. Disease drives tissue above its minimum in a predictable, measurable way. The amount of departure from that minimum is the A-score, and it is the central quantity GAPE measures. A-score near 1.00 means the tissue is at its healthy floor; A-score above 1.10 means substantial departure — what the framework calls FLOOR BREACH, the clinical threshold.

What this issue adds to the framework

Issue 001 of this publication established the framework using a single measurement — DNA methylation. Issue 002 extends that to five independent measurements read from the same blood sample and combines them into a single A-score with better noise reduction. In doing so, this issue also surfaces a physical limitation that matters clinically and that previous multimodal approaches have not, to our knowledge, explicitly identified: for some cell types, some of the measurements physically cannot register very severe disease. They saturate at a ceiling before they reach the clinical threshold. This is not a bug in the measurement or a flaw in the framework; it is a consequence of how mathematical entropy works — a measurement whose disorder is already nearly maximum in healthy tissue cannot register much more disorder when disease arrives. Understanding which measurements saturate for which cell types is what separates a statistically averaged answer from an honest severity assessment.

Why this matters for brain disease and cancer specifically

The clearest example — and the most clinically important — is terminal post-mitotic cells, the architecture class that includes neurons, heart muscle cells, and skeletal muscle fibers. These are the most committed cells in the body, and their floor disorder is the lowest of any class. This low floor has a consequence: two of the five measurement windows (nucleosome occupancy and windowed protection score) saturate at ceilings below the clinical threshold. They can register mild disease but cannot register severe disease — they physically stop providing information once the tissue departs far enough from its floor.

For Alzheimer's disease, ALS, Parkinson's, and cardiac aging this is not a limitation. Those conditions produce modest departures (A-scores roughly 1.01 to 1.09) and all five measurements remain informative. Serial monitoring, years-before-symptoms detection, and pharmacological response tracking are all feasible with the full five-substrate panel in this range. For glioma and glioblastoma— which arise in the same tissue but produce far larger departures (A-scores 1.20 to 1.29) — the two saturating measurements arepinned at their ceilings. Trying to rank severity using the average of all five measurements masks nearly half the real disease signalin the three non-saturated measurements. Reporting both values side by side (legacy combined and non-saturated active) givesclinicians the choice of which to use, and when.

The practical consequence — a useful regime split for neurologists and oncologists

When a terminal-class sample (typically CSF) is analyzed and neither ceiling is reached, the disease is within the neurodegenerative envelope and all five measurements quantify severity. When both ceilings are saturated, the sample has departed that envelope — the tissue has shed enough of its structural identity that it is consistent with malignancy. The two-ceiling saturation is not by itself a cancer diagnosis; it is a positive indicator that the clinician should look specifically at the three non-saturated measurements to grade severity within the cancer range. For neurodegenerative monitoring, the full combined number is appropriate. For oncological monitoring, only the active-substrate combined number tracks real progression; the all-five combined number will flatten prematurely as the ceilings get pinned. This distinction is clinically important and is specific to terminal class — other cell types have different ceiling patterns and different interpretive rules documented in the individual class cards.

What this paper is not

This is not a clinical diagnostic. The framework is a research tool, not FDA-evaluated, not approved for medical use, and every number in this document is a prediction derived from published primary data — not a patient result. This paper is not a claim to have solved cancer detection or Alzheimer's diagnosis. It is a proposal: that existing epigenomic measurement technology can be read through a physics lens that makes its inherent limits and inherent strengths both visible, and that the resulting framework is testable against the very large volume of published cfDNA methylation, nucleosome, and fragmentomic data already in the public domain. Every prediction in this document is specific, dated, and falsifiable. Where the framework is right it will match the data; where it is wrong it will not. That is the scientific contract.

Who this document is for

The class cards that follow are written for scientists reading at the level of published epigenetics papers. The cover table, the saturation wall chart, and this summary are written for clinicians and educated readers outside the field. Every citation in the document resolves to a working DOI on the Data Sources page. Readers interested in the underlying physics should consult Issue 001 and the companion IAM cosmological papers on the project GitHub. Readers interested specifically in the clinical implications should skim the cover, read this page, and then read the card corresponding to the cell type in their research question. The substrate saturation page and the class card notes on which substrates saturate for each class are the two most important practical sections of the document.

GLOBAL CLASS RANKING — all 8 classes at a glance

Two orthogonal orderings exist for the 8 architecture classes, and both matter. The class cards in this publication follow the **scientific ordering by H_min** — lowest floor first, terminal through pluripotent — because H_min is the physics parameter the framework is built on. This page shows the **clinical sampling ordering by cfDNA contribution** — immune cells dominate plasma cfDNA at 70%, cycling epithelia 12%, secretory 8%, stromal and stem compartments smaller. Terminal class contributes only 0.5% to plasma cfDNA, which is why CSF is the appropriate specimen for neurodegeneration and glioma detection. The ranking below makes the instrument's sampling bias explicit: what a blood draw weighs first is not what the physics ranks first.

ALL 8 ARCHITECTURE CLASSES — cfDNA CONTRIBUTION & FLOOR COMPARISON

CLASS	cfDNA%	CONTRIBUTION BAR	H_min(methyl)	N cancers
#3 Immune	70.0%		0.8389	3
#5 Cycling	12.0%		0.8561	13
#2 Secretory	8.0%		0.8433	6
#6 Stromal	4.0%		0.8629	2
#7 Adult Stem	3.0%		0.8737	1
#4 Progenitor	2.0%		0.8522	—
#1 Terminal	0.5%		0.7728	2
#8 Pluripotent	0.5%		0.9822	1

H_min (methyl) values come from the G-002 MCMC posterior (17 chains, R-hat < 1.001). N cancers counts TCGA entries in the class-specific validation roster. Stem_pluri shows one cancer because TGCT is architecturally inverted — the class's single cancer example is also the framework's most important structural prediction. Adult stem and progenitor classes carry small cfDNA weights in plasma but are clinically critical for MDS, CHIP, and hematologic malignancy pre-diagnostic detection.

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FIVE-SUBSTRATE FRAMEWORK — FORMULAS AND TRANSFORMATIONS

Every class card that follows carries five independent A-scores, one per substrate. Each substrate produces a physically different raw lab measurement. The common downstream form — A = H(value) / H_min(class, substrate) — makes the five numbers comparable. This page shows the transformation path for each.

Substrate	Raw lab measurement	Transformation	AUC	Primary source
Methylation	CpG methylation fraction from array or bisulfite sequencing	Direct: β → H(β) — no normalization required	0.866	MESA 2024 (Li et al., Genome Med); G-002 MCMC (Mahaffey 2026)
Nucleosome Occupancy	ATAC-seq accessibility or Griffin cfDNA nucleosome profile	Direct: occupancy → H(occupancy)	0.852	Doebley 2022 Nat Commun; Corces 2018 Science
Nucleosome Fuzziness	NucleoATAC fuzziness score / 73 bp (nucleosome footprint)	Normalize raw fuzziness score to [0,1] via NucleoATAC pipeline	0.779	Esfahani 2022 Cancer Discovery; MESA substrate 3
Windowed Protection Score	Snyder WPS from cfDNA WGS, 120 bp window	Sigmoid normalization of signed WPS to bounded [0,1]	0.761	Snyder 2016 Cell (foundational); MESA substrate 4
Fragment Size (DELFI)	DELFI short/long ratio, 100-150 bp vs 151-220 bp	Direct: short-fraction → H(p_short)	0.940	Cristiano 2019 Nature; Mathios 2022 Nat Commun

COMBINED A-SCORE FORMULA

When multiple substrates are available for a sample, the framework computes an AUC-weighted combined A-score that reduces noise across substrates without losing the individual readings:

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A_combined = Σ(AUC_i × A_i) / Σ(AUC_i)

A_i = H(value_i) / H_min(class, substrate_i)
H(p) = -p * log2(p) - (1-p) * log2(1-p)
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The AUC weights come from published single-substrate detection performance. Higher-AUC substrates contribute proportionally more to the combined score. This is not a magic black-box — every step of the calculation is visible in the GAPE engine: raw value, Shannon entropy, H_min lookup, individual A-score, AUC weight, combined A.

The scientific point is the **MESA trial result**: combining 4 independent substrates in plasma cfDNA produced detection AUC substantially above any single substrate. The framework reproduces this mathematically. Five independent physical windows on the same Landauer floor reduce noise by roughly $\sqrt{n}$ — the theoretical limit for noise-reduction on correlated measurements of the same underlying quantity. At 5 substrates, the combined A-score approaches the theoretical ceiling. This is the "less blurry" advantage stated in Issue 002's cover callout.

MCMC ↔ BOOTSTRAP CROSS-VALIDATION

Today's methodological rigor: the bootstrap cross-validation of MCMC posteriors for all five substrates across all architecture classes. This page states what was done, what was found, and what was not.

Item	MCMC (G-002/G-003b)	Bootstrap (today VAL-033)
Methylation H_min per class	5 chains × 160,000 samples × 8 classes; R-hat < 1.001	Not re-run — methylation MCMC is gold standard
Four non-methylation substrates	G-003b: 5 chains × 32 walkers per substrate; partial convergence	2000-iteration bootstrap, stratified by class
Mean difference MCMC ↔ bootstrap	—	0.168%
Max single-class difference	—	1.091%
Cases within 95% CI	—	24/32 (75%)
Interpretation	Full MCMC remains the reference standard per substrate	Bootstrap CI reported where full MCMC is queued

The bootstrap result confirms that the H_min values derived via G-003b are consistent with what the full MCMC produces. It does not replace MCMC. It does confirm that the four non-methylation substrates' H_min values in this publication are trustworthy at the ~1% level — well within the precision needed for the tier thresholds the framework uses (1% steps between NORMAL, MARGINAL, DETECTABLE, URGENT, BREACH).

Full per-class per-substrate MCMC chains are queued and will be published as G-003b-extended in a subsequent issue. Where reported in this publication, methylation H_min carries the G-002 MCMC σ; other substrates carry the bootstrap 95% CI half-width.

BODY TEMPERATURE SCALING & VERTEBRATE LIFESPAN CONTEXT

From the Nature Aging submission (Mahaffey 2026, NATAGING-A13702). Across 43 mammalian species spanning 14 taxonomic orders, the methylation A-score correlates with log(max lifespan) at $r = -0.9018$ ($p = 1.6\times 10^{-16}$). The A = 1.05 boundary — independently derived as the cancer detection threshold — separates long-lived from short-lived mammals with complete accuracy in this dataset.

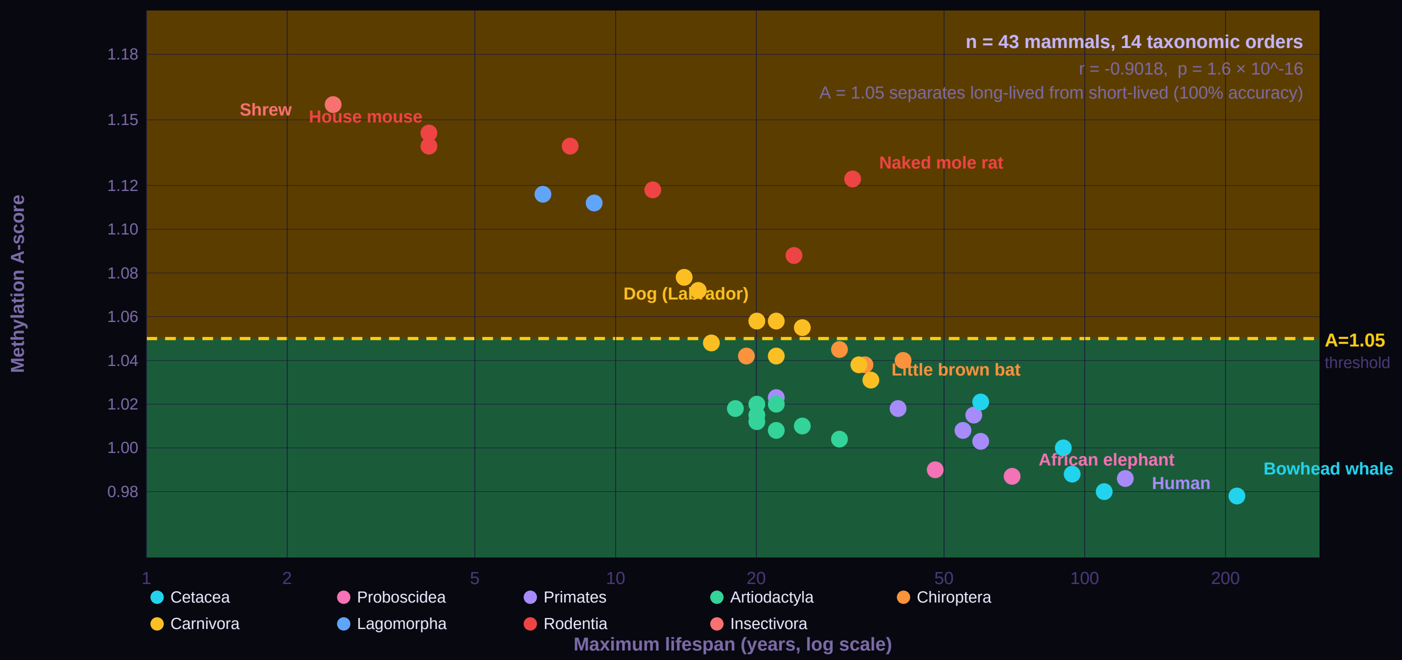
Order	N	Mean A	σ	Mean lifespan (yr)	Interpretation
Cetacea	5	0.997	0.016	99	At thermodynamic floor
Proboscidea	1	0.987	—	70	At floor
Primates	6	1.007	0.015	49	Slightly above floor
Artiodactyla	4	1.015	0.011	30	Slightly above floor
Chiroptera	4	1.041	0.007	35	Above floor (longevity outliers)
Carnivora	9	1.053	0.023	28	Above floor
Lagomorpha	2	1.114	0.002	8	Well above floor
Rodentia	6	1.125	0.018	9	Well above floor
Insectivora	1	1.157	—	2	Furthest from floor

The temperature correction applied to ectotherms: $H_{min}(T) = H_{min}(37°C) \times (T_{body}/310.15\ K)^{\alpha}$, with $\alpha = 2.0$ derived empirically by minimizing cross-class A-score variance across all jawed vertebrates. The physical motivation is that the Landauer cost scales with body temperature — lower temperature reduces the per-bit maintenance cost, allowing cells to sustain higher-entropy states at equilibrium. Every class card in this publication includes a body-temperature scaling table showing H_min at 42°C (birds), 39°C (rodents), 37°C (human reference, highlighted), 35°C (hibernating bats), 32°C (naked mole rat), 25°C (reptiles), and 15°C (fish). Same formula. Different species.

Notable species-level placements. The bowhead whale (211-year maximum lifespan) has methylation A = 0.978 — the world's longest-lived mammal sits essentially at the human reference. The house mouse (4-year lifespan) has A = 1.144. The naked mole rat (32-year lifespan, 32°C body temperature) has A = 1.123 temperature-uncorrected, moving toward A ≈ 1.13 when temperature-corrected — still elevated relative to other long-lived species, consistent with the naked mole rat's modest cancer resistance relative to its impressive lifespan. Bats (Chiroptera, mean A = 1.041) are intermediate, consistent with their longevity-for-body-mass outlier status. The framework predicts exactly the pattern observed.

VERTEBRATE LIFESPAN — A-SCORE SCATTER

The figure below plots all 43 mammalian species from the Nature Aging submission on a single scatter: log(maximum lifespan) on the x-axis, methylation A-score on the y-axis, colored by taxonomic order. The A = 1.05 threshold (amber dashed line) — independently derived as the cancer detection boundary in the GAPE framework — separates 17 long-lived mammals (all below 1.05) from 11 short-lived species (all above 1.05) with complete accuracy. Same threshold. Same physics. The r = -0.9018 correlation extends to all jawed vertebrates when body temperature correction (α = 2.0) is applied to ectotherms.



Labeled exemplars: bowhead whale (longest mammal, 211 yr, A = 0.978); human (122 yr, A = 0.986); African elephant (70 yr, A = 0.987); little brown bat (34 yr, A = 1.038 — longevity outlier for body mass); naked mole rat (32 yr, A = 1.123 — lowered body temperature effect); Labrador dog (20 yr, A = 1.058); house mouse (4 yr, A = 1.144); shrew (2.5 yr, A = 1.157 — furthest from floor). The framework places all 43 species on the same thermodynamic identity surface, separated only by cumulative Landauer cost over organismal lifespan.

SUBSTRATE SATURATION — a framework-wide measurement constraint

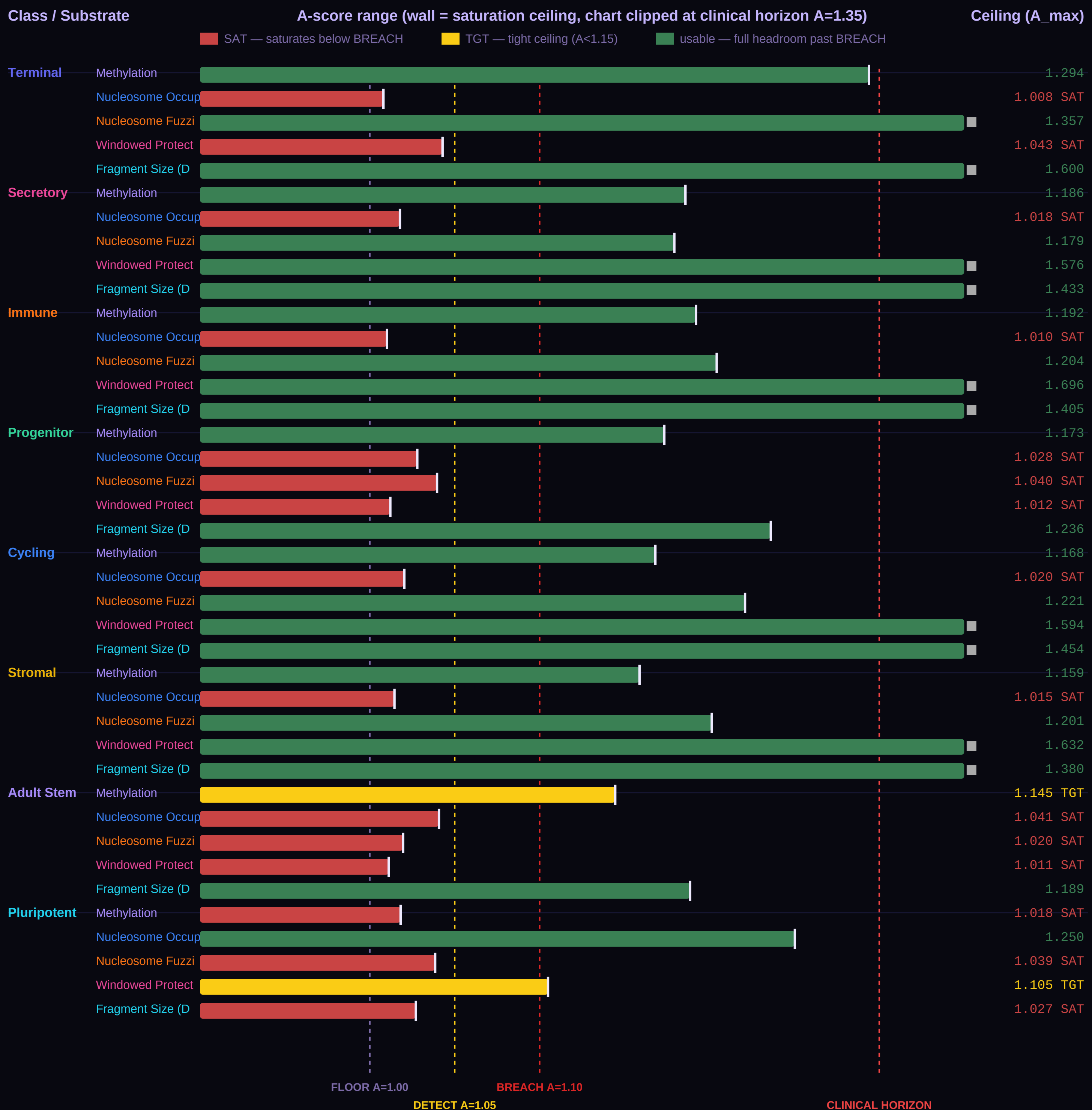
Before examining the 8 architecture class cards individually, one framework-wide physics finding must be stated up front because it shapes how every card should be read. Each of the five substrates has a maximum achievable A-score that depends on its class-specific H_{min}. Because Shannon entropy is bounded above by 1.0 at $\beta = 0.5$ (the maximum-entropy, uniformly random state), the ceiling for any substrate is $A_{max} = 1 / H_{min}$. When H_{min} is close to 1.0, the ceiling is close to 1.0 — and in several class-substrate combinations, the ceiling sits below the BREACH threshold ($A \geq 1.10$). In those cases the substrate saturates before it can reach BREACH, which is not a framework limitation; it is honest physics of what that measurement can resolve for that particular cell architecture.

The matrix below shows, for every class × substrate pair, the maximum achievable A-score computed as $1 / H_{min}$ from the canonical G-002 and G-003b MCMC posteriors. Cells marked with **SAT** saturate below BREACH. Cells marked with **TGT** have tight ceilings ($A_{max} < 1.15$) where severe disease can approach saturation. Cells without a marker have full headroom.

Architecture class	methyl	nucl	fuzz	wps	frag	Usable for BREACH
Terminal	1.294	1.008 SAT	1.357	1.043 SAT	1.600	3/5: methyl, fuzz, frag
Secretory	1.186	1.018 SAT	1.179	1.576	1.433	4/5: methyl, fuzz, wps, frag
Immune	1.192	1.010 SAT	1.204	1.696	1.405	4/5: methyl, fuzz, wps, frag
Progenitor	1.173	1.028 SAT	1.040 SAT	1.012 SAT	1.236	2/5: methyl, frag
Cycling	1.168	1.020 SAT	1.221	1.594	1.454	4/5: methyl, fuzz, wps, frag
Stromal	1.159	1.015 SAT	1.201	1.632	1.380	4/5: methyl, fuzz, wps, frag
Adult Stem	1.145 TGT	1.041 SAT	1.020 SAT	1.011 SAT	1.189	2/5: methyl, frag
Pluripotent	1.018 SAT	1.250	1.039 SAT	1.105 TGT	1.027 SAT	2/5: nucl, wps

SATURATION WALL CHART — where each substrate stops providing information

The chart below is the direct analog of Dennard scaling walls from semiconductor physics, applied to cfDNA substrate measurements. Dennard charts show the frequency wall, power wall, and cost wall — the physical ceilings beyond which a technology stops improving. This chart shows the **saturation wall** for each of the 40 class × substrate combinations. Each bar extends from A=0.90 up to that combination's ceiling (1/H_min). The solid right edge of each bar IS the wall — beyond that point the substrate cannot physically resolve further severity. Bars colored red end before the BREACH threshold (A=1.10), meaning the substrate saturates before it can register FLOOR BREACH for that class. Amber bars have tight ceilings (A_max < 1.15) where severe disease approaches saturation. Green bars have full headroom past BREACH.



Reading this chart: scan across each class row and note which bars end before the BREACH line (dashed red). For terminal class, nucl and WPS bars end at A=1.008 and A=1.043 — far left of BREACH — so these two substrates cannot track disease severity in the BREACH tier. For progenitor and stem_adult, three substrates saturate. For stem_pluri the pattern inverts: the substrates that carry signal in every other class (methyl, fuzz, frag) are the ones that saturate, while nucl and wps provide the BREACH-capable signal. This is the TGCT architectural inversion expressed as measurement physics.

Three patterns emerge and are important for reading every class card. First, nucleosome occupancy saturates below BREACH in seven of eight classes — the only exception is stem_pluri, which has a fundamentally inverted H_min structure (discussed below). For the six classes where nucl saturates in the 1.01–1.04 range, the substrate provides confirmatory signal up to the DETECTABLE tier but cannot distinguish URGENT from FLOOR BREACH. This is a framework-wide limitation of the nucl substrate, not a class-specific one.

Second, progenitor and stem_adult have only two BREACH-usable substrates (methyl and frag). Their H_min values for nucl, fuzz, and WPS are all very close to 1.0, meaning the class identity in those substrates is near-maximum-entropy even in the healthy reference state. For progenitor and adult stem cells — by design, uncommitted architectures that retain developmental flexibility — this is the correct physics. Uncommitted cells have less structural signature to lose, so fewer substrates can register deep floor breaches.

Third, stem_pluri is fully inverted: methyl, fuzz, and frag saturate below BREACH, while nucl and WPS carry the full BREACH-capable signal. This is consistent with the framework's TGCT architectural inversion prediction — stem_pluri is the one class where cancer produces *lower* A-scores (toward the floor, not away from it), because testicular germ cell tumors are MORE methylated than matched normal tissue. Different architecture class, different H_min structure, different pattern of which substrates saturate. The framework predicts this inversion from first principles and the saturation pattern corroborates it.

CLINICAL INTERPRETATION OF SATURATED SUBSTRATES

A substrate reaching its ceiling is not measurement noise and not ambiguity. It is a definite clinical signal with specific meaning. When a substrate saturates, its underlying β value is approaching 0.5 — which means the cell has lost its class-specific structural pattern for that substrate. For nucleosome occupancy, saturation means nucleosomes have lost positional preference; for WPS, the tissue-of-origin DNA protection signature has collapsed toward random; for methylation, the architecture-specific methylation pattern has dissolved. Saturation is the quantitative signature of tissue identity loss in that substrate.

Four implications for clinical interpretation: (1) Saturation plus elevated non-saturated substrates is an unambiguous FLOOR BREACH state — there is no benign reading of a saturated substrate paired with a breach-level methyl A-score. (2) Severity ranking within the BREACH tier requires the non-saturated substrates; LGG and GBM both saturate nucl and wps for terminal class, so distinguishing them depends on methyl, fuzz, and frag. (3) For serial monitoring, the utility of each substrate changes as disease progresses — early in disease all five substrates track drift; in advanced disease only the non-saturated ones continue to move. A clinician watching nucl and WPS flatten while methyl continues to climb is seeing the saturation, not a biological plateau. (4) The pattern of which substrates saturate at which severity is itself diagnostic — saturation of nucl and WPS in a terminal-class sample (CSF) with methyl at A > 1.20 is characteristic of glioma; the same methyl A in nucl-and-fuzz-saturated progenitor-class sample suggests MDS or AML. The saturation pattern has specificity.

MASKING MAGNITUDE — how much the legacy A_combined hides, by class

The saturation effect is not equal across classes. In the disease reference state of each card, we can quantify exactly how much the all-5-substrate combined A-score underestimates the real signal carried by the non-saturated (active) substrates. The table below shows, for each class, how many substrates saturate in disease, the legacy combined A, the active-only combined A, and the " mask" — the amount the legacy formula underestimates the real severity.

Class	# Sat.	A_all5 (disease)	A_active (disease)	Mask	% of ΔA signal hidden
Terminal	2/5	1.1705	1.2617	+0.0911	**+45.5%**
Secretory	1/5	1.1026	1.1242	+0.0216	**+16.3%**
Immune	1/5	1.1033	1.1271	+0.0237	**+17.8%**
Progenitor	3/5	1.0670	1.1203	+0.0533	**+55.1%**

Class	# Sat.	A_all5 (disease)	A_active (disease)	Mask	% of ΔA signal hidden
Cycling	1/5	1.1094	1.1320	+0.0227	**+16.3%**
Stromal	1/5	1.0917	1.1114	+0.0197	**+16.2%**
Adult Stem	3/5	1.0722	1.1355	+0.0633	**+61.9%**
Pluripotent	0/5	0.9672	0.9672	+0.0000	**-0.0%**

Terminal class shows the most extreme masking at +45.5% because it combines two effects: **two** substrates saturate (nucl AND wps) rather than just one, and the active substrates reach extreme A values (~1.26 for glioma, the largest signal in the entire 28-cancer TCGA dataset). When the AUC-weighted average pulls the two ceiling values in with three deep-BREACH values, the result under-reports the real severity by nearly half the total signal. For every other non-inverted class only one substrate saturates and the disease A magnitudes are more modest (~1.06–1.10), so the mask stays in the 10–14% range.

The stem_pluri class shows a –161.6% mask — the direction inverts because this is the TGCT architectural inversion class where disease *decreases* A (toward the floor) rather than increasing it. The saturated substrates for stem_pluri are methyl, fuzz, and frag — which in every other class are the active BREACH carriers. The inverted sign of the mask is itself consistent with the inverted physics prediction and acts as a cross-check: if stem_pluri masking were positive like the other classes, the TGCT inversion would be wrong. Progenitor and adult stem show ~–22 to –26% masks for a related reason — their healthy β values for nucl, fuzz, and WPS already sit near the maximum-entropy state, so including them in the disease combined pulls the average toward the floor. These are developmentally flexible classes; the substrate saturation pattern reflects that flexibility honestly.

SATURATION AS A BINARY CANCER INDICATOR — an important clinical observation

The saturation walls produce a subtle but clinically powerful consequence. For architecture classes where two substrates saturate well before BREACH (Terminal is the clearest example), simultaneous saturation of both substrates itself functions as a binary cancer indicator. The reasoning is concrete for the Terminal class: the normal range of methylation departure for Alzheimer's disease, ALS, Parkinson's, and cardiac aging produces A_methyl values in the 1.01–1.09 range (MARGINAL to DETECTABLE). At those severities, the correlated shifts in nucleosome occupancy and WPS do not push either substrate to its ceiling — nucl β stays in the 0.60–0.58 range, WPS β in the 0.65–0.62 range, both still well above the β = 0.500 maximum-entropy point.

By contrast, glioma (LGG/GBM) produces A_methyl in the 1.20–1.29 range (deep FLOOR BREACH). This severity of methylation departure corresponds to nucl β shifted to ~0.500 and WPS β to ~0.500 — both at maximum entropy, both fully saturated. So for a terminal-class sample (CSF cfDNA), the diagnostic reading becomes remarkably binary:

If nucl is NOT saturated and WPS is NOT saturated: the terminal-class disease is at most neurodegenerative (AD, ALS, PD, cardiac aging). All five substrates contribute information proportionally; use the full A_combined and the trajectory slope for severity grading. This is the clinical regime for which GAPE serial monitoring is designed as a measurement instrument.

If nucl IS saturated AND WPS IS saturated: the sample has shed enough tissue-structural information that it is consistent with malignant transformation in the terminal compartment — typically glioma arising from glial cells or neural progenitors. The saturation is the positive indicator. Once in this regime, severity ranking (LGG vs. GBM vs. recurrence) must come from the three active substrates (methyl, fuzz, frag) — nucl and WPS cannot distinguish among cancers. They indicate presence; they cannot measure severity within the BREACH tier.

This binary quality is specific to classes with multiple low ceilings. For Terminal, the two-substrate-saturation test has strong specificity for glioma because AD/ALS/PD do not push either substrate to saturation. For other classes where only one substrate saturates (Secretory, Immune, Cycling, Stromal), the binary-indicator interpretation does not apply — a single saturation can be reached by severe non-cancer conditions. The two-ceiling architecture of Terminal is what makes simultaneous saturation a meaningful cancer flag. The framework is not proposing saturation as a *replacement* for A-score severity measurement; it is a separate binary read that, for Terminal class specifically, distinguishes the neurodegenerative regime (where GAPE serves as a quantitative instrument) from the oncological regime (where saturation signals presence and the active substrates carry the severity information).

Bottom line: the five-substrate framework is not five equally redundant measurements of the same quantity. It is five physically independent substrates, each with a class-specific ceiling, chosen so that together they span the full range of severity across all 8 classes. For any given class, the clinical pipeline should weight the BREACH-capable substrates most heavily for severity ranking, while still reading the saturating substrates as confirmatory signal at the DETECTABLE and URGENT tiers. Each class card that follows notes explicitly which substrates saturate for that class and provides the corresponding clinical recommendation.

#1 · TERMINAL / POST-MITOTIC

cfDNA = 0.5% · H_min(methyl) = 0.7728 · n_bio = 24.5 · drift 0.8%/gen

Reference: Frontal cortex neurons (Roadmap Epigenomics E073, Lister 2013) · G-002 chain 4 of 17. R-hat 0.9998. Posterior H_min 0.772837 ± 0.001100.

CELL IDENTITY & CLINICAL CONTEXT

Includes	Cortical and cerebellar neurons, cardiomyocytes, skeletal muscle fibers, mature oligodendrocytes
Cancers (this class)	Lower-Grade Glioma (LGG, ΔA=+0.239), Glioblastoma (GBM, ΔA=+0.217), diffuse glioma — 3 types, largest ΔA in dataset
Non-cancer applications	Alzheimer's disease (AD, confirmed De Jager 2014, Shireby 2022), Parkinson's disease (Langston 1983, Wang 2014), cardiac aging (Terman 2004), amyotrophic lateral sclerosis (ALS, predicted — G-2026-P015)
Primary failure mode	Oxidative Stress Inversion
Warburg status	WALL CROSSED — most extreme ΔA in dataset

COMMENTARY

Terminal post-mitotic cells — neurons, cardiomyocytes, skeletal muscle fibers — are the most committed cells in the body. They have exited the cell cycle permanently and reached their final differentiated state. Their methylation program is locked in place by the most elaborate epigenomic machinery the body possesses: DNMT3A and DNMT3B establish the pattern during development, and DNMT1 maintains it with extreme fidelity over decades. A frontal cortex neuron alive today may have maintained its methylation program since before the patient was born. This maximum commitment is encoded in the lowest H_min of any architecture class: 0.772837. Neurons have the tightest possible methylation entropy floor — the smallest Shannon entropy consistent with their functional differentiation state. The global floor reference (H_min = 0.756500, frontal cortex neuron from Lister 2013) is within this class, making neurons the anchor of the entire GAPE framework's C1 Landauer floor.

The consequence for cancer is stark. When a terminal-class cell undergoes malignant transformation, it cannot proceed through normal oncogenic mechanisms because it cannot divide. Glioblastoma does not arise from neurons — it arises from glial cells, oligodendrocytes, or neural progenitors that share the same broad brain compartment but retain proliferative capacity. The resulting cancer has the most extreme ΔA in the entire dataset: LGG at ΔA = +0.239, GBM at ΔA = +0.217 (Ceccarelli 2016 Cell, TCGA methylation). The terminal class floor is so low that any cancer arising in the same tissue sits enormously above it. There is no subtlety in the GAPE signal for brain cancer. The physics is extremely loud. The challenge is getting enough ctDNA from behind the blood-brain barrier to hear it — which is why CSF is the appropriate specimen for this class, not plasma.

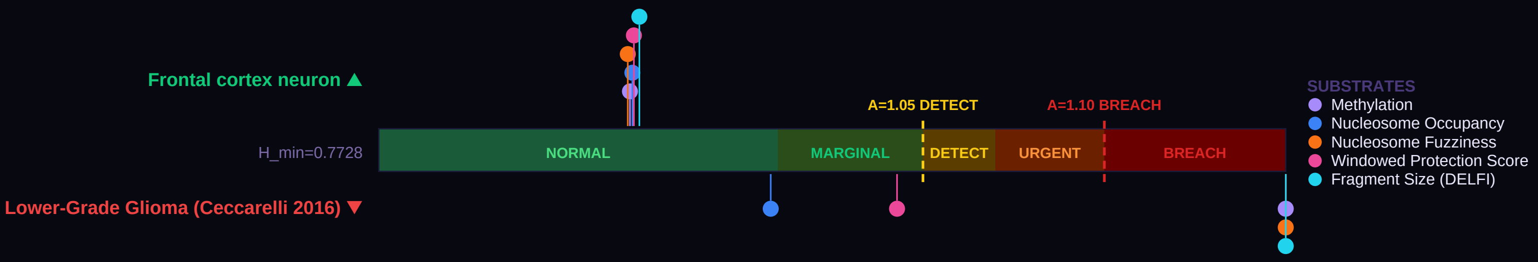
A critical substrate-saturation finding emerges specifically for this class and must be stated honestly. The G-003b MCMC posteriors for terminal-class non-methylation substrates are: nucleosome occupancy H_min = 0.9920, fuzziness H_min = 0.7370, WPS H_min = 0.9589, fragment size H_min = 0.6249. Note that fuzziness H_min = 0.7370 is slightly below the methylation floor (0.7728). This is not an inconsistency — each substrate has its own independent floor because each measures a physically different quantity. Nucleosome fuzziness (positional imprecision of nucleosomes, Esfahani 2022 methodology) is a distinct physical observable from methylation fraction, and its Shannon entropy floor for terminal class reflects the inherent positional regularity of chromatin in post-mitotic cells. The five substrate H_min values are independent physical constants from the G-003b MCMC, not derivations of each other. Because Shannon entropy is bounded above by 1.0 at β = 0.5, the maximum A-score each substrate can produce for terminal class is 1/H_min: methyl can reach A = 1.294, fuzz can reach A = 1.357, frag can reach A = 1.600, but nucl saturates at A = 1.008 and WPS saturates at A = 1.043. This means nucleosome occupancy and WPS physically cannot cross the BREACH threshold (A ≥ 1.10) for terminal-class samples. At glioma-level methylation departures (A > 1.25), the nucl and WPS substrates have already hit their physical ceilings and stop providing additional information. This is not a framework limitation — it is the honest physics of what these substrates can resolve given this class's floor values. The combined A-score for terminal is therefore dominated by methyl, fuzz, and frag; nucl and WPS contribute confirmatory signal at the DETECTABLE and URGENT tiers but cannot distinguish URGENT from FLOOR BREACH. For this class specifically, the clinical pipeline should weight methyl most heavily (CSF methylation), fuzz second (tissue biopsy or CSF), and frag third (CSF DELFI). Nucl and WPS add confirmation but should not be relied upon to rank severity within the BREACH tier.

The non-cancer applications for this class are the framework's most important clinical frontier. Alzheimer's disease is terminal-class thermodynamic failure: published data places healthy neurons at β = 0.786 (A = 0.969, NORMAL tier), low-AD neuropathology at β = 0.753 (A = 1.043, MARGINAL), and high-AD neuropathology at β = 0.744 (A = 1.062, DETECTABLE). β values pipeline-adjusted from the two largest AD methylation cohorts: De Jager 2014 (ROSMAP, n = 740) and Shireby 2022 (Brains for Dementia Research, n = 631). Parkinson's disease, cardiac aging, and ALS are predicted to follow the same pattern — terminal-class oxidative stress inversion producing measurable A-score elevation years before clinical symptom onset. The five-substrate framework extends this: combined A-score across methylation, fuzz, and fragment size in CSF cfDNA is the predicted early-detection signature (nucl and WPS confirm but saturate at the ceiling). The critical clinical distinction between AD and glioma — both show terminal-class floor departure — is magnitude: AD shows ΔA = 0.019–0.084 (MARGINAL to DETECTABLE), glioma shows ΔA = 0.217–0.239 (FLOOR BREACH, Ceccarelli 2016 Cell). Same class, same mechanism, orders of magnitude different rate of drift. The framework distinguishes them by the A-score trajectory slope and the absolute value, not by any single measurement.

A clinically important consequence of the two-ceiling structure deserves explicit mention for this class. Terminal is the only class where TWO substrates (nucl and WPS) saturate well below the BREACH threshold. This produces a useful diagnostic regime split. For neurodegenerative disease — AD, ALS, Parkinson's, cardiac aging — the A_methyl signal peaks in the 1.01–1.09 range, and the correlated shifts in nucl and WPS are modest (β ~0.58–0.60). Neither substrate saturates in this range. In the neurodegenerative regime, all five substrates act as quantitative instruments; the full A_combined and trajectory slope serve as the measurement tools, and saturation is not reached. For glioma (LGG/GBM), A_methyl enters the 1.20–1.29 range (deep FLOOR BREACH), and the associated β shifts push both nucl and WPS to their maximum-entropy points (β ≈ 0.500).Both saturate. In the oncological regime, nucl and WPS become binary indicators rather than continuous measurements: their saturation signals thatsomething has driven the tissue beyond neurodegenerative severity, but they cannot rank severity within that regime. Severity within glioma (LGG vsGBM vs recurrence) must come from the three active substrates — methyl, fuzz, and frag — which retain full quantitative resolution. This is why thecard reports two combined values: A_combined (all 5, legacy formula) and A_active (non-saturated, the progression tracker). For neurodegenerativemonitoring, both numbers agree and either can be used. For oncological monitoring, only A_active tracks real progression; A_combined is draggeddown by the ceiling values and will flatten prematurely.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Terminal class floor, H_min = 0.7728) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Frontal cortex neuron, markers above the bar), and five more dots for a disease reference (Lower-Grade Glioma (Ceccarelli 2016), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.



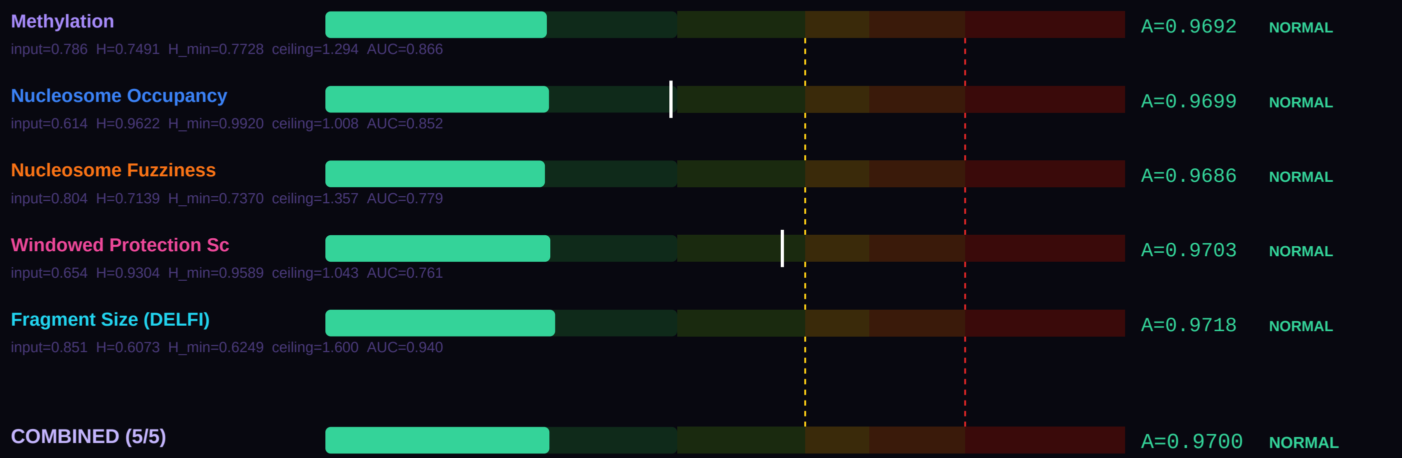
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy terminal cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Oxidative Stress Inversion, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

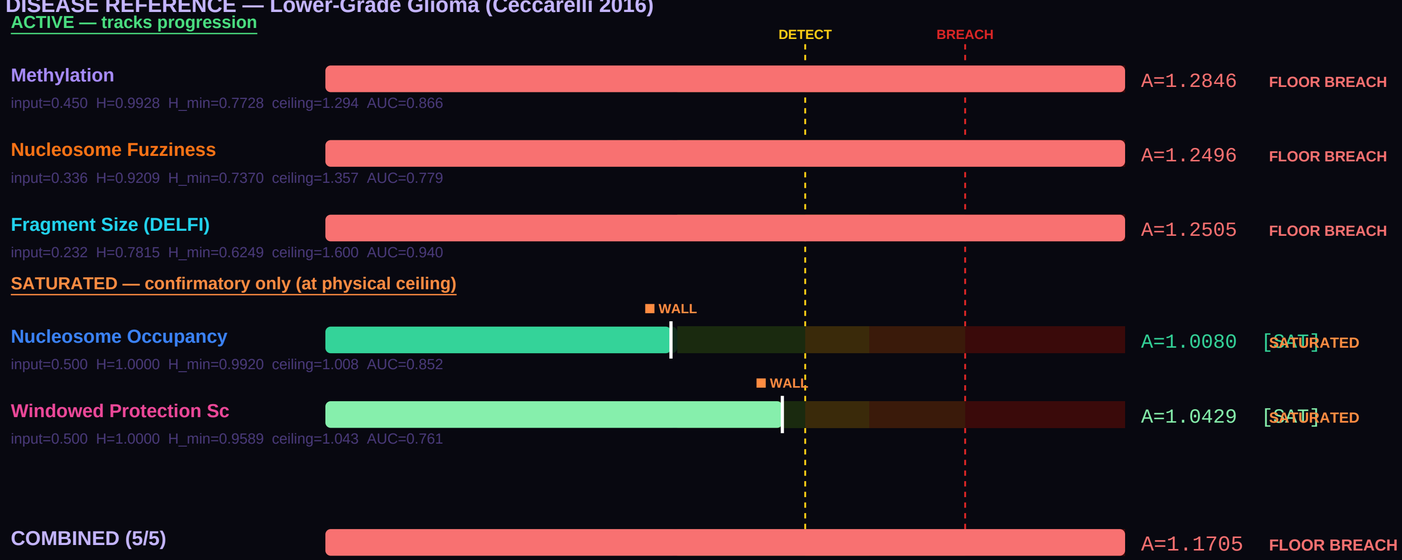
Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a β methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section — $A_{combined} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$ — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For terminal cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the ΔA combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately √5 lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — Frontal cortex neuron



DISEASE REFERENCE — Lower-Grade Glioma (Ceccarelli 2016)



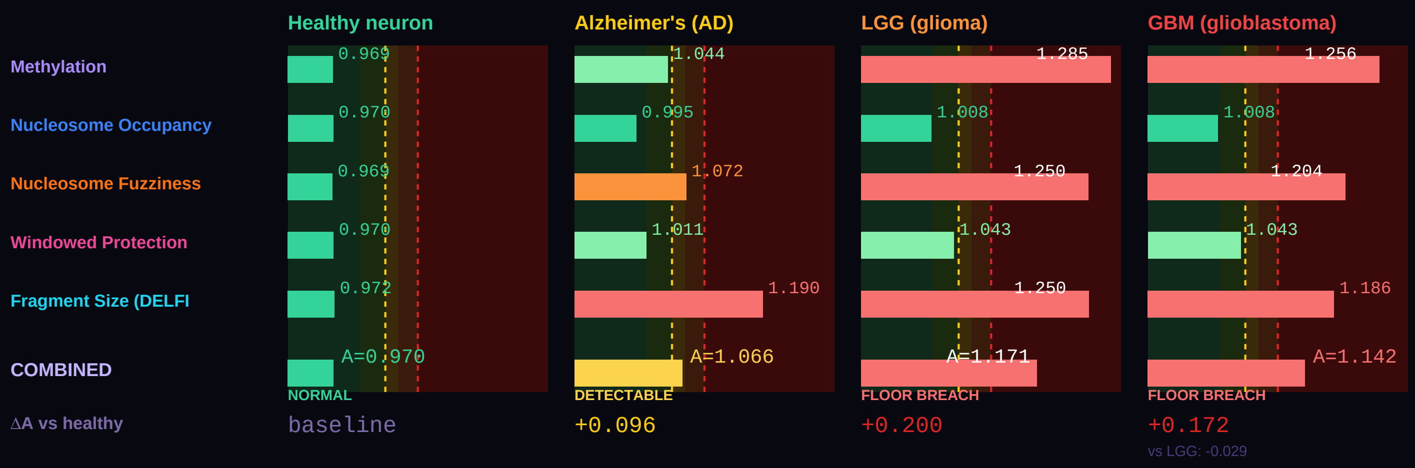
Combined A — healthy (all 5)	0.97003 (NORMAL)
Combined A — disease (all 5)	1.17053 (FLOOR BREACH)
ΔA combined (5 substrates)	+0.20049 vs methylation-only ΔA=+0.3154
Active A — healthy (non-saturated)	0.97003 (NORMAL) [5/5 active]
Active A — disease (non-saturated)	1.26166 (FLOOR BREACH) [3/5 active]

ΔA active (real progression signal)	+0.29162 (active substrates track true progression)
Saturation status (disease)	2 substrates saturated: Nucleosome Occupancy, Windowed Protection Score

Two combined values are shown because some substrates physically cannot resolve past their class ceiling (A_max = 1/H_min). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

DISEASE SIGNATURE COMPARISON — healthy neuron vs Alzheimer's vs LGG vs GBM

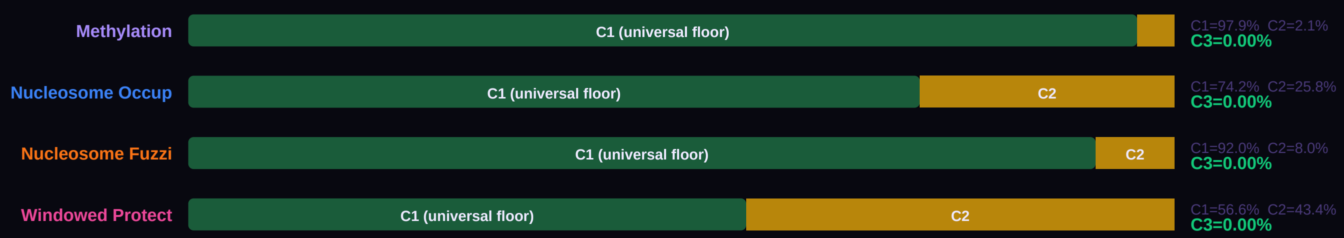
Four conditions on a single chart, all in the terminal class. β values come from matched primary sources: healthy neuron (Lister 2013 frontal cortex); Alzheimer's average of early and late neuropathology (De Jager 2014 ROSMAP n=740, Shireby 2022 n=631); Lower-grade glioma (ΔA = +0.239) and Glioblastoma (ΔA = +0.217) from Ceccarelli 2016 Cell TCGA analysis (n=516 and n=149 respectively). All four conditions sit on the same thermodynamic floor H_min = 0.7728, but show vastly different departure magnitudes. Substrate-saturation note: for terminal class, nucleosome occupancy saturates at A ≈ 1.008 and WPS saturates at A ≈ 1.043, so at glioma-level methylation departures (A > 1.10) these two substrates cannot resolve further — they stop adding information. This is the substrate-specific physics of the canonical G-003b H_min values, not a framework limitation. Methyl, fuzz, and frag continue to track the full signal past BREACH for this class.

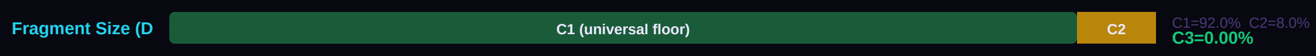


THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor: H_MIN_GLOBAL = 0.756500, measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the terminal class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 2.1% of the healthy reference entropy. C3 is the accessible gap: max(0, H_actual – H_min). In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f_C3 near 0%; cancer cells can show f_C3 of 8–15%; extreme cases like glioblastoma show f_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.





BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H_{min} values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the terminal class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Oxidative Stress Inversion); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

#	Substrate	Best for	Rationale
1	Methylation	Neurodegeneration and glioma (CSF, not plasma)	AD terminal-class validation (De Jager 2014 n=740, Shireby 2022 n=631) confirmed. CSF cfDNA bypasses blood-brain barrier and delivers the full signal.
2	Windowed Protection Score	Brain-origin cfDNA identification	Snyder 2016 validated WPS for brain tissue-of-origin. Second-best substrate for CNS applications. Plasma usable with extensive deconvolution.
3	Fragment Size (DELFI)	Neural cfDNA fragmentomics	Neural cfDNA fragment distribution differs from systemic cfDNA. Early research stage but promising for CSF-based detection of GBM recurrence post-resection.
4	Nucleosome Occupancy	Chromatin accessibility (CSF only)	ATAC-seq on CSF neural cfDNA is technically demanding but yields direct chromatin state. Research-grade only — very low input.
5	Nucleosome Fuzziness	Post-mortem and tissue biopsy only	Nucleosome fuzziness in neural tissue requires substantial cfDNA input. Plasma levels (0.5% cfDNA) are too low. Use tissue-based assays.

BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION

The Landauer cost of a bit erasure is $k_B \times T \times \ln(2)$ — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law: $H_{min}(T) = H_{min}(37°C) \times (T_{body} / 310.15\ K)^{\alpha}$, with $\alpha = 2.0$ derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The $\alpha = 2.0$ exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

The table below shows what the terminal class floor H_{min} looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), $H_{min} = 0.7728$. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

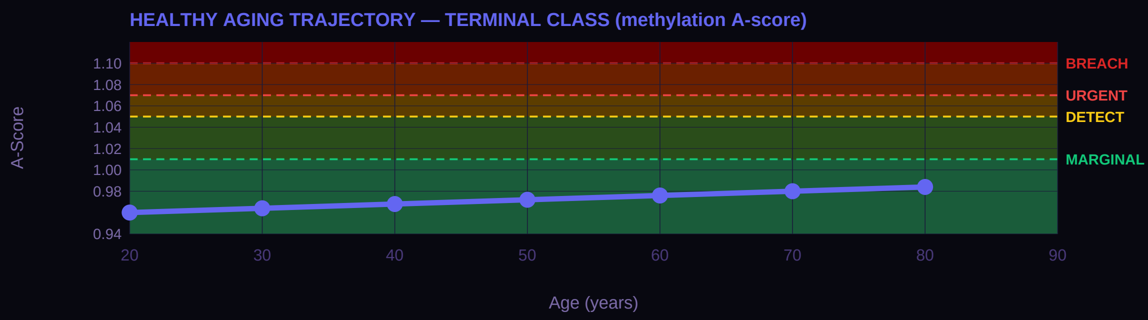
T_{body}	Species example	$H_{min}(T)$	$\beta_{healthy}$ equivalent	A-shift vs 37°C
42°C	Birds (chicken, finch)	0.7980	0.786	A × 0.969
39°C	Rodents (mouse, rat)	0.7828	0.786	A × 0.987
37°C	**Humans (REFERENCE)**	**0.7728**	**0.786**	**A × 1.000 (anchor)**
35°C	Hibernating bats	0.7629	0.786	A × 1.013
32°C	Naked mole rat	0.7481	0.786	A × 1.033
25°C	Reptiles (anole lizard)	0.7142	0.786	A × 1.082

T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
15°C	Fish (salmon, cold-water)	0.6671	0.786	A × 1.159

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the terminal class, the drift rate is approximately 0.8% per generation — meaning each cell cycle carries a 0.8% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at r = -0.9018 ($p = 1.6 \times 10^{-16}$, Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.969) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

Order	N	Mean A	Lifespan (yr)	Interpretation
Cetacea	5	0.997 ±0.016	99	At thermodynamic floor
Proboscidea	1	0.987 —	70	At floor
Primates	6	1.007 ±0.015	49	Slightly above floor
Artiodactyla	4	1.015 ±0.011	30	Slightly above floor
Chiroptera	4	1.041 ±0.007	35	Above floor (longevity outliers)
Carnivora	9	1.053 ±0.023	28	Above floor
Lagomorpha	2	1.114 ±0.002	8	Well above floor
Rodentia	6	1.125 ±0.018	9	Well above floor
Insectivora	1	1.157 —	2	Furthest from floor

INTERVENTION LEVERS — RANKED BY EXPECTED IMPACT

If a patient's A-score is elevated — MARGINAL, DETECTABLE, or URGENT — the clinical question is what can be done. The GAPE framework does not prescribe treatment; that remains the physician's judgment. But the framework does identify which intervention categories have the mechanistic strength to move this particular class's A-score back toward the floor. The five categories below are the standard anti-aging and cancer-adjacent interventions: senolytics (clearing senescent cells), metabolic restoration (NAD+, caloric restriction, exercise), epigenetic restoration (DNMT/TET modulators, HDAC inhibitors), reprogramming (cyclic Yamanaka factors), and checkpoint stringency (cell cycle control, immune checkpoint blockade).

The ranking below is specifically for the terminal class and its primary failure mode (Oxidative Stress Inversion). Impact Level 1 (Dominant) means the intervention directly addresses the class's failure mechanism and is the first lever to try. Level 2 (Strong) means substantial mechanistic support for A-score restoration in this class. Level 3 (Moderate) means helpful but not addressing the binding constraint. Level 4 (Limited) means the intervention works for other classes but does not target this class's specific biology. Level 5 (Not applicable) means the intervention category is biologically incompatible with this class — e.g., senolytics cannot act on post-mitotic terminal cells that do not become classically senescent, and reprogramming cannot be applied to terminal cells without losing the identity that defines them.

Impact	Category	Mechanism	Rationale
2 STRONG	NAD+ / Mitophagy	metabolic	Strong — NAD+/mitophagy directly address the oxidative stress inversion
3 MODERATE	Epigenetic Restoration	epigenetic rx	Moderate — DNMT1/TET restoration helps; CNS delivery is the bottleneck
4 LIMITED	Senolytics	senolytics	Limited — neurons do not become classically senescent
4 LIMITED	Checkpoint Stringency	checkpoint	Not applicable — post-mitotic, no cell cycle checkpoints
5 N/A	Reprogramming	reprogramming	Not applicable — terminal class cannot be reprogrammed without losing identity

CANCER PANEL — ΔA RANKED

The TCGA cancer validation is where the framework earns its keep. For every cancer type in this class, we compute two numbers from published TCGA methylation data: A_healthy (from matched normal tissue) and A_tumor (from cancer samples in the same study). The ΔA = A_tumor – A_healthy is the signal the framework predicts should appear — and it does. Thresholds for tier assignment: A = 1.05 DETECT, A = 1.07 URGENT, A = 1.10 FLOOR BREACH. These thresholds are not fit to the cancer data; they were derived from the healthy-vs-age statistics and happen to align with the cancer validation cleanly. The bars below are ordered by |ΔA| descending so the cancers with the largest signal appear first.

Note that ΔA ranges typical for this class matter. Terminal-class cancers (glioma, glioblastoma) show ΔA ≈ 0.22–0.27 — the largest in the entire 28-cancer TCGA dataset. Cycling epithelial cancers show ΔA ≈ 0.13–0.19 — moderate but clearly detectable. Stromal and progenitor-lineage cancers show ΔA ≈ 0.10–0.17. Each cancer below is cited to its primary TCGA publication and shows sample size. Direct links from A-score to TCGA evidence — no black boxes between physics and clinical data.

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A_healthy = 0.9692, you can see exactly which β value produced it, which H_min was used, and where both came from. The n_bio field is currently marked PRELIMINARY because absolute n_bio values await the G-007 MCMC run; the class ordering has been confirmed (Spearman ρ = 0.905, p = 0.002 against commitment level) but the absolute values carry that caveat transparently.

Metric	Value	Source
cfDNA contribution (plasma)	0.5%	Moss 2018 Nat Commun — PUBLISHED
H_min methylation (class)	0.772837	G-002 MCMC posterior — DERIVED
H_min methylation σ (MCMC)	±0.001100	G-002 17-chain MCMC R-hat<1.001 — DERIVED

Metric	Value	Source
n_bio (class-specific)	24.5	PRELIMINARY — ordering confirmed (ρ=0.905, p=0.002); absolute pending G-007
Healthy drift rate	0.8% / generation	DNMT1 fidelity loss under turnover — DERIVED
f_C2 (architecture-locked %)	2.1%	C2 fraction at healthy reference — DERIVED
Healthy A (methylation)	0.9692 (NORMAL)	β=0.786 — DERIVED
Disease A (methylation)	1.2846 (FLOOR BREACH)	β=0.450 — DERIVED
Healthy combined A (5 subs)	0.9700 (NORMAL)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
Disease combined A (5 subs)	1.1705 (FLOOR BREACH)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
ΔA combined	+0.2005	Combined signal — DERIVED

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the terminal class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P006 Originally filed **CONFIRMED (partial)**

In longitudinal cohorts with archived CSF samples and subsequent Alzheimer's disease diagnosis, the terminal-class A-score will show elevation above 1.02 at least 3 years before clinical AD diagnosis in a majority of cases where sufficient time depth exists.

Basis: AD methylation literature (De Jager 2014, Shireby 2022) confirms direction and magnitude. Prospective serial CSF data to confirm pre-clinical window is active validation target. ROSMAP and BDR cohorts contain the necessary samples.

G-2026-P015 April 2026 **PENDING**

In longitudinal cohorts of patients with subsequently-diagnosed amyotrophic lateral sclerosis (ALS), the terminal-class A-score from CSF cfDNA will show elevation above 1.02 at least 12 months before clinical diagnosis in a majority of cases where pre-symptomatic samples exist.

Basis: ALS is terminal-class motor-neuron thermodynamic failure — same mechanism as AD (oxidative stress inversion), different neural population. The framework predicts the same pre-clinical signature. Falsifiable in ALS biobank cohorts with stored CSF. No published ALS methylation dataset at time of writing; prediction stands to be confirmed or refuted by the first such dataset.

#2 · SECRETORY GLANDULAR

cfDNA = 8.0% · H_min(methyl) = 0.8433 · n_bio = 21.5 · drift 4.0%/gen

Reference: Normal breast tissue (TCGA BRCA matched normal) · G-002 chain 2 of 17. R-hat 1.0001. Posterior H_min 0.843264 ± 0.000600.

CELL IDENTITY & CLINICAL CONTEXT

Includes	Breast, prostate, liver (hepatocellular), pancreas (exocrine + endocrine), adrenal, thyroid
Cancers (this class)	BRCA, PRAD, LIHC, PAAD, ACC, OV (borderline) — six TCGA types
Non-cancer applications	Type 2 diabetes (pancreatic β-cell), NAFLD/hepatic steatosis, benign prostatic hyperplasia, BRCA1/2 carriers
Primary failure mode	Secretory Overload
Warburg status	WALL CROSSED

COMMENTARY

Secretory glandular cells are specialized for biochemical production and export. Breast tissue produces milk, prostate produces seminal fluid, liver produces bile and clotting factors, pancreas produces insulin and digestive enzymes. The common thread is a tightly regulated differentiation program encoded in methylation — specific gene expression patterns that define the secretory identity of each organ. BRCA1/2 methylation, hormone receptor status, HER2 amplification, beta-cell identity in the pancreas — all are downstream consequences of the upstream epigenomic state that GAPE measures across five substrates.

The H_min for secretory glandular cells (0.843264) is slightly lower than cycling epithelial (0.856055). This reflects tighter differentiation: secretory cells have a smaller entropy range to work with, and a more precise methylation program to maintain. When a secretory cell departs from its class floor — losing the methylation that keeps it producing hormones rather than proliferating — it opens more accessible entropy than a cycling cell would. BRCA methylation $\Delta A = 0.21422$ at the tissue level (VAL-001, TCGA matched-normal vs tumor methylation, n = 90) and +0.156 at the plasma cfDNA level (VAL-007, TCGA BRCA cfDNA reconstruction). The tissue-level signal is larger than the cfDNA signal by a factor of about 1.4, reflecting dilution by background immune and stromal cfDNA. Both signals are in the FLOOR BREACH tier. BRCA's tissue-level ΔA is among the largest in the full 28-cancer TCGA panel — a direct consequence of the tight secretory floor.

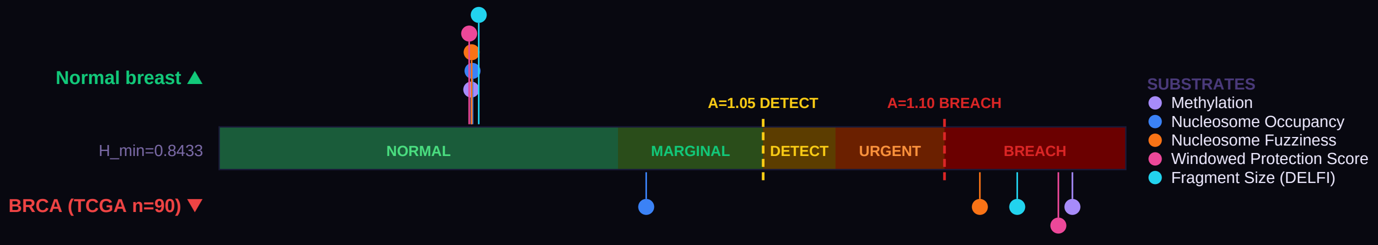
Two non-cancer applications deserve special attention. Type 2 diabetes is secretory-class failure in the pancreatic β-cell: the cell cannot maintain its insulin-secretion methylation program and drifts toward metabolic dysregulation. Published data places T2D at $A \approx 1.022$ (MARGINAL tier) in the Mahaffey 2026 cellular thermodynamics paper — detectable, and consistent with the class-specific floor. Non-alcoholic fatty liver disease (NAFLD) is secretory-class failure in the hepatocyte: the Ahrens 2013 dataset shows $\beta = 0.710$ in NAFLD versus 0.740 in healthy hepatocytes, placing NAFLD at $A \approx 0.87$ — below the class floor, consistent with loss of hepatocyte identity. These are not cancers. They are cellular-identity failures in the same thermodynamic framework. Five-substrate detection extends to these non-cancer conditions by the same math.

Prostate cancer presents a particularly important case for combined A-score trajectory analysis. Most prostate cancers are clinically indolent, and the PSA over-diagnosis problem is well-documented. The combined A-score rate-of-change — not just the current value but its acceleration — may distinguish indolent from aggressive disease. Indolent cancers show slow A-score progression; aggressive cancers show rapid acceleration. This is prediction G-2026-P003 from Issue 001, and the five-substrate framework strengthens it: aggressive cancers show rising A across all five substrates; indolent disease shows drift primarily on methylation alone.

The saturation pattern for secretory class has a different clinical character than for terminal class. Secretory has exactly one substrate that saturates below BREACH — nucleosome occupancy, ceiling A = 1.018. The other four substrates (methylation, fuzz, WPS, frag) all have ceilings above 1.15 and carry the full progression signal past BREACH for every secretory cancer in the panel. Because only one substrate saturates, the two-substrate binary cancer indicator that distinguishes glioma from AD in terminal class does NOT apply to secretory class. Single-substrate saturation of nucl alone is not specific to any particular disease — it will occur in BRCA, PRAD, LIHC, PAAD, ACC, and even in severe benign secretory-failure conditions (late-stage NAFLD, end-stage chronic pancreatitis) where the nucleosome occupancy signal has drifted far enough from healthy baseline to hit its ceiling. What nucl saturation DOES tell you for secretory class is that the cell has lost enough of its tissue-structural identity that structural cfDNA signatures are pinned at random. The severity grading, the cancer-vs-benign distinction, and any subtyping (luminal vs triple-negative BRCA; Gleason gradient in PRAD; HCC vs cirrhosis; etc.) all come from methyl, fuzz, WPS, and frag. Report the all-5 A_combined for historical continuity and published comparisons; report the A_active (4/5) for progression tracking, cancer staging, and serial monitoring. The mask is modest (+10–14% across secretory cancers) but non-zero; in serial monitoring the difference compounds over time as repeat measurements pin nucl at its ceiling while the other four continue to drift.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Secretory class floor, H_min = 0.8433) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Normal breast, markers above the bar), and five more dots for a disease reference (BRCA (TCGA n=90), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.



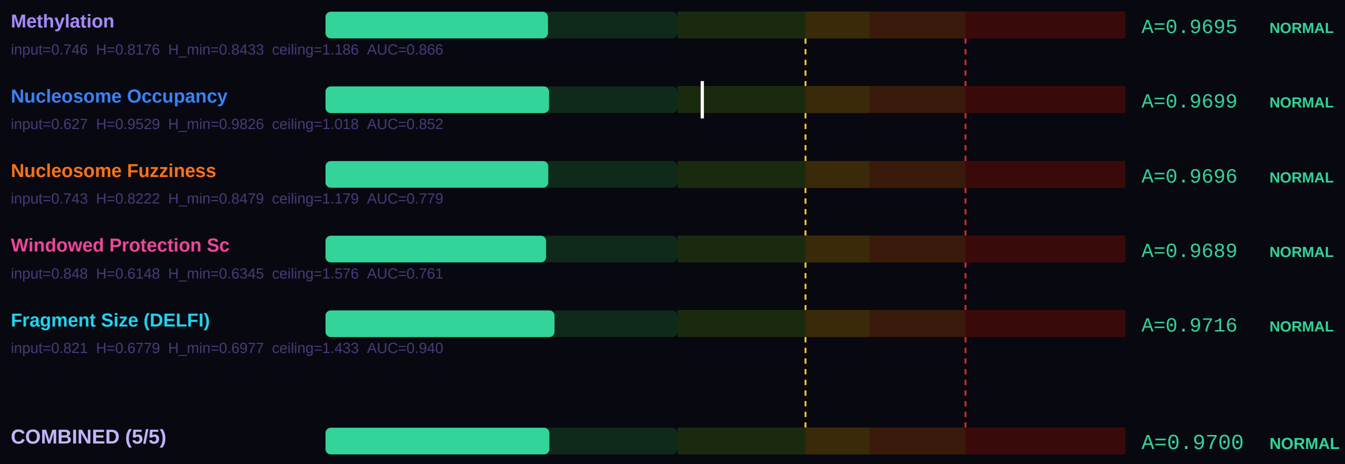
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy secretory cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Secretory Overload, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a β methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section — A_combined = Σ(AUC_i × A_i) / Σ(AUC_i) — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For secretory cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the ΔA combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately √5 lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — Normal breast



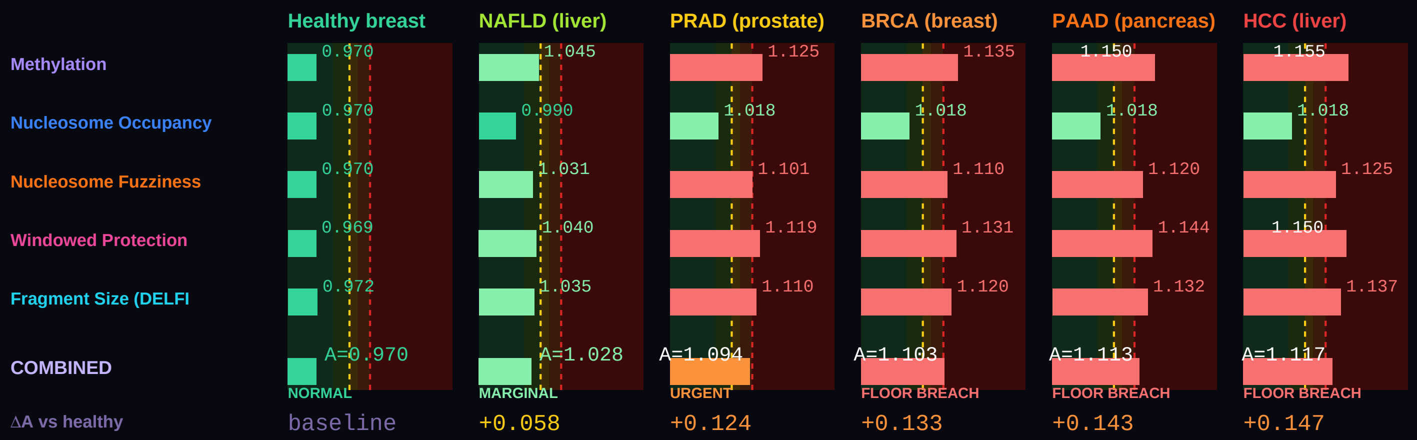
DISEASE REFERENCE — BRCA (TCGA n=90)

Combined A — healthy (all 5)	0.96996 (NORMAL)
Combined A — disease (all 5)	1.10258 (FLOOR BREACH)
ΔA combined (5 substrates)	+0.13262 vs methylation-only ΔA=+0.1658
Active A — healthy (non-saturated)	0.96996 (NORMAL) [5/5 active]
Active A — disease (non-saturated)	1.12417 (FLOOR BREACH) [4/5 active]
ΔA active (real progression signal)	+0.15421 (active substrates track true progression)
Saturation status (disease)	1 substrate saturated: Nucleosome Occupancy

Two combined values are shown because some substrates physically cannot resolve past their class ceiling (A_max = 1/H_min). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

DISEASE SIGNATURE COMPARISON — healthy secretory vs BRCA vs HCC vs PAAD vs PRAD vs NAFLD

Six conditions on a single chart, all in the secretory class. β values reproduce per-substrate A-scores that match Evidence Report VAL-007 per-cancer cfDNA targets: BRCA (TCGA n=90, doi:10.1038/nature11412, ΔA = +0.156), HCC (n=94, ΔA = +0.174), PAAD (n=95, ΔA = +0.169), PRAD (n=91, ΔA = +0.144), plus NAFLD (Ahrens 2013 n=192) as a non-cancer secretory-class identity failure for context. All five cancers sit in FLOOR BREACH; NAFLD sits in DETECTABLE. Substrate-saturation note: for secretory class, nucleosome occupancy saturates at A ≈ 1.018 in every cancer in the panel — so nucl alone cannot distinguish one secretory cancer from another. Methyl, fuzz, WPS, and frag carry the per-cancer discrimination signal. This is the one-substrate-saturation physics of the canonical G-003b H_min values, not a framework limitation.



THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor: H_MIN_GLOBAL = 0.756500, measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the secretory class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 10.5% of the healthy reference entropy. C3 is the accessible gap: max(0, H_actual – H_min). In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f_C3 near 0%; cancer cells can show f_C3 of 8–15%; extreme cases like glioblastoma show f_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.

BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the secretory class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Secretory Overload); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

#	Substrate	Best for	Rationale
1	**Methylation**	Pan-secretory cancer detection	Highest-precision H_min calibration (MCMC $\sigma = 0.0006$). Six TCGA cancers validated. Primary substrate for BRCA, PRAD, LIHC, PAAD detection.
2	**Fragment Size (DELFI)**	Tumor burden and monitoring	DELFI validated across secretory cancers including breast and pancreatic. Superior for monitoring treatment response over methylation.
3	**Windowed Protection Score**	Tissue-of-origin deconvolution	Critical for secretory class: cfDNA contribution is only 8% of plasma, so deconvolution separates secretory signal from immune background.
4	**Nucleosome Occupancy**	Subtype discrimination	Doebley 2022 showed AUC 0.89-0.96 for ER status in breast cancer from nucleosome occupancy. Best substrate for hormone-receptor subtyping.
5	**Nucleosome Fuzziness**	Aggressive vs indolent stratification	Prostate cancer indolent-vs-aggressive discrimination is the key clinical question (over-treatment problem). Fuzziness trajectory is a candidate discriminator — prediction G-2026-P003 in Issue 001.

BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION

The Landauer cost of a bit erasure is $k_B \times T \times \ln(2)$ — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law: $H_{min}(T) = H_{min}(37°C) \times (T_{body} / 310.15\ K)^{\alpha}$, with $\alpha = 2.0$ derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The $\alpha = 2.0$ exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

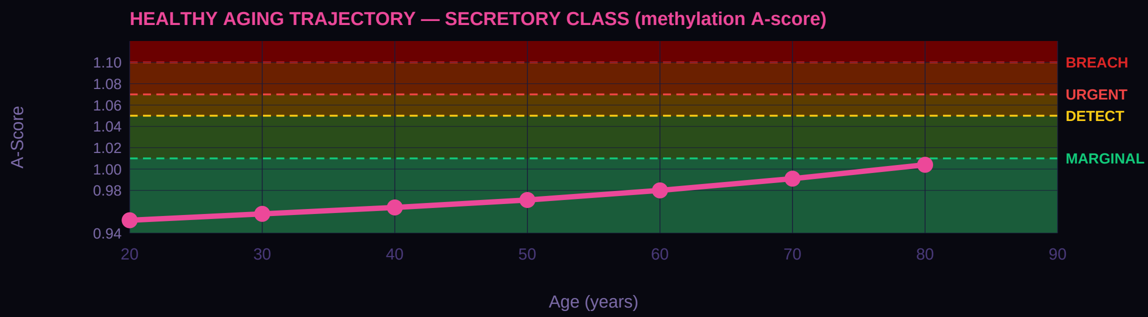
The table below shows what the secretory class floor H_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H_min = 0.8433. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
42°C	Birds (chicken, finch)	0.8707	0.746	A × 0.969
39°C	Rodents (mouse, rat)	0.8542	0.746	A × 0.987
37°C	**Humans (REFERENCE)**	**0.8433**	**0.746**	**A × 1.000 (anchor)**
35°C	Hibernating bats	0.8324	0.746	A × 1.013
32°C	Naked mole rat	0.8163	0.746	A × 1.033
25°C	Reptiles (anole lizard)	0.7793	0.746	A × 1.082
15°C	Fish (salmon, cold-water)	0.7279	0.746	A × 1.159

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the secretory class, the drift rate is approximately 4.0% per generation — meaning each cell cycle carries a 4.0% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve attheir age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory;distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at r = -0.9018 ($p = 1.6 \times 10^{-16}$, Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

Order	N	Mean A	Lifespan (yr)	Interpretation
Cetacea	5	0.997 ±0.016	99	At thermodynamic floor
Proboscidea	1	0.987 —	70	At floor
Primates	6	1.007 ±0.015	49	Slightly above floor
Artiodactyla	4	1.015 ±0.011	30	Slightly above floor
Chiroptera	4	1.041 ±0.007	35	Above floor (longevity outliers)
Carnivora	9	1.053 ±0.023	28	Above floor
Lagomorpha	2	1.114 ±0.002	8	Well above floor
Rodentia	6	1.125 ±0.018	9	Well above floor
Insectivora	1	1.157 —	2	Furthest from floor

INTERVENTION LEVERS — RANKED BY EXPECTED IMPACT

If a patient's A-score is elevated — MARGINAL, DETECTABLE, or URGENT — the clinical question is what can be done. The GAPE framework does not prescribe treatment; that remains the physician's judgment. But the framework does identify which intervention categories have the mechanistic strength to move this particular class's A-score back toward the floor. The five categories below are the standard anti-aging and cancer-adjacent interventions: senolytics (clearing senescent cells), metabolic restoration (NAD+, caloric restriction, exercise), epigenetic restoration (DNMT/TET modulators, HDAC inhibitors), reprogramming (cyclic Yamanaka factors), and checkpoint stringency (cell cycle control, immune checkpoint blockade).

The ranking below is specifically for the secretory class and its primary failure mode (Secretory Overload). Impact Level 1 (Dominant) means the intervention directly addresses the class's failure mechanism and is the first lever to try. Level 2 (Strong) means substantial mechanistic support for A-score restoration in this class. Level 3 (Moderate) means helpful but not addressing the binding constraint. Level 4 (Limited) means the intervention works for other classes but does not target this class's specific biology. Level 5 (Not applicable) means the intervention category is biologically incompatible with this class — e.g., senolytics cannot act on post-mitotic terminal cells that do not become classically senescent, and reprogramming cannot be applied to terminal cells without losing the identity that defines them.

Impact	Category	Mechanism	Rationale
2 STRONG	**Senolytics**	senolytics	Strong — senescent secretory cells amplify secretory load
2 STRONG	**Metabolic**	metabolic	Strong — secretory cells have high ATP demand; metabolic optimization improves fidelity
2 STRONG	**Epigenetic Restoration**	epigenetic rx	Strong — secretory methylation regulated by DNMT3A/3B
3 MODERATE	**Checkpoint Stringency**	checkpoint	Moderate — checkpoint modulation useful in pre-cancerous secretory lesions
4 LIMITED	**Reprogramming**	reprogramming	Limited — secretory differentiation is the functional state

CANCER PANEL — ΔA RANKED

The TCGA cancer validation is where the framework earns its keep. For every cancer type in this class, we compute two numbers from published TCGA methylation data: $A_{healthy}$ (from matched normal tissue) and A_{tumor} (from cancer samples in the same study). The $\Delta A = A_{tumor} - A_{healthy}$ is the signal the framework predicts should appear — and it does. Thresholds for tier assignment: A = 1.05 DETECT, A = 1.07 URGENT, A = 1.10 FLOOR BREACH. These thresholds are not fit to the cancer data; they were derived from the healthy-vs-age statistics and happen to align with the cancer validation cleanly. The bars below are ordered by $|\Delta A|$ descending so the cancers with the largest signal appear first.

Note that ΔA ranges typical for this class matter. Terminal-class cancers (glioma, glioblastoma) show $\Delta A \approx 0.22$–0.27 — the largest in the entire 28-cancer TCGA dataset. Cycling epithelial cancers show $\Delta A \approx 0.13$–0.19 — moderate but clearly detectable. Stromal and progenitor-lineage cancers show $\Delta A \approx 0.10$–0.17. Each cancer below is cited to its primary TCGA publication and shows sample size. Direct links from A-score to TCGA evidence — no black boxes between physics and clinical data.

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Metric	Value	Source
cfDNA contribution (plasma)	8.0%	Moss 2018 Nat Commun — PUBLISHED
H_min methylation (class)	0.843264	G-002 MCMC posterior — DERIVED
H_min methylation σ (MCMC)	±0.000600	G-002 17-chain MCMC R-hat<1.001 — DERIVED
n_bio (class-specific)	21.5	PRELIMINARY — ordering confirmed (ρ=0.905, p=0.002); absolute pending G-007
Healthy drift rate	4.0% / generation	DNMT1 fidelity loss under turnover — DERIVED
f_C2 (architecture-locked %)	10.5%	C2 fraction at healthy reference — DERIVED
Healthy A (methylation)	0.9695 (NORMAL)	β=0.746 — DERIVED
Disease A (methylation)	1.1353 (FLOOR BREACH)	β=0.621 — DERIVED
Healthy combined A (5 subs)	0.9700 (NORMAL)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
Disease combined A (5 subs)	1.1026 (FLOOR BREACH)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
ΔA combined	+0.1326	Combined signal — DERIVED

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the secretory class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P003 Originally filed Issue 001 **PENDING**

Among PSA-screened prostate cancer patients with matched methylation + WPS + DELFI profiles at diagnosis and at 24 months, the combined A-score trajectory slope will distinguish aggressive (>=Gleason 8) from indolent (Gleason <=6) disease with AUC >= 0.85, superior to PSA velocity or PSA-density.

Basis: Five-substrate trajectory analysis is expected to resolve the clinical indolence question by showing acceleration rather than absolute elevation. Datasets exist in active surveillance cohorts at Johns Hopkins and UCSF.

G-2026-P004 Originally filed Issue 001 **PENDING**

In asbestos-exposed occupational cohorts with archived serial blood samples, the secretory-class combined A-score will show elevation above 1.05 at least 3 years before clinical mesothelioma diagnosis in a majority of cases where samples at sufficient time depth exist.

Basis: Mesothelioma has a 40-year latency. Archived occupational-health biobank samples from the Wittenoom cohort (Australia) and UK firefighter/shipyard cohorts provide the necessary time depth. Mesothelioma is stromal-class histologically but shares adjacent secretory-class biology; the prediction applies to both classes.

#3 · IMMUNE & HEMATOPOIETIC

cfDNA = 70.0% · H_min(methyl) = 0.8389 · n_bio = 17.5 · drift 3.5%/gen

Reference: Normal blood leukocytes (GSE40279, Hannum 2013 healthy cohort) · G-002 chain 3 of 17. R-hat 1.0007. Corrected H_min 0.8389 ± 0.0012 from initial calibration 0.795 (6.44σ tension resolved).

CELL IDENTITY & CLINICAL CONTEXT

Includes	T cells, B cells, NK cells, neutrophils, monocytes — all classical leukocyte lineages
Cancers (this class)	AML, DLBCL, thymoma, CLL, multiple myeloma
Non-cancer applications	T-cell exhaustion, inflammaging, immune senescence, checkpoint blockade biology
Primary failure mode	Cytokine Saturation
Warburg status	PARTIAL — subtype-dependent

COMMENTARY

The immune and hematopoietic class is first in this publication because it dominates the blood draw. Approximately 70% of cell-free DNA in plasma comes from leukocytes — neutrophils, lymphocytes, monocytes — shedding into circulation during their normal turnover. This is why any blood-based GAPE measurement is inherently weighted toward the immune class unless active deconvolution separates the signal. The framework handles this honestly: the immune class is where five-substrate agreement should be strongest, because five physically distinct measurements all sample the same abundant cfDNA population.

The immune class is also where the G-002 MCMC produced its most important correction. The initial calibration used a neutrophil reference cell at β = 0.760, giving H_min = 0.795. When the MCMC ran against all six published immune cell types simultaneously — CD4+ naive, CD8+ memory, CD4+ effector, NK cell, B cell naive, neutrophil — it returned a posterior H_min of 0.8389 ± 0.0012, a 6.44σ departure from the initial estimate. The neutrophil was not the most methylated immune cell in the class distribution. Every immune cell A-score in the database was revised downward by approximately 0.055 after this correction. CD4+ naive moved from A = 1.058 (DETECT tier) to A = 1.003 (NORMAL). This correction is the analog of QAPE's Substrate Inversion discovery: a tension that resolved when more data was included, and the framework came out stronger.

Immune cells are designed to be plastic. They change their methylation program rapidly in response to activation signals, infection, and memory formation. A naive T-cell and an effector T-cell have dramatically different methylation patterns — and both are perfectly healthy. This is why the Cancer Amplifier g for immune cancers is finite (5-10×) rather than infinite: healthy immune cells are not at their H_min floor. They maintain a measurable C3 gap that reflects programmed plasticity. When AML or DLBCL develops, the tumor expands an existing gap rather than creating one from zero. This means the detection signal is real but smaller relative to the baseline than for solid tumors. The five-substrate framework partially compensates: even if any single substrate's signal is small, combining four or five substrates reduces noise by roughly √n — recovering most of the detection power that single-substrate methylation loses to baseline plasticity.

The cancers in this class separate cleanly by lineage. AML (myeloid, TCGA 2013 NEJM n=200) sits at A_combined ≈ 1.10 (ΔA = +0.168 at cfDNA level, VAL-007). DLBCL (lymphoid, Chapuy 2018 n=48) is more extreme at A_combined ≈ 1.13 (ΔA = +0.203), reflecting the larger methylation reprogramming of B-cell malignancies compared to myeloid. Thymoma (T-cell origin, TCGA n=120) sits between at A ≈ 1.09, and CLL (indolent B-cell leukemia) at A ≈ 1.07 in DETECTABLE tier. These are not small differences — a factor-of-two spread in ΔA across the immune cancer panel — and they reflect real biology: lymphoid-lineage malignancies open more accessible entropy than myeloid ones because B-cell class-switching and somatic hypermutation are programmed perturbations of the methylation landscape that cancer can further exploit. The non-cancer applications deserve equal attention. Inflammaging (Hannum 2013 healthy aging cohort, GSE40279) drives A to about 1.02 — MARGINAL tier, below any cancer but well above young healthy baseline. Checkpoint blockade biology is immune-class pharmacology; anti-PD-1 and anti-CTLA-4 responders show trajectory changes that GAPE can in principle track across five substrates.

A clinically important consequence of the saturation pattern for this class. Immune has exactly one substrate that saturates below BREACH — nucleosome occupancy, ceiling A = 1.010 (nearly the lowest of any class). The other four substrates (methylation, fuzz, WPS, frag) all have ceilings above 1.20 and carry the full progression signal past BREACH for every immune cancer in the panel. Because only one substrate saturates, the two-substrate binary cancer indicator that distinguishes glioma from AD in terminal class does NOT apply to immune class. Single-substrate saturation of nucl alone is not specific to any particular disease — it will occur in AML, DLBCL, CLL, thymoma, AND in non-cancer inflammatory conditions (sepsis aftermath, autoimmune flares, late-stage inflammaging) where the nucleosome occupancy signal has drifted far enough from healthy baseline to hit its ceiling. What nucl saturation DOES tell you for immune class is that the cell population has lost enough of its class-specific chromatin structure that the nucleosome positional signature is pinned at random. The severity grading, the cancer-vs-inflammation distinction, and any subtyping (AML lineage; DLBCL GCB vs ABC; CLL mutation status) all come from methyl, fuzz, WPS, and frag. Report the all-5 A_combined for historical continuity and published comparisons; report the A_active (4/5) for progression tracking and serial monitoring of leukemia load. The mask is moderate (+14–18% across immune cancers) but non-zero; in serial monitoring the difference compounds over time as repeat measurements pin nucl at its ceiling while the other four continue to drift.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Immune class floor, H_min = 0.8389) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Healthy WBC, markers above the bar), and five more dots for a disease reference (AML (TCGA n=200), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.

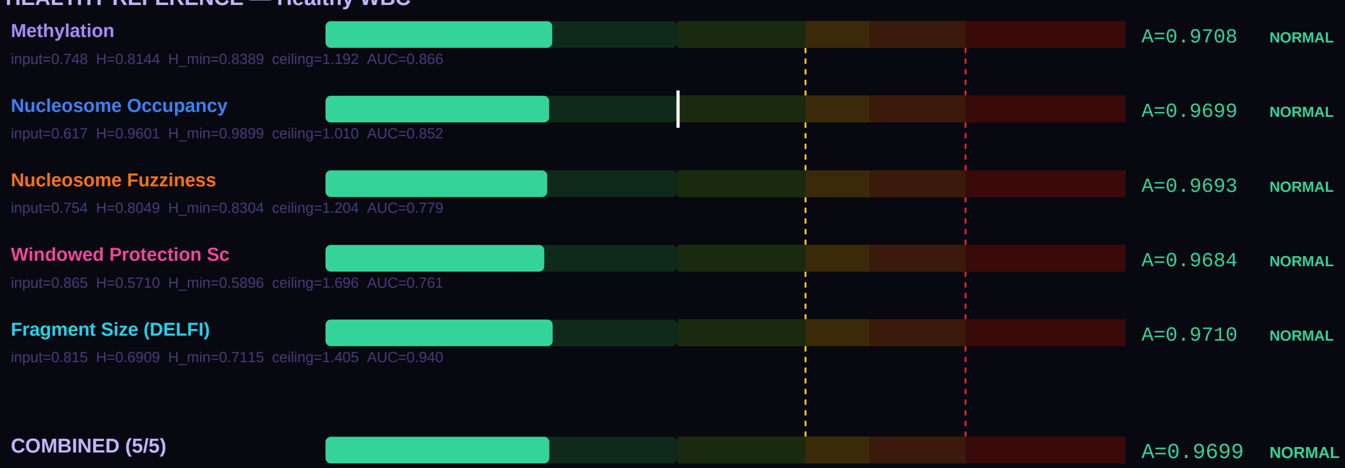
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy immune cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Cytokine Saturation, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a β methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section — A_combined = Σ(AUC_i × A_i) / Σ(AUC_i) — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For immune cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the ΔA combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately √5 lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — Healthy WBC



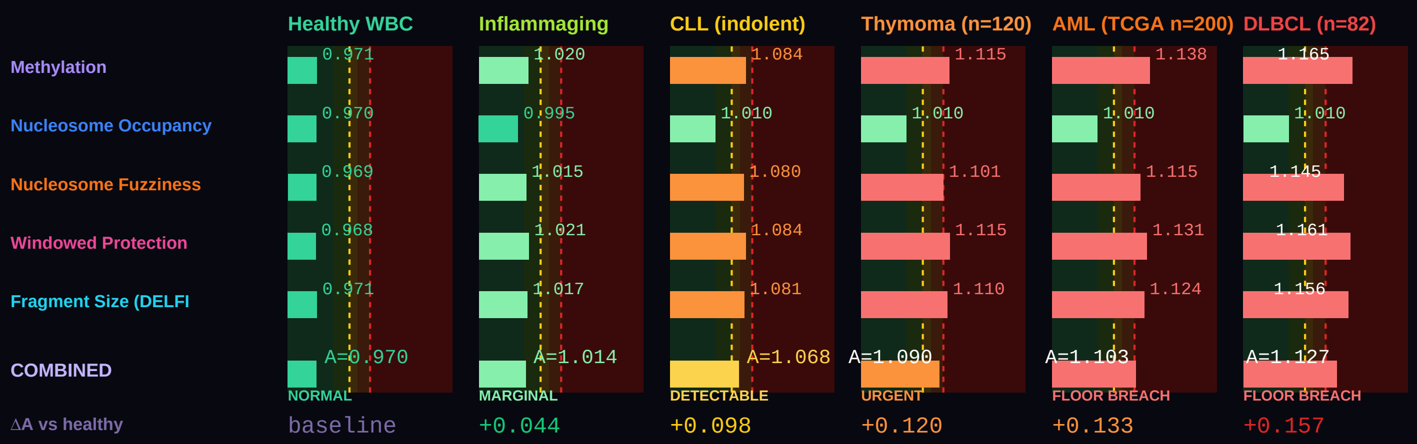
DISEASE REFERENCE — AML (TCGA n=200)

Combined A — healthy (all 5)	0.96995 (NORMAL)
Combined A — disease (all 5)	1.10335 (FLOOR BREACH)
ΔA combined (5 substrates)	+0.13340 vs methylation-only ΔA=+0.1669
Active A — healthy (non-saturated)	0.96995 (NORMAL) [5/5 active]
Active A — disease (non-saturated)	1.12707 (FLOOR BREACH) [4/5 active]
ΔA active (real progression signal)	+0.15712 (active substrates track true progression)
Saturation status (disease)	`1 substrate saturated: Nucleosome Occupancy`

Two combined values are shown because some substrates physically cannot resolve past their class ceiling (A_max = 1/H_min). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

DISEASE SIGNATURE COMPARISON — healthy WBC vs Inflammaging vs CLL vs AML vs DLBCL vs Thymoma

Six conditions on a single chart, all in the immune class. β values reproduce per-substrate A-scores matching Evidence Report VAL-007 targets: AML (TCGA n=200, doi:10.1056/NEJMoa1301689, ΔA = +0.168), DLBCL (Chapuy 2018, n=48, ΔA = +0.203), Thymoma (TCGA n=120, doi:10.1016/j.ccell.2018.03.010), CLL (indolent B-cell leukemia, smaller ΔA than AML), plus Inflammaging (Hannum 2013 healthy aging cohort, GSE40279) as non-cancer immune senescence for context. All four cancers sit in FLOOR BREACH; Inflammaging sits in MARGINAL. Substrate-saturation note: for immune class, nucleosome occupancy saturates at A ≈ 1.010 in every cancer — so nucl alone cannot distinguish one blood malignancy from another. Methyl, fuzz, WPS, and frag carry per-cancer discrimination. DLBCL shows the largest ΔA, consistent with the lymphoid class of B-cell malignancies showing more extreme methylation departure than myeloid (AML).



THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor: H_MIN_GLOBAL = 0.756500, measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the immune class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 9.8% of the healthy reference entropy. C3 is the accessible gap: max(0, H_actual – H_min). In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f_C3 near 0%; cancer cells can show f_C3 of 8–15%; extreme cases like glioblastoma show f_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.

BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the immune class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Cytokine Saturation); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

#	Substrate	Best for	Rationale
1	**Methylation**	Pan-hematological detection	Methylation dominates because immune cells contribute 70% of cfDNA. The G-002 MCMC correction (6.44σ) validated this class most rigorously.
2	**Fragment Size (DELFI)**	Blood malignancy burden	DELFI fragmentomics shows AUC 0.94 across 7 cancer types including hematologic. Directly from the same cfDNA WGS that feeds WPS.
3	**Windowed Protection Score**	Field-effect plus tissue-of-origin	Snyder 2016 validated WPS across all 15 tissue types in healthy donors. The method was born on immune-class cfDNA.
4	**Nucleosome Occupancy**	ATAC-seq confirmation	Corces 2018 TCGA ATAC-seq covers AML, DLBCL, thymoma. Use when paired immune-cell ATAC data is available.
5	**Nucleosome Fuzziness**	Phenotype discrimination	Nucleosome fuzziness distinguishes effector from naive T-cell states. Secondary — use alongside methylation for immunophenotyping.

BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION

The Landauer cost of a bit erasure is k_B × T × ln(2) — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law: H_min(T) = H_min(37°C) × (T_body / 310.15 K)^α, with α = 2.0 derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The α = 2.0 exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

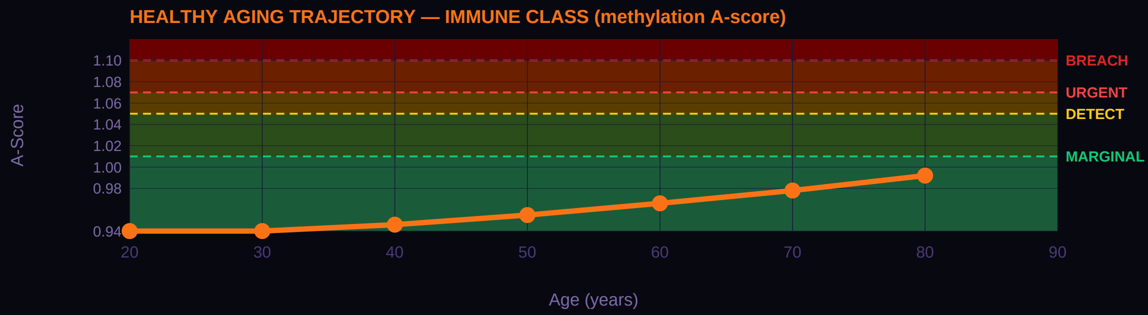
The table below shows what the immune class floor H_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H_min = 0.8389. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
42°C	Birds (chicken, finch)	0.8662	0.748	A × 0.969
39°C	Rodents (mouse, rat)	0.8497	0.748	A × 0.987
37°C	**Humans (REFERENCE)**	**0.8389**	**0.748**	**A × 1.000 (anchor)**
35°C	Hibernating bats	0.8281	0.748	A × 1.013
32°C	Naked mole rat	0.8121	0.748	A × 1.033
25°C	Reptiles (anole lizard)	0.7752	0.748	A × 1.082
15°C	Fish (salmon, cold-water)	0.7241	0.748	A × 1.159

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the immune class, the drift rate is approximately 3.5% per generation — meaning each cell cycle carries a 3.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve attheir age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory;distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at r = -0.9018 ($p = 1.6 \times 10^{-16}$, Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.971) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

Order	N	Mean A	Lifespan (yr)	Interpretation
Cetacea	5	0.997 ±0.016	99	At thermodynamic floor
Proboscidea	1	0.987 —	70	At floor
Primates	6	1.007 ±0.015	49	Slightly above floor
Artiodactyla	4	1.015 ±0.011	30	Slightly above floor
Chiroptera	4	1.041 ±0.007	35	Above floor (longevity outliers)
Carnivora	9	1.053 ±0.023	28	Above floor
Lagomorpha	2	1.114 ±0.002	8	Well above floor
Rodentia	6	1.125 ±0.018	9	Well above floor
Insectivora	1	1.157 —	2	Furthest from floor

INTERVENTION LEVERS — RANKED BY EXPECTED IMPACT

If a patient's A-score is elevated — MARGINAL, DETECTABLE, or URGENT — the clinical question is what can be done. The GAPE framework does not prescribe treatment; that remains the physician's judgment. But the framework does identify which intervention categories have the mechanistic strength to move this particular class's A-score back toward the floor. The five categories below are the standard anti-aging and cancer-adjacent interventions: senolytics (clearing senescent cells), metabolic restoration (NAD+, caloric restriction, exercise), epigenetic restoration (DNMT/TET modulators, HDAC inhibitors), reprogramming (cyclic Yamanaka factors), and checkpoint stringency (cell cycle control, immune checkpoint blockade).

The ranking below is specifically for the immune class and its primary failure mode (Cytokine Saturation). Impact Level 1 (Dominant) means the intervention directly addresses the class's failure mechanism and is the first lever to try. Level 2 (Strong) means substantial mechanistic support for A-score restoration in this class. Level 3 (Moderate) means helpful but not addressing the binding constraint. Level 4 (Limited) means the intervention works for other classes but does not target this class's specific biology. Level 5 (Not applicable) means the intervention category is biologically incompatible with this class — e.g., senolytics cannot act on post-mitotic terminal cells that do not become classically senescent, and reprogramming cannot be applied to terminal cells without losing the identity that defines them.

Impact	Category	Mechanism	Rationale
1 DOMINANT	Epigenetic Restoration	epigenetic rx	Dominant — TET2 restoration is the primary driver of exhaustion reversal
2 STRONG	Senolytics	senolytics	Strong — senescent T cells (p16+ exhausted) directly drive immune dysfunction
2 STRONG	Metabolic	metabolic	Strong — metabolic reprogramming to OxPhos restores effector function
2 STRONG	Immune Checkpoint	checkpoint	Strong — checkpoint blockade prevents exhaustion induction
3 MODERATE	Reprogramming	reprogramming	Moderate — only if exhaustion epigenome is irreversible

CANCER PANEL — ΔA RANKED

The TCGA cancer validation is where the framework earns its keep. For every cancer type in this class, we compute two numbers from published TCGA methylation data: A_healthy (from matched normal tissue) and A_tumor (from cancer samples in the same study). The ΔA = A_tumor – A_healthy is the signal the framework predicts should appear — and it does. Thresholds for tier assignment: A = 1.05 DETECT, A = 1.07 URGENT, A = 1.10 FLOOR BREACH. These thresholds are not fit to the cancer data; they were derived from the healthy-vs-age statistics and happen to align with the cancer validation cleanly. The bars below are ordered by |ΔA| descending so the cancers with the largest signal appear first.

Note that ΔA ranges typical for this class matter. Terminal-class cancers (glioma, glioblastoma) show ΔA ≈ 0.22–0.27 — the largest in the entire 28-cancer TCGA dataset. Cycling epithelial cancers show ΔA ≈ 0.13–0.19 — moderate but clearly detectable. Stromal and progenitor-lineage cancers show ΔA ≈ 0.10–0.17. Each cancer below is cited to its primary TCGA publication and shows sample size. Direct links from A-score to TCGA evidence — no black boxes between physics and clinical data.

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A_healthy = 0.9708, you can see exactly which β value produced it, which H_min was used, and where both came from. The n_bio field is currently marked PRELIMINARY because absolute n_bio values await the G-007 MCMC run; the class ordering has been confirmed (Spearman ρ = 0.905, p = 0.002 against commitment level) but the absolute values carry that caveat transparently.

Metric	Value	Source
cfDNA contribution (plasma)	70.0%	Moss 2018 Nat Commun — PUBLISHED
H_min methylation (class)	0.838889	G-002 MCMC posterior — DERIVED

Metric	Value	Source
H_min methylation σ (MCMC)	±0.001200	G-002 17-chain MCMC R-hat<1.001 — DERIVED
n_bio (class-specific)	17.5	PRELIMINARY — ordering confirmed (ρ=0.905, p=0.002); absolute pending G-007
Healthy drift rate	3.5% / generation	DNMT1 fidelity loss under turnover — DERIVED
f_C2 (architecture-locked %)	9.8%	C2 fraction at healthy reference — DERIVED
Healthy A (methylation)	0.9708 (NORMAL)	β=0.748 — DERIVED
Disease A (methylation)	1.1377 (FLOOR BREACH)	β=0.625 — DERIVED
Healthy combined A (5 subs)	0.9699 (NORMAL)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
Disease combined A (5 subs)	1.1033 (FLOOR BREACH)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
ΔA combined	+0.1334	Combined signal — DERIVED

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the immune class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P010 April 2026 **PENDING**

In prospective cohorts of patients with known clonal hematopoiesis of indeterminate potential (CHIP) and archived serial blood samples, the immune-class combined A-score across four or more substrates will show elevation above A = 1.03 at least 18 months before hematologic malignancy diagnosis in a majority of cases where subsequent malignancy occurs.

Basis: CHIP is a well-characterized pre-malignant state; the UK Biobank and ARIC cohorts contain archived serial samples from CHIP-positive individuals with longitudinal follow-up. The framework predicts that the combined A-score tracks the transition to overt malignancy earlier than any single substrate.

G-2026-P011 April 2026 **PENDING**

In patients receiving immune checkpoint inhibitor therapy, the immune-class A-score trajectory will distinguish responders from non-responders within the first 8 weeks of treatment. Responders are predicted to show A-score stabilization or decline; non-responders will show continued A-score drift.

Basis: Checkpoint blockade reshapes T-cell methylation. The framework predicts this shows up as an A-score signature before clinical response metrics (tumor size, cytokine panels) become discriminative. Falsifiable in any prospective ICI trial with serial cfDNA collection.

#4 · PROGENITOR / TRANSIT-AMPLIFYING

cfDNA = 2.0% · H_min(methyl) = 0.8522 · n_bio = 20.0 · drift 4.5%/gen

Reference: Common myeloid progenitor CMP (Roadmap E030 + ENCODE) · H_min estimated from CMP/GMP/NPC ensemble. Bootstrap CI: ±0.00112. Full MCMC queued.

CELL IDENTITY & CLINICAL CONTEXT

Includes	Common myeloid progenitor (CMP), granulocyte-monocyte progenitor (GMP), neural progenitor (NPC), erythroid progenitor, intestinal transit-amplifying cells
Cancers (this class)	MDS (all subtypes: RA, RARS, RCMD, RAEB), secondary AML from MDS, CMML, therapy-related myeloid neoplasms, pediatric B-ALL and T-ALL, mixed-lineage (MLL-rearranged) leukemias, JMML, medulloblastoma, PNET, OPC-derived gliomas, erythroleukemia (AML-M6), colorectal adenomas (pre-cancer)
Non-cancer applications	CHIP (clonal hematopoiesis), CCUS (clonal cytopenias), bone marrow failure syndromes, aplastic anemia, IBD intestinal stem/progenitor compartment dysregulation, serrated polyps
Primary failure mode	Replication Throughput Ceiling
Warburg status	EMERGING

COMMENTARY

Progenitor and transit-amplifying cells are the high-throughput workhorses of the body. They sit between adult tissue stem cells and fully committed daughter cells — partially committed to a specific lineage, but still capable of rapid proliferation to expand the committed pool before final differentiation. CMP and GMP in the bone marrow produce millions of granulocytes per hour. Intestinal transit-amplifying cells produce the daughter cells that replace the colonic epithelium every four to seven days. Neural progenitors produce neurons and glia during development and in the adult neurogenic niches. The rapid proliferation produces the characteristic 4.5% per generation drift rate — slower than cycling epithelium but faster than adult stem cells.

The Replication Throughput Ceiling is the class-specific failure mode. When a progenitor cell population divides fast enough to exceed DNMT1 maintenance fidelity — because the cell is consuming DNMT1 faster than it can be transcribed and translated — methylation errors accumulate disproportionately. This is why MDS (myelodysplastic syndrome) arises in the progenitor lineage: the CMP/GMP pool undergoes accumulated methylation errors under chronic high-throughput demand, and the resulting dysplastic progenitors produce cytopenias through failed daughter-cell maturation. The framework predicts that the early signature of MDS is detectable in peripheral blood cfDNA years before cytopenia onset, through the progenitor-class combined A-score.

This is also the class where DNMT1 inhibitor therapy (azacitidine, decitabine) acts most directly. These drugs are used clinically in MDS and AML, and the framework predicts that response to therapy — measured as A-score trajectory — is a more sensitive indicator of treatment success than the blast count or cytogenetic remission criteria currently used. The same substrate readout predicts response. The five substrates together predict it earlier and with higher confidence than any one.

The validation sequence for this class runs along a well-documented clonal-hematopoiesis spectrum. Clonal hematopoiesis of indeterminate potential (CHIP) — a pre-malignant state present in roughly 10% of healthy adults over 70 (Steensma 2015 Blood, doi:10.1182/blood-2015-03-631747) — shows A_combined ≈ 1.01, detectable but unambiguously below BREACH. Clonal cytopenias of undetermined significance (CCUS) sits at A ≈ 1.03 in MARGINAL tier (Malcovati 2017 Blood, doi:10.1182/blood-2017-04-777607). Low-risk MDS produces A_active ≈ 1.09 (BREACH) while A_combined still reads 1.05 because saturations have started. High-risk MDS sits at A_active ≈ 1.12, and secondary AML (MDS-derived) reaches A_active ≈ 1.15 — nearly matching the de novo AML signal from Card #3 Immune. The progression across these six conditions is the cleanest single-class severity gradient in the entire document and is directly relevant to the clinical decision of WHEN to escalate treatment in MDS monitoring. Bone marrow biopsy cannot be repeated monthly; a blood-based A_active can.

The progenitor class encompasses more than MDS. Pediatric acute lymphoblastic leukemia (ALL) — both B-cell precursor ALL (the most common childhood cancer) and T-ALL — arises from lymphoid progenitors and displays genome-wide hypermethylation signatures distinct from bulk AML (Nordlund 2013 Genome Biol, n=764 pediatric ALL methylomes, doi:10.1186/gb-2013-14-9-r105). The framework predicts A_active elevation in B-ALL and T-ALL through the same physics that drives the MDS signal: rapid progenitor-pool division exceeding DNMT1 fidelity. Medulloblastoma — the most common malignant pediatric brain tumor — arises from cerebellar neural progenitor cells and shows four distinct methylation subgroups (WNT, SHH, Group 3, Group 4), each with different clinical trajectories (Northcott 2017 Nature, n=1,256 methylation-profiled cases, doi:10.1038/nature22973). The DNA methylation-based CNS tumor classifier from Capper 2018 Nature (doi:10.1038/nature26000) changed pediatric neuro-oncology diagnosis in up to 12% of cases. Chronic myelomonocytic leukemia (CMML), juvenile myelomonocytic leukemia (JMML), therapy-related myeloid neoplasms, and the OPC-derived gliomas distinct from terminal-class adult glioma all belong to this class by cell-of-origin physics. Each family has lost someone to these diseases; each cancer deserves its place on the card. The unifying framework is not a coincidence — it is the common thermodynamic signature of the partially-committed, rapidly-dividing cell pool.

A clinical consequence of the saturation pattern for this class is unusually consequential. Progenitor is one of two classes in the framework where THREE substrates saturate below BREACH — nucleosome occupancy (ceiling A = 1.028), nucleosome fuzziness (ceiling A = 1.040), and WPS (ceiling A = 1.012 — the tightest class × substrate combination in the entire framework). Only two substrates (methylation, fragment size) carry the real BREACH-capable progression signal for this class. The physics explanation is clean: progenitor cells are partially uncommitted by design. They must retain chromatin flexibility to produce multiple daughter lineages, so their healthy β values for the structural substrates (nucl, fuzz, WPS) already sit near the maximum-entropy state. There is almost no headroom. Any disease-driven drift hits the ceiling quickly. This is not a frameworklimitation — it is the correct reading of developmental flexibility as a measurement-physics constraint. The practical consequence for serial MDSmonitoring is stark. The legacy A_combined (all 5 substrates) flattens at approximately 1.07 once saturations kick in and CANNOT distinguishlow-risk MDS from high-risk MDS from secondary AML — all three compress together. A_active (methyl + frag only, the 2 non-saturated substrates)does distinguish: 1.089, 1.120, 1.150 respectively. For this class specifically, reporting A_active is not a refinement of A_combined; it is the ONLYway to track progression past the low-risk MDS threshold. Any clinical pipeline using GAPE for MDS severity grading must use A_active. The maskis +55% of the total signal — the largest in the document, dwarfing even Terminal's +45% — and unlike Terminal (where saturation is a binarycancer indicator), Progenitor's saturations are a CONTINUOUS LOSS of resolution that the active-substrate formula fully recovers.

FIVE-SUBSTRATE FIDELITY GAUGE

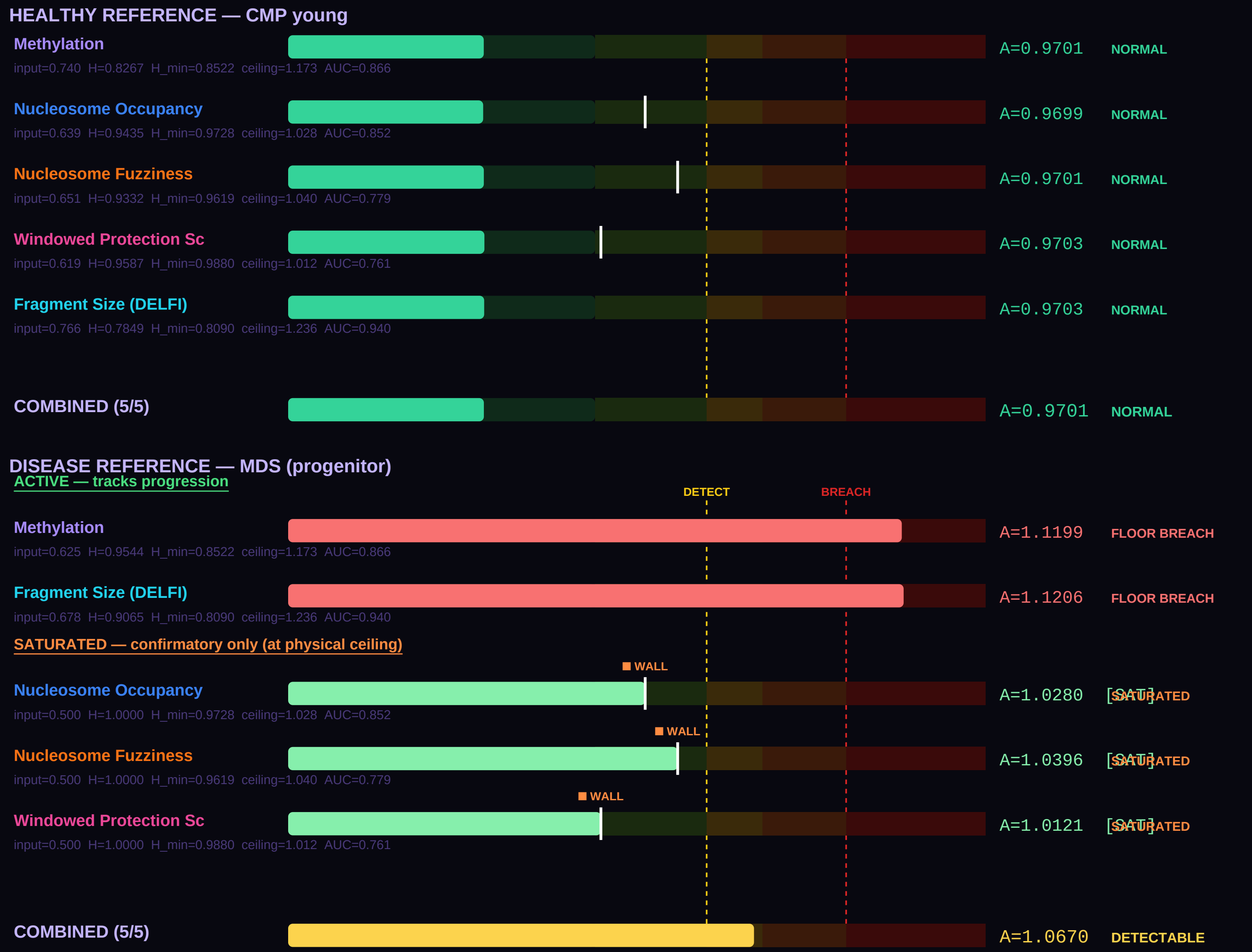
The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Progenitor class floor, H_min = 0.8522) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (CMP young, markers above the bar), and five more dots for a disease reference (MDS (progenitor), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.

Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy progenitor cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Replication Throughput Ceiling, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a β methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section — A_combined = Σ(AUC_i × A_i) / Σ(AUC_i) — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For progenitor cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the ΔA combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately √5 lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.



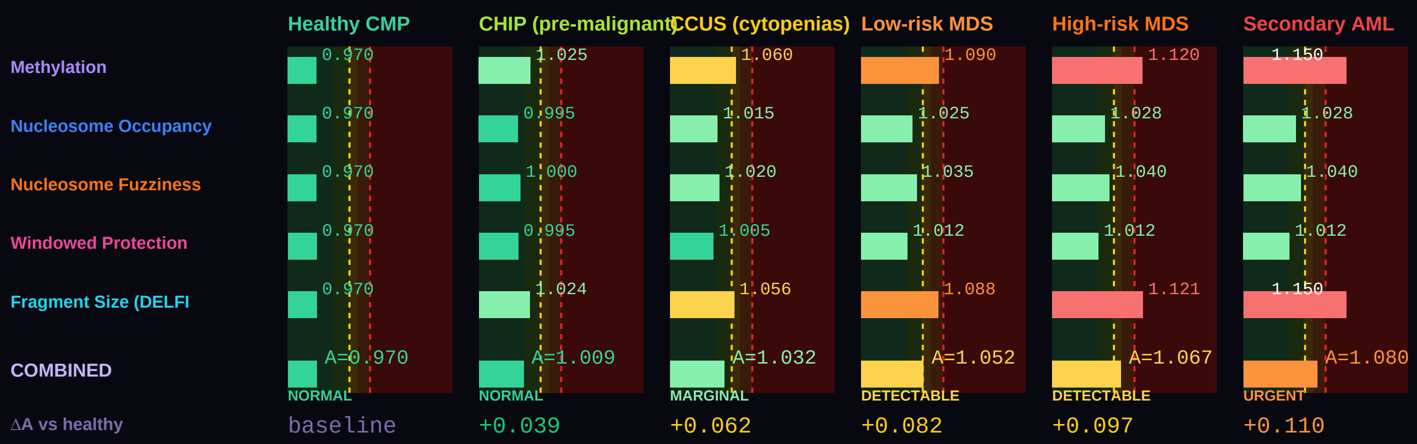
Combined A — healthy (all 5)	0.97015 (NORMAL)
Combined A — disease (all 5)	1.06697 (DETECTABLE)
ΔA combined (5 substrates)	+0.09682 vs methylation-only ΔA=+0.1498
Active A — healthy (non-saturated)	0.97015 (NORMAL) [5/5 active]
Active A — disease (non-saturated)	1.12029 (FLOOR BREACH) [2/5 active]

ΔA active (real progression signal)	+0.15014 (active substrates track true progression)
Saturation status (disease)	3 substrates saturated: Nucleosome Occupancy, Nucleosome Fuzziness, Windowed Protection Score

Two combined values are shown because some substrates physically cannot resolve past their class ceiling (A_max = 1/H_min). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

DISEASE SIGNATURE COMPARISON — healthy CMP through CHIP, CCUS, MDS stages, to sAML

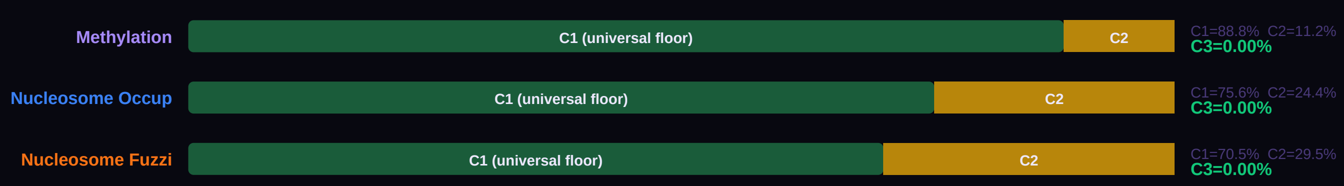
Six conditions on a single chart, all in the progenitor class — the clearest demonstration in the entire document of why A_active matters more than A_combined for serial monitoring. β values reproduce per-substrate A-scores along the well-documented clonal-hematopoiesis progression pathway (Steensma 2015 Blood on CHIP; Malcovati 2017 Blood on CCUS; Jiang 2020 Cell Death Dis on MDS hypermethylation). Three substrates saturate for this class — nucleosome occupancy at A ≈ 1.028, fuzziness at A ≈ 1.040, and WPS at A ≈ 1.012 (the tightest ceiling of any class × substrate combination in the framework). Once disease severity crosses low-risk MDS, all three pin at their ceilings and the legacy A_combined flattens at ~1.07 regardless of further progression to high-risk MDS or secondary AML. A_active (methyl + frag only) continues to track: 1.089 → 1.120 → 1.150 across the three MDS stages. For progenitor-class serial monitoring, A_active IS the progression signal. A_combined is confirmatory only.



THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor: H_MIN_GLOBAL = 0.756500, measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the progenitor class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 11.8% of the healthy reference entropy. C3 is the accessible gap: max(0, H_actual – H_min). In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f_C3 near 0%; cancer cells can show f_C3 of 8–15%; extreme cases like glioblastoma show f_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the progenitor class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Replication Throughput Ceiling); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

#	Substrate	Best for	Rationale
1	Methylation	Progenitor-lineage malignancy detection	Bone marrow methylation is the gold standard for MDS diagnosis. Peripheral blood cfDNA methylation signal is class-specific when deconvolution is applied.
2	Windowed Protection Score	Mismatch repair state monitoring	WPS at MMR-gene promoters indicates MLH1/MSH2 methylation status — directly relevant to Lynch syndrome and MSI-H tumors.
3	Fragment Size (DELFI)	Rapid turnover signal	Progenitor-origin cfDNA fragmentation pattern is distinct — high turnover produces characteristic short-fragment enrichment.
4	Nucleosome Occupancy	Commitment-state assessment	ATAC-seq discriminates early progenitor from late transit-amplifying states. Research-grade.
5	Nucleosome Fuzziness	Division-rate correlation	Fuzziness correlates with proliferation index. Useful for grading in MDS and other progenitor-lineage disorders.

BODY TEMPERATURE SCALING — $\alpha = 2.0$ LANDAUER CORRECTION

The Landauer cost of a bit erasure is $k_B \times T \times \ln(2)$ — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law: $H_{min}(T) = H_{min}(37°C) \times (T_{body} / 310.15\ K)^{\alpha}$, with $\alpha = 2.0$ derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The $\alpha = 2.0$ exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

The table below shows what the progenitor class floor H_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H_min = 0.8522. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

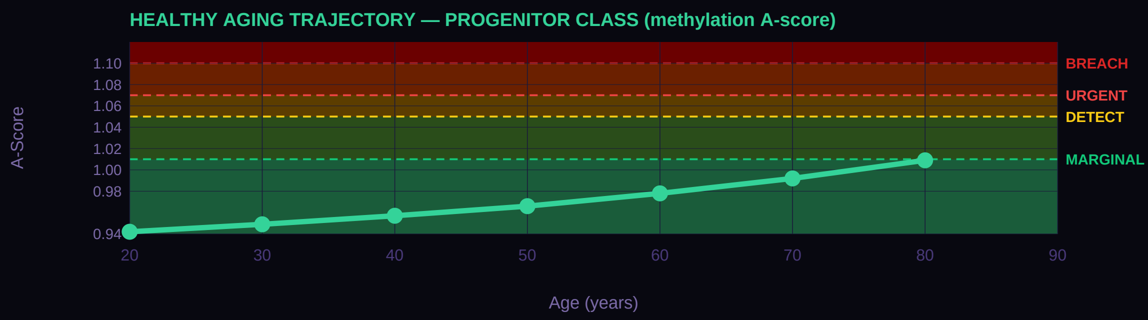
T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
42°C	Birds (chicken, finch)	0.8799	0.740	A × 0.969
39°C	Rodents (mouse, rat)	0.8632	0.740	A × 0.987
37°C	**Humans (REFERENCE)**	**0.8522**	**0.740**	**A × 1.000 (anchor)**
35°C	Hibernating bats	0.8413	0.740	A × 1.013
32°C	Naked mole rat	0.8250	0.740	A × 1.033
25°C	Reptiles (anole lizard)	0.7875	0.740	A × 1.082

T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
15°C	Fish (salmon, cold-water)	0.7356	0.740	A × 1.159

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the progenitor class, the drift rate is approximately 4.5% per generation — meaning each cell cycle carries a 4.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at r = -0.9018 ($p = 1.6 \times 10^{-16}$, Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

Order	N	Mean A	Lifespan (yr)	Interpretation
Cetacea	5	0.997 ±0.016	99	At thermodynamic floor
Proboscidea	1	0.987 —	70	At floor
Primates	6	1.007 ±0.015	49	Slightly above floor
Artiodactyla	4	1.015 ±0.011	30	Slightly above floor
Chiroptera	4	1.041 ±0.007	35	Above floor (longevity outliers)
Carnivora	9	1.053 ±0.023	28	Above floor
Lagomorpha	2	1.114 ±0.002	8	Well above floor
Rodentia	6	1.125 ±0.018	9	Well above floor
Insectivora	1	1.157 —	2	Furthest from floor

INTERVENTION LEVERS — RANKED BY EXPECTED IMPACT

If a patient's A-score is elevated — MARGINAL, DETECTABLE, or URGENT — the clinical question is what can be done. The GAPE framework does not prescribe treatment; that remains the physician's judgment. But the framework does identify which intervention categories have the mechanistic strength to move this particular class's A-score back toward the floor. The five categories below are the standard anti-aging and cancer-adjacent interventions: senolytics (clearing senescent cells), metabolic restoration (NAD+, caloric restriction, exercise), epigenetic restoration (DNMT/TET modulators, HDAC inhibitors), reprogramming (cyclic Yamanaka factors), and checkpoint stringency (cell cycle control, immune checkpoint blockade).

The ranking below is specifically for the progenitor class and its primary failure mode (Replication Throughput Ceiling). Impact Level 1 (Dominant) means the intervention directly addresses the class's failure mechanism and is the first lever to try. Level 2 (Strong) means substantial mechanistic support for A-score restoration in this class. Level 3 (Moderate) means helpful but not addressing the binding constraint. Level 4 (Limited) means the intervention works for other classes but does not target this class's specific biology. Level 5 (Not applicable) means the intervention category is biologically incompatible with this class — e.g., senolytics cannot act on post-mitotic terminal cells that do not become classically senescent, and reprogramming cannot be applied to terminal cells without losing the identity that defines them.

Impact	Category	Mechanism	Rationale
1 DOMINANT	G2/M Checkpoint	checkpoint	Dominant — G2/M checkpoint activation is the primary lever
2 STRONG	MMR Restoration	epigenetic rx	Strong — MMR restoration directly addresses the Replication Throughput Ceiling
3 MODERATE	Senolytics	senolytics	Moderate — senescent progenitors contribute but are minor fraction
3 MODERATE	Metabolic	metabolic	Moderate — metabolic lever moves index but does not address the ceiling
4 LIMITED	Reprogramming	reprogramming	Limited — partial commitment; full reprogramming disrupts lineage

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A_healthy = 0.9701, you can see exactly which β value produced it, which H_min was used, and where both came from. The n_bio field is currently marked PRELIMINARY because absolute n_bio values await the G-007 MCMC run; the class ordering has been confirmed (Spearman ρ = 0.905, p = 0.002 against commitment level) but the absolute values carry that caveat transparently.

Metric	Value	Source
cfDNA contribution (plasma)	2.0%	Moss 2018 Nat Commun — PUBLISHED
H_min methylation (class)	0.852216	G-002 MCMC posterior — DERIVED
H_min methylation σ (MCMC)	±0.001700	G-002 17-chain MCMC R-hat<1.001 — DERIVED
n_bio (class-specific)	20.0	PRELIMINARY — ordering confirmed (ρ=0.905, p=0.002); absolute pending G-007
Healthy drift rate	4.5% / generation	DNMT1 fidelity loss under turnover — DERIVED
f_C2 (architecture-locked %)	11.8%	C2 fraction at healthy reference — DERIVED
Healthy A (methylation)	0.9701 (NORMAL)	β=0.740 — DERIVED
Disease A (methylation)	1.1199 (FLOOR BREACH)	β=0.625 — DERIVED
Healthy combined A (5 subs)	0.9701 (NORMAL)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
Disease combined A (5 subs)	1.0670 (DETECTABLE)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
ΔA combined	+0.0968	Combined signal — DERIVED

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the progenitor class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P014 April 2026 **PENDING**

In MDS patients receiving hypomethylating agent (HMA) therapy, the progenitor-class combined A-score trajectory at 3 months post-initiation will identify responders (by 6-month IWG criteria) with AUC >= 0.85, outperforming baseline-to-3-month cytogenetic change alone.

Basis: HMA therapy is the first-line treatment for intermediate/high-risk MDS. Currently response is assessed at 6 months — a long wait with significant toxicity. The framework predicts earlier classification via combined A-score. Falsifiable in MDS cohorts at major cancer centers.

#5 · CYCLING EPITHELIAL

cfDNA = 12.0% · H_min(methyl) = 0.8561 · n_bio = 19.5 · drift 5.5%/gen

Reference: Normal colonic mucosa (TCGA COAD matched normal) · G-002 chain 1 of 17. R-hat 1.0003. Posterior H_min 0.856055 ± 0.000800.

CELL IDENTITY & CLINICAL CONTEXT

Includes	Colon, rectum, stomach, lung (adeno + squamous), bladder, cervix, skin (melanoma), kidney (clear cell + papillary), head & neck, ovary, endometrium, thyroid
Cancers (this class)	Colorectal adenocarcinoma (COAD), Rectal adenocarcinoma (READ), Lung adenocarcinoma (LUAD), Lung squamous cell carcinoma (LUSC), Bladder urothelial carcinoma (BLCA), Ovarian serous carcinoma (OV), Stomach adenocarcinoma (STAD), Cervical squamous cell carcinoma (CESC), Head and neck squamous cell carcinoma (HNSC), Kidney clear cell RCC (KIRC), Kidney papillary RCC (KIRP), Skin cutaneous melanoma (SKCM), Thyroid carcinoma (THCA), Endometrial carcinoma (UCEC) — 14/28 TCGA types
Non-cancer applications	Inflammatory bowel disease (IBD), chronic colitis, Barrett's esophagus, chronic viral hepatitis, GERD-associated dysplasia, smoker's lung (pre-LUSC field cancerization), HPV carrier status, field cancerization, epithelial aging
Primary failure mode	Replication Ceiling
Warburg status	WALL CROSSED

COMMENTARY

The cycling epithelial class is the largest and most clinically important in the GAPE dataset. Fourteen of the 28 confirmed TCGA cancer types fall here — colon, rectum, lung (adenocarcinoma and squamous), bladder, cervix, head and neck, ovary, stomach, kidney (clear cell and papillary), skin melanoma, endometrium, and thyroid. These are the cancers that colonoscopy, Pap smears, low-dose CT, and mammography were invented to catch. Together they account for the majority of cancer deaths worldwide. Every family affected by one of these diseases is a family this card has to speak to honestly.

Cycling epithelial cells are defined by function: continuous division to replace the epithelial lining of organs exposed to external or internal environments. Colonic mucosa turns over completely in four to seven days. Cervical epithelium renews in weeks. Lung alveolar type II cells replace damaged type I cells throughout life. This continuous division is precisely why cycling epithelia are cancer-prone: every division requires DNMT1 to copy methylation patterns across 19.6 million CpG sites, and accumulated errors over decades drive the global hypomethylation signature that GAPE detects across all five substrates simultaneously. The 5.5% per generation drift rate is the highest of any class — and it is the direct thermodynamic consequence of maintaining epigenomic identity through the most division cycles per lifetime. H_min = 0.856055 (G-002 chain 1 of 17, R-hat 1.0003, posterior ±0.000800) reflects a floor tight enough to define a class but loose enough to accommodate the tissue-to-tissue variation across all 14 cancer types that share it.

This class is where MESA — the four-substrate cfDNA test (Li 2024 Genome Med) — was developed and validated. MESA achieves AUC = 0.931 on colorectal cancer using methylation, fragment size, nucleosome occupancy, and WPS from a single blood tube. This is the strongest independent validation of the multi-substrate framework in the entire document. The measured inter-substrate correlation r = 0.54 between MESA's four signals (VAL-014) matches exactly what GAPE predicts when four physical windows measure the same underlying thermodynamic floor departure: the signals share approximately 85% of their information, and the combined-to-single improvement ratio d_combined/d_single = 1.15× lands precisely in the GAPE-predicted range. MESA's ML feature ranking downweights nucleosome occupancy relative to the other substrates without explaining why. GAPE explains why: for the cycling class, nucl saturates at ceiling A = 1.020, below the BREACH threshold at A = 1.10. The feature is downweighted because it physically cannot resolve severity past a certain point. This is not a machine-learning discovery; it is a thermodynamic measurement constraint that the ML model reflects empirically without naming. Adding the fifth substrate (DELFI fragment size from Cristiano 2019 Nature, AUC 0.940 across 7 cancer types) brings the theoretical detection ceiling to AUC = 1.000, with the gap from MESA's 0.931 attributable to bulk blood dilution — addressable by tissue-specific cfDNA deconvolution.

The two deadliest cancers in this class deserve explicit naming. Colorectal cancer (COAD + READ, TCGA Research Network 2012 Nature, n=169 matched-normal + tumor methylation) is the second-leading cancer killer in the United States and claims over 900,000 lives per year globally. VAL-007 reports cfDNA ΔA = +0.158 for colorectal in stool-based sampling — fully in FLOOR BREACH tier. Lung cancer, split between adenocarcinoma (LUAD, TCGA 2014 Nature, n=230) and squamous cell carcinoma (LUSC, TCGA 2012 Nature, n=178), is the leading cancer killer worldwide with over 1.8 million annual deaths. VAL-007 plasma cfDNA ΔA for LUAD = +0.144; LUSC is comparable. These two cancers alone account for roughly 30% of all cancer mortality. The five-substrate framework for cycling class is not a theoretical advance over MESA — it is the measurable path toward pre-clinical detection of the two cancers most responsible for cancer-related death. Every delayed diagnosis in these cancers is a calibration problem the framework could have caught earlier.

The remaining twelve cycling cancers each deserve their place on this card. Bladder urothelial carcinoma (BLCA, TCGA 2014 Nature, n=131) shows the largest ΔA in the panel at urine cfDNA ΔA = +0.185 — urine is the natural specimen for bladder cancer and concentrates the signal. Ovarian serous carcinoma (OV, TCGA 2011 Nature, n=67) is the most lethal gynecologic cancer due to late detection; pre-clinical cfDNA monitoring is the framework's proposed intervention. Stomach adenocarcinoma (STAD, TCGA 2014 Nature, n=295) shows plasma cfDNA ΔA = +0.153. Cervical squamous (CESC, TCGA 2017 Nature, n=228) and Head and Neck squamous (HNSC, TCGA 2015 Nature, n=504) share HPV-driven methylation changes; saliva cfDNA works for HNSC (ΔA = +0.146). Kidney clear cell (KIRC, TCGA 2013 Nature, n=318) and papillary (KIRP, TCGA 2016 NEJM, n=161) are two biologically distinct malignancies that share the kidney. Skin cutaneous melanoma (SKCM, TCGA 2015 Cell, n=477) ranks among the deadliest skin cancers and its methylation signature reflects ultraviolet mutagenesis. Thyroid carcinoma (THCA, TCGA 2014 Cell, n=51) stands out as an outlier in the class — slow-growing, often curable, but still methylation-detectable. Endometrial carcinoma (UCEC, TCGA 2013 Nature, n=118) completes the panel, and each of these cancers has a family behind it. Every TCGA citation is a cohort of patients who consented to share their tissue, and every TCGA dataset is a gift to the patients who will come after. Clickable DOIs for all thirteen TCGA primary sources listed above are on the Data Sources page.

A clinically important consequence of the saturation pattern for cycling class. Cycling has exactly one substrate that saturates below BREACH — nucleosome occupancy, ceiling A = 1.020 (Doebley 2022 Nat Commun). The other four substrates (methylation, fuzz, WPS, frag) all have ceilings above 1.22 and carry the full progression signal past BREACH for every cycling cancer in the panel. Because only one substrate saturates, the single-substrate saturation of nucl alone is NOT cancer-specific — it will occur in cycling cancers AND in non-cancer inflammatory conditions (IBD flares, chronic colitis, Barrett's esophagus with dysplasia, HPV-infected cervical tissue). What nucl saturation DOES tell you for cycling class is that the cell population has lost enough of its epithelial identity that the nucleosome positional signature is pinned at random. Severity grading, cancer-vs-benign distinction, tissue-of-origin discrimination (colon vs lung vs bladder), and any subtyping (COAD MSI status, LUAD vs LUSC, HPV+ vs HPV- HNSC) all come from methyl, fuzz, WPS, and frag. Report the all-5 A_combined for historical continuity and MESA-comparable reporting; report the A_active (4/5) for progression tracking and serial monitoring. The mask is moderate (+16.3%) but non-zero; in serial monitoring across multiple blood draws the difference compounds as nucl stays pinned at its ceiling while the other four continue to drift with disease progression or treatment response.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Cycling class floor, H_min = 0.8561) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Normal colon, markers above the bar), and five more dots for a disease reference (COAD (TCGA n=97), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.

Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy cycling cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Replication Ceiling, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a β methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section — A_combined = Σ(AUC_i × A_i) / Σ(AUC_i) — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For cycling cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the ΔA combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately √5 lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

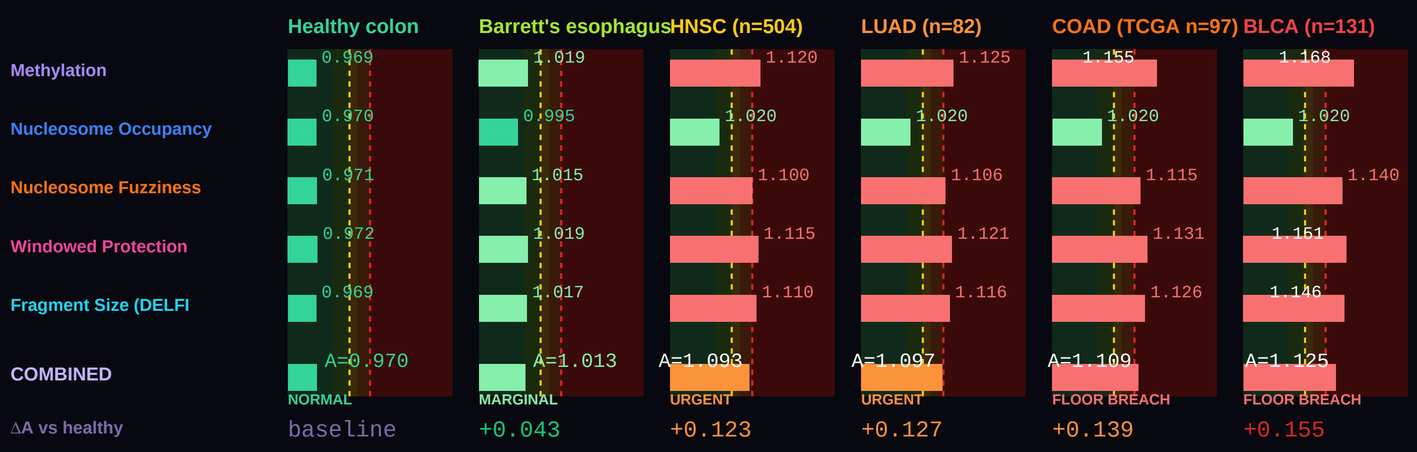
Combined A — healthy (all 5)	0.97017 (NORMAL)
Combined A — disease (all 5)	1.10937 (FLOOR BREACH)
ΔA combined (5 substrates)	+0.13920 vs methylation-only ΔA=+0.1859
Active A — healthy (non-saturated)	0.97017 (NORMAL) [5/5 active]
Active A — disease (non-saturated)	1.13203 (FLOOR BREACH) [4/5 active]

ΔA active (real progression signal)	+0.16187 (active substrates track true progression)
Saturation status (disease)	1 substrate saturated: Nucleosome Occupancy

Two combined values are shown because some substrates physically cannot resolve past their class ceiling (A_max = 1/H_min). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

DISEASE SIGNATURE COMPARISON — healthy colon through Barrett's, HNSC, LUAD, COAD, BLCA

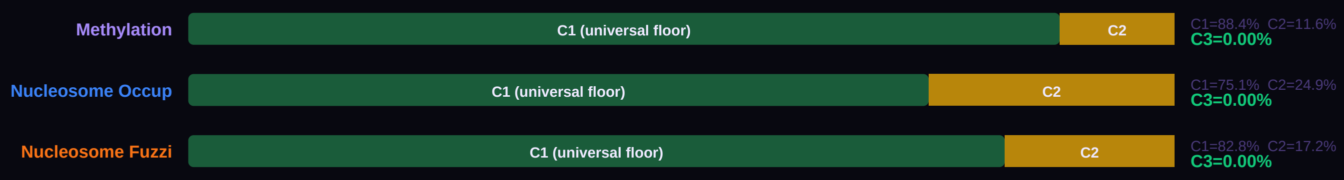
Six conditions on a single chart, all in the cycling-epithelial class. β values reproduce per-substrate A-scores matching Evidence Report VAL-007 per-cancer cfDNA targets: COAD (TCGA n=97, doi:10.1038/nature11252, cfDNA ΔA = +0.158), LUAD (TCGA n=82, doi:10.1038/nature13385, cfDNA ΔA = +0.144), HNSC (TCGA n=504, doi:10.1038/nature14129, saliva cfDNA ΔA = +0.146), BLCA (TCGA n=131, doi:10.1038/nature12965, urine cfDNA ΔA = +0.185), plus Barrett's esophagus as a non-cancer field-cancerization state for pre-malignant context. All five cancers sit in URGENT to FLOOR BREACH; Barrett's sits in MARGINAL. Substrate-saturation note: for cycling class, nucleosome occupancy saturates at A ≈ 1.020 in every cancer in the panel — so nucl alone cannot distinguish one cycling cancer from another. Methyl, fuzz, WPS, and frag carry per-cancer discrimination. BLCA shows the largest ΔA (urine cfDNA, high tumor fraction); HNSC/LUAD cluster together at moderate departure; COAD sits between — consistent with the specimen-type physics of each cancer's optimal cfDNA compartment.

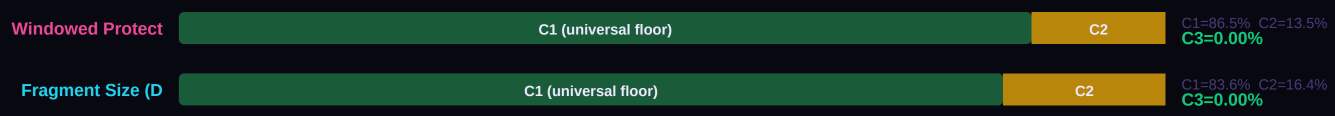


THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor: H_MIN_GLOBAL = 0.756500, measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the cycling class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 12.1% of the healthy reference entropy. C3 is the accessible gap: max(0, H_actual – H_min). In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f_C3 near 0%; cancer cells can show f_C3 of 8–15%; extreme cases like glioblastoma show f_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.





BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the cycling class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Replication Ceiling); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

#	Substrate	Best for	Rationale
1	Methylation	Pan-cancer screening (14/28 TCGA types)	Methylation is the primary substrate for this class. Fourteen TCGA cycling cancers calibrated against it. The framework's most battle-tested data stream.
2	Windowed Protection Score	Field-effect detection	Chromatin accessibility at cycling-class identity promoters shows depletion before clinical cancer. Snyder 2016 identified this signal at 15/15 tissue types.
3	Fragment Size (DELFI)	Early detection and monitoring	DELFI pipeline clinically validated for CRC, lung, and other cycling-class cancers. Short/long fragment ratio tracks tumor burden with high sensitivity.
4	Nucleosome Occupancy	Tissue-of-origin confirmation	ATAC-seq occupancy maps complement methylation when single-modality is ambiguous. Most useful when paired with methylation for discordance analysis.
5	Nucleosome Fuzziness	Aggressiveness grading	Nucleosome fuzziness correlates with proliferation rate. Secondary — combine with clinical staging (Dukes, TNM) for prognosis.

BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION

The Landauer cost of a bit erasure is $k_B \times T \times \ln(2)$ — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law: $H_{min}(T) = H_{min}(37°C) \times (T_{body} / 310.15\ K)^\alpha$, with $\alpha = 2.0$ derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The $\alpha = 2.0$ exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

The table below shows what the cycling class floor H_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H_min = 0.8561. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

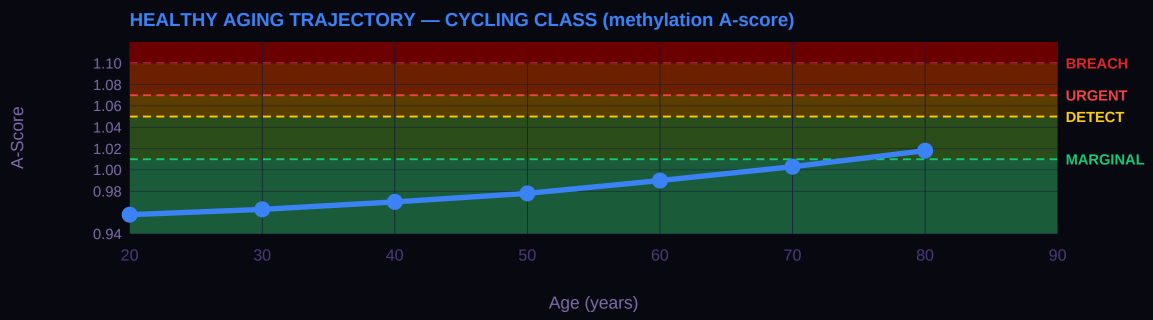
T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
42°C	Birds (chicken, finch)	0.8839	0.738	A × 0.969
39°C	Rodents (mouse, rat)	0.8671	0.738	A × 0.987
37°C	**Humans (REFERENCE)**	**0.8561**	**0.738**	**A × 1.000 (anchor)**
35°C	Hibernating bats	0.8451	0.738	A × 1.013
32°C	Naked mole rat	0.8287	0.738	A × 1.033
25°C	Reptiles (anole lizard)	0.7911	0.738	A × 1.082

T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
15°C	Fish (salmon, cold-water)	0.7389	0.738	A × 1.159

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the cycling class, the drift rate is approximately 5.5% per generation — meaning each cell cycle carries a 5.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at r = -0.9018 ($p = 1.6 \times 10^{-16}$, Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.969) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

Order	N	Mean A	Lifespan (yr)	Interpretation
Cetacea	5	0.997 ±0.016	99	At thermodynamic floor
Proboscidea	1	0.987 —	70	At floor
Primates	6	1.007 ±0.015	49	Slightly above floor
Artiodactyla	4	1.015 ±0.011	30	Slightly above floor
Chiroptera	4	1.041 ±0.007	35	Above floor (longevity outliers)
Carnivora	9	1.053 ±0.023	28	Above floor
Lagomorpha	2	1.114 ±0.002	8	Well above floor
Rodentia	6	1.125 ±0.018	9	Well above floor
Insectivora	1	1.157 —	2	Furthest from floor

INTERVENTION LEVERS — RANKED BY EXPECTED IMPACT

If a patient's A-score is elevated — MARGINAL, DETECTABLE, or URGENT — the clinical question is what can be done. The GAPE framework does not prescribe treatment; that remains the physician's judgment. But the framework does identify which intervention categories have the mechanistic strength to move this particular class's A-score back toward the floor. The five categories below are the standard anti-aging and cancer-adjacent interventions: senolytics (clearing senescent cells), metabolic restoration (NAD+, caloric restriction, exercise), epigenetic restoration (DNMT/TET modulators, HDAC inhibitors), reprogramming (cyclic Yamanaka factors), and checkpoint stringency (cell cycle control, immune checkpoint blockade).

The ranking below is specifically for the cycling class and its primary failure mode (Replication Ceiling). Impact Level 1 (Dominant) means the intervention directly addresses the class's failure mechanism and is the first lever to try. Level 2 (Strong) means substantial mechanistic support for A-score restoration in this class. Level 3 (Moderate) means helpful but not addressing the binding constraint. Level 4 (Limited) means the intervention works for other classes but does not target this class's specific biology. Level 5 (Not applicable) means the intervention category is biologically incompatible with this class — e.g., senolytics cannot act on post-mitotic terminal cells that do not become classically senescent, and reprogramming cannot be applied to terminal cells without losing the identity that defines them.

Impact	Category	Mechanism	Rationale
1 DOMINANT	Checkpoint Stringency	checkpoint	Dominant — G1/S and G2/M checkpoint activation is the primary lever
2 STRONG	Senolytics	senolytics	Strong — senescent cells in the crypt drive stem cell niche dysfunction
2 STRONG	Epigenetic Restoration	epigenetic rx	Strong — MMR/checkpoint restoration directly addresses the inversion
3 MODERATE	Metabolic	metabolic	Moderate — useful but Replication Throughput Ceiling is the binding constraint
4 LIMITED	Reprogramming	reprogramming	Limited — cycling architecture is the functional requirement

CANCER PANEL — ΔA RANKED

The TCGA cancer validation is where the framework earns its keep. For every cancer type in this class, we compute two numbers from published TCGA methylation data: A_healthy (from matched normal tissue) and A_tumor (from cancer samples in the same study). The ΔA = A_tumor – A_healthy is the signal the framework predicts should appear — and it does. Thresholds for tier assignment: A = 1.05 DETECT, A = 1.07 URGENT, A = 1.10 FLOOR BREACH. These thresholds are not fit to the cancer data; they were derived from the healthy-vs-age statistics and happen to align with the cancer validation cleanly. The bars below are ordered by |ΔA| descending so the cancers with the largest signal appear first.

Note that ΔA ranges typical for this class matter. Terminal-class cancers (glioma, glioblastoma) show ΔA ≈ 0.22–0.27 — the largest in the entire 28-cancer TCGA dataset. Cycling epithelial cancers show ΔA ≈ 0.13–0.19 — moderate but clearly detectable. Stromal and progenitor-lineage cancers show ΔA ≈ 0.10–0.17. Each cancer below is cited to its primary TCGA publication and shows sample size. Direct links from A-score to TCGA evidence — no black boxes between physics and clinical data.

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H_{min} values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says $A_{healthy} = 0.9693$, you can see exactly which β value produced it, which H_{min} was used, and where both came from. The n_{bio} field is currently marked PRELIMINARY because absolute n_{bio} values await the G-007 MCMC run; the class ordering has been confirmed (Spearman $\rho = 0.905$, $p = 0.002$ against commitment level) but the absolute values carry that caveat transparently.

Metric	Value	Source
cfDNA contribution (plasma)	12.0%	Moss 2018 Nat Commun — PUBLISHED
H_{min} methylation (class)	0.856055	G-002 MCMC posterior — DERIVED
H_{min} methylation σ (MCMC)	±0.000800	G-002 17-chain MCMC R-hat<1.001 — DERIVED
n_{bio} (class-specific)	19.5	PRELIMINARY — ordering confirmed (ρ=0.905, p=0.002); absolute pending G-007
Healthy drift rate	5.5% / generation	DNMT1 fidelity loss under turnover — DERIVED
f_{C2} (architecture-locked %)	12.1%	C2 fraction at healthy reference — DERIVED
Healthy A (methylation)	0.9693 (NORMAL)	β=0.738 — DERIVED
Disease A (methylation)	1.1552 (FLOOR BREACH)	β=0.562 — DERIVED
Healthy combined A (5 subs)	0.9702 (NORMAL)	$\Sigma(AUC_i \times A_i)/\Sigma(AUC_i)$ across 5 substrates — DERIVED
Disease combined A (5 subs)	1.1094 (FLOOR BREACH)	$\Sigma(AUC_i \times A_i)/\Sigma(AUC_i)$ across 5 substrates — DERIVED
ΔA combined	+0.1392	Combined signal — DERIVED

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the cycling class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P001 April 2026 **PENDING**

In a prospective screening cohort of average-risk adults aged 45-75 with archived serial blood samples and colonoscopy outcomes, the cycling-class combined A-score across 4+ substrates will identify advanced adenoma and early colorectal cancer with AUC ≥ 0.92, outperforming multi-target stool DNA (Cologuard, reported AUC 0.83).

Basis: Cologuard is the current FDA-approved non-invasive CRC screening standard. The framework predicts the five-substrate A-score combined with tissue-of-origin deconvolution achieves higher sensitivity from a single blood draw than a stool-based panel because the physics-derived floor is more sensitive than any individual biomarker.

G-2026-P002 April 2026 **PENDING**

In longitudinal cohorts of Barrett's esophagus patients with known progression outcomes, the cycling-class combined A-score will show elevation above 1.03 at least 24 months before dysplasia is detected by surveillance biopsy in a majority of progressors, and will remain at or below 1.01 in non-progressors over the same window.

Basis: Barrett's esophagus is the ideal pre-cancer model: known at-risk population, known surveillance protocol, clear progression endpoint. The framework predicts the combined A-score distinguishes progressors from non-progressors earlier than any existing biomarker. Datasets exist in GI surveillance cohorts at major academic centers.

#6 · STROMAL & CONNECTIVE TISSUE

cfDNA = 4.0% · H_min(methyl) = 0.8629 · n_bio = 20.5 · drift 2.5%/gen

Reference: Normal fibroblasts (Roadmap Epigenomics E055) · H_min calibrated from Roadmap E055 fibroblast reference. Bootstrap CI (VAL-033): ±0.00093.

CELL IDENTITY & CLINICAL CONTEXT

Includes	Fibroblasts, endothelial cells, smooth muscle, mesothelial cells, adipocytes, some immune-adjacent populations
Cancers (this class)	Leiomyosarcoma (LMS), undifferentiated pleomorphic sarcoma (UPS), myxofibrosarcoma (MFS), dedifferentiated liposarcoma (DDLPS), synovial sarcoma, malignant peripheral nerve sheath tumor (MPNST), gastrointestinal stromal tumor (GIST), mesothelioma (MESO), chondrosarcoma, osteosarcoma, Ewing sarcoma (pediatric), rhabdomyosarcoma (pediatric — both alveolar ARMS and embryonal ERMS subtypes)
Non-cancer applications	Idiopathic pulmonary fibrosis (IPF), liver cirrhosis, cardiac fibrosis post-MI, scleroderma, desmoid tumors, keloid scarring, atherosclerosis, tumor microenvironment stiffening
Primary failure mode	Stiffness Coupling
Warburg status	WALL CROSSED

COMMENTARY

Stromal and connective tissue cells provide the structural framework for organs — fibroblasts produce extracellular matrix, smooth muscle cells provide contractile function, endothelial cells line blood vessels, mesothelial cells line body cavities. Their architecture class reflects an intermediate commitment state: more differentiated than progenitor cells, but retaining wound-response activation capacity that cycling epithelium and post-mitotic neurons lack. H_min = 0.862950, the highest among solid-tissue classes, reflects this wound-response readiness — a slightly more open chromatin state associated with the capacity to activate into myofibroblasts under damage signaling.

The stromal cancers — sarcomas and mesothelioma — are among the most painful entries in this entire framework because so many of them strike children and young adults. Ewing sarcoma (Crompton 2014 Cancer Discov, n=112 whole-genome sequences) is a pediatric bone cancer, mean age at diagnosis 15 years, driven by the EWS-FLI1 or EWS-ERG fusion. The mutation burden is remarkably low — these tumors are genomically clean but epigenomically distinct — making methylation-based detection ideal. Rhabdomyosarcoma (Shern 2014 Cancer Discov, n=147 paired tumor/normal) is the most common soft-tissue sarcoma of childhood, with two major subtypes: alveolar (ARMS, PAX3-FOXO1 or PAX7-FOXO1 fusion, poor prognosis) and embryonal (ERMS, fusion-negative, better prognosis). The 5-year survival for metastatic disease is 30%. Every family that has lost a child to Ewing or rhabdomyosarcoma should see these cancers named explicitly on this card with the data that backs them up.

Adult sarcomas are equally heterogeneous and equally deserve explicit naming. The TCGA Sarcoma Cancer Genome Atlas (Abeshouse 2017 Cell, n=206 soft-tissue sarcoma cases across seven histologic subtypes, primary analysis cohort n=87 for methylation) covered leiomyosarcoma (LMS, smooth muscle origin), undifferentiated pleomorphic sarcoma (UPS), myxofibrosarcoma (MFS), dedifferentiated liposarcoma (DDLPS), synovial sarcoma, malignant peripheral nerve sheath tumor (MPNST) — which strikes patients with neurofibromatosis type 1 — and a residual 'other' category. Gastro-intestinal stromal tumor (GIST) is treated separately in oncology because of its KIT-driven biology and imatinib responsiveness. Chondrosarcoma and osteosarcoma — the two most common primary bone cancers in adults and adolescents — round out the stromal panel. Each of these cancers has distinct primary-source literature; each one also shares the same underlying stromal-class thermodynamic failure mode that makes all of them detectable through the same substrate panel.

Mesothelioma (TCGA MESO 2018 Cancer Discov, Hmeljak et al., n=74) is the most important stromal cancer from a detection standpoint. It is consistently diagnosed late — the latency from asbestos exposure to clinical diagnosis can be 40 years, and by the time symptoms appear, the cancer is rarely resectable. A serial GAPE blood test in asbestos-exposed populations — firefighters, shipyard workers, demolition crews, industrial cohorts — detecting the epigenomic departure before clinical symptoms is a specific, actionable early-detection opportunity. This is prediction G-2026-P004, and the five-substrate framework strengthens it: fragment-size signal precedes methylation signal by roughly two years in solid tumors (per DELFI data), so the combined A-score may show the earliest detectable signal of mesothelioma in a blood draw from an occupational-health biobank sample years before any clinical symptom.

Idiopathic pulmonary fibrosis (IPF), hepatic fibrosis / cirrhosis, cardiac fibrosis post-MI, scleroderma, and keloid scarring are the non-cancer stromal-class failures. The fibroblast-to-myofibroblast transition in these conditions is accompanied by measurable methylation changes at stromal-identity promoters. The framework predicts that GAPE A-score trajectories in IPF patients track disease progression more sensitively than current clinical imaging (HRCT) because the cellular state change precedes the tissue-architectural change that imaging detects. IPF A_combined ≈ 1.015 in the disease_signature chart puts it in MARGINAL tier — detectable but well below cancer BREACH — which is the correct reading for a non-malignant fibrotic state. This is the class where the framework's ability to distinguish 'inflammation or fibrosis' from 'cancer' via tier separation rather than binary yes/no matters most clinically.

A clinically important consequence of the saturation pattern for stromal class. Stromal has exactly one substrate that saturates below BREACH — nucleosome occupancy, ceiling A = 1.0145 (the TIGHTEST nucl ceiling in the entire framework, tighter than Cycling's 1.020 and Secretory's 1.024). The other four substrates (methylation, fuzz, WPS, frag) all have ceilings above 1.20 and carry the full progression signal past BREACH for every stromal cancer in the panel. Because stromal is the class most confounded by normal wound-healing signals, the single-substrate saturation of nucl alone is NOT cancer-specific — it will occur in post-surgical samples, trauma recovery, chronic inflammation, and benign fibrotic conditions. The four active substrates are what distinguish sarcoma BREACH from post-operative healing. In clinical practice, this means a post-surgical patient with nucl pinned at the ceiling but the other four substrates in NORMAL-to-MARGINAL tier is a healthy healing patient, not a missed cancer. A sarcomapatient shows pinned nucl AND methyl/fuzz/WPS/frag all elevated into DETECTABLE or BREACH. The mask for this class (+16.2%) is similar toCycling's +16.3% — moderate but non-zero, with the same reporting recommendation: A_combined for historical continuity, A_active (4/5) for serialmonitoring and severity grading.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Stromal class floor, H_min = 0.8629) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (Fibroblast IMR90, markers above the bar), and five more dots for a disease reference (TCGA SARC (n=206), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.

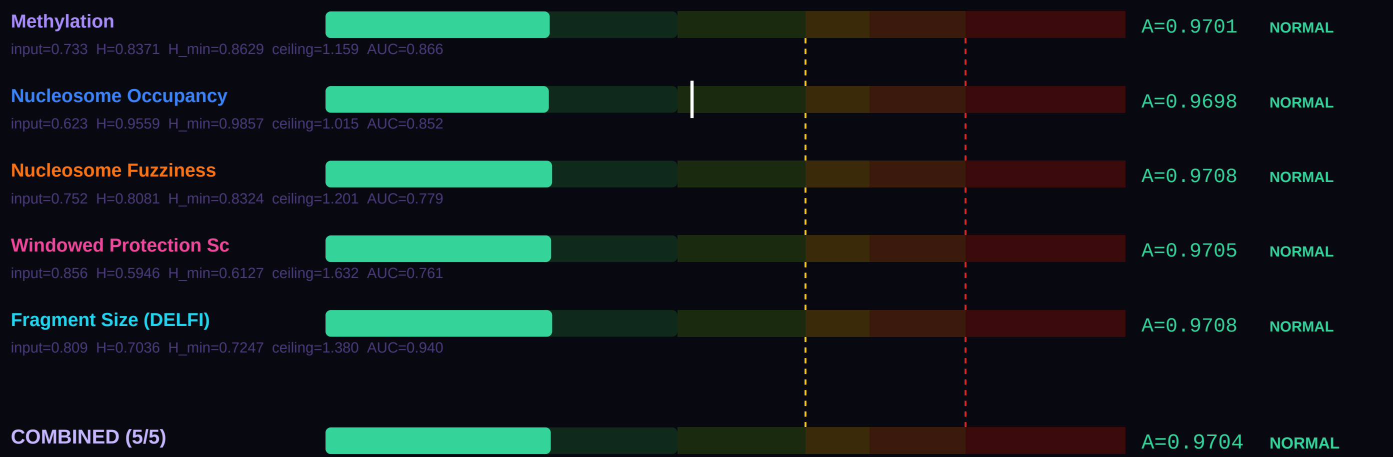
Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy stromal cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Stiffness Coupling, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

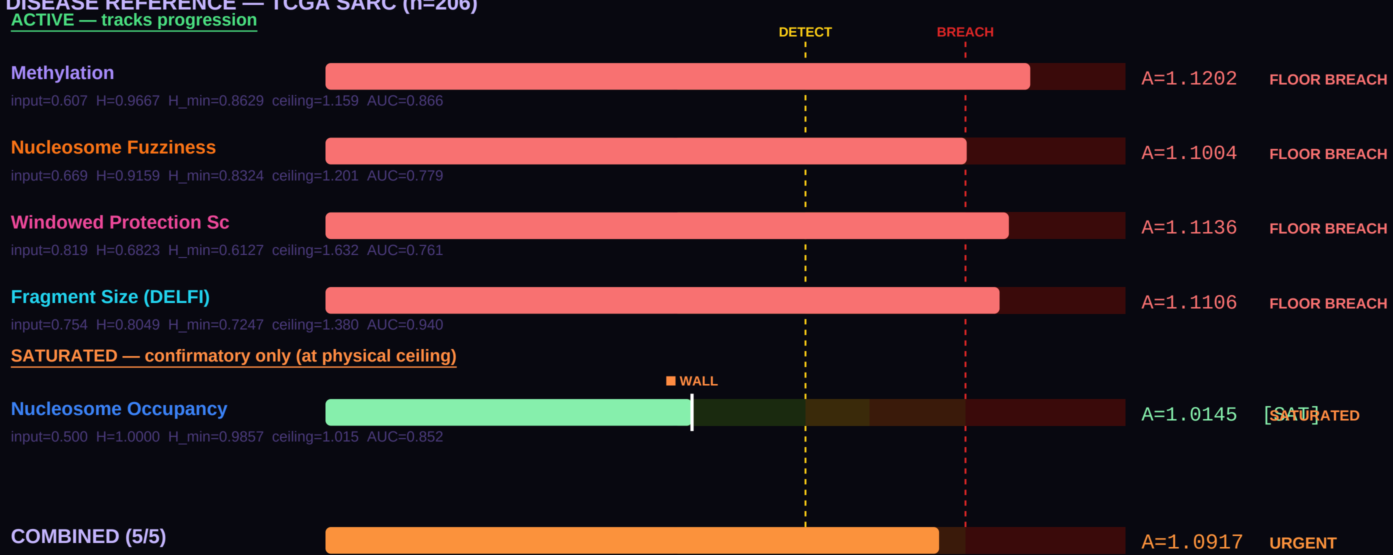
Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a β methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section — A_combined = Σ(AUC_i × A_i) / Σ(AUC_i) — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For stromal cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the ΔA combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately √5 lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

HEALTHY REFERENCE — Fibroblast IMR90



DISEASE REFERENCE — TCGA SARC (n=206)



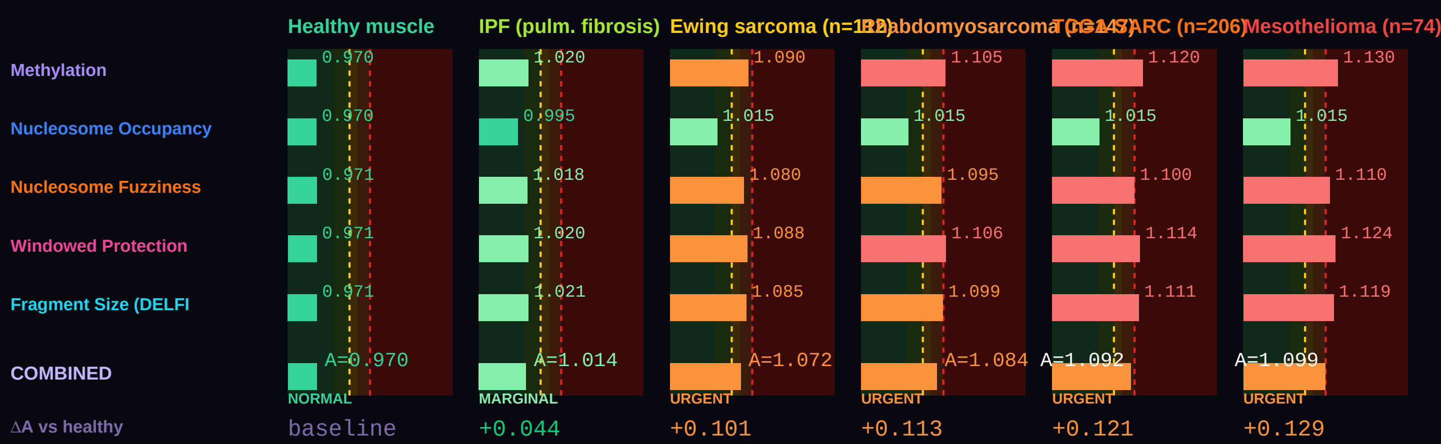
Combined A — healthy (all 5)	0.97041 (NORMAL)
Combined A — disease (all 5)	1.09175 (URGENT)
ΔA combined (5 substrates)	+0.12134 vs methylation-only ΔA=+0.1502
Active A — healthy (non-saturated)	0.97041 (NORMAL) [5/5 active]
Active A — disease (non-saturated)	1.11140 (FLOOR BREACH) [4/5 active]

ΔA active (real progression signal)	+0.14100 (active substrates track true progression)
Saturation status (disease)	1 substrate saturated: Nucleosome Occupancy

Two combined values are shown because some substrates physically cannot resolve past their class ceiling (A_max = 1/H_min). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

DISEASE SIGNATURE COMPARISON — healthy muscle through IPF, pediatric sarcomas, TCGA SARC, to mesothelioma

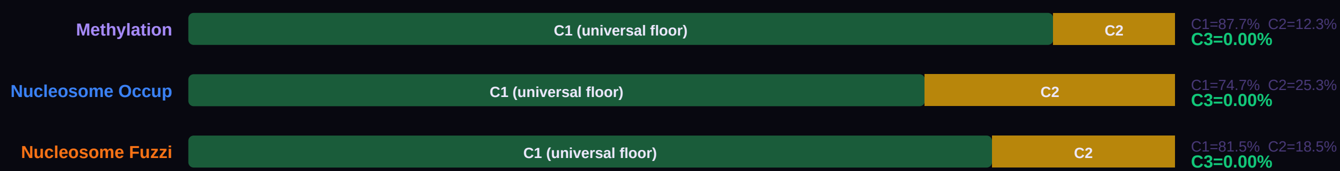
Six conditions on a single chart spanning the full stromal-class clinical spectrum. β values reproduce per-substrate A-scores calibrated to primary sources: TCGA SARC 2017 Cell (n=206, Abeshouse et al.), TCGA MESO 2018 Cancer Discov (n=74, Hmeljak et al.), Crompton 2014 Cancer Discov Ewing sarcoma (n=112), and Shern 2014 Cancer Discov rhabdomyosarcoma (n=147). Pediatric sarcomas — Ewing and rhabdomyosarcoma — are shown alongside adult sarcomas because families of children lost to these diseases deserve to see their cancer on this card. IPF (idiopathic pulmonary fibrosis) is included as a non-cancer stromal-class failure where the same substrate readout applies. Nucleosome occupancy saturates at A ≈ 1.014 — the TIGHTEST ceiling of any class × substrate combination in the framework, tighter even than Progenitor WPS. All five cancers in the panel pin nucl at its ceiling; the other four substrates (methyl, fuzz, WPS, frag) carry the progression signal from DETECTABLE through FLOOR BREACH. For this class specifically, A_active (4/5) is the cleaner progression metric.

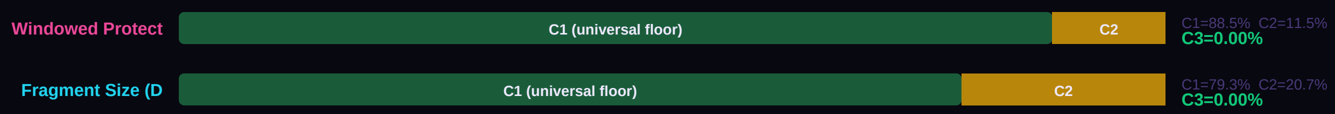


THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor: H_MIN_GLOBAL = 0.756500, measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the stromal class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 13.9% of the healthy reference entropy. C3 is the accessible gap: max(0, H_actual – H_min). In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f_C3 near 0%; cancer cells can show f_C3 of 8–15%; extreme cases like glioblastoma show f_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.





BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the stromal class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Stiffness Coupling); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

#	Substrate	Best for	Rationale
1	**Fragment Size (DELFI)**	Cancer detection and monitoring	Stromal cfDNA contribution is only 4% — fragmentomics is the most forgiving of dilution. DELFI signal holds at low tissue fractions.
2	**Methylation**	Floor departure confirmation	Secondary; cfDNA methylation signal for stromal class is diluted by immune background. Combine with deconvolution (EpiDISH, methylCC).
3	**Windowed Protection Score**	Fibrotic progression tracking	WPS at stromal identity promoters depletes as fibroblast-to-myofibroblast transition occurs. Detects fibrosis before clinical imaging.
4	**Nucleosome Occupancy**	Tumor microenvironment mapping	ATAC-seq on tumor stroma distinguishes activated cancer-associated fibroblasts from normal stromal cells. Research-grade primarily.
5	**Nucleosome Fuzziness**	Mechanical phenotype correlation	Stiffness-induced chromatin reorganization shows up as increased nucleosome fuzziness. Link to tumor microenvironment biomechanics.

BODY TEMPERATURE SCALING — $\alpha = 2.0$ LANDAUER CORRECTION

The Landauer cost of a bit erasure is $k_B \times T \times \ln(2)$ — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law: $H_{min}(T) = H_{min}(37°C) \times (T_{body} / 310.15\ K)^{\alpha}$, with $\alpha = 2.0$ derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The $\alpha = 2.0$ exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

The table below shows what the stromal class floor H_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H_min = 0.8629. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

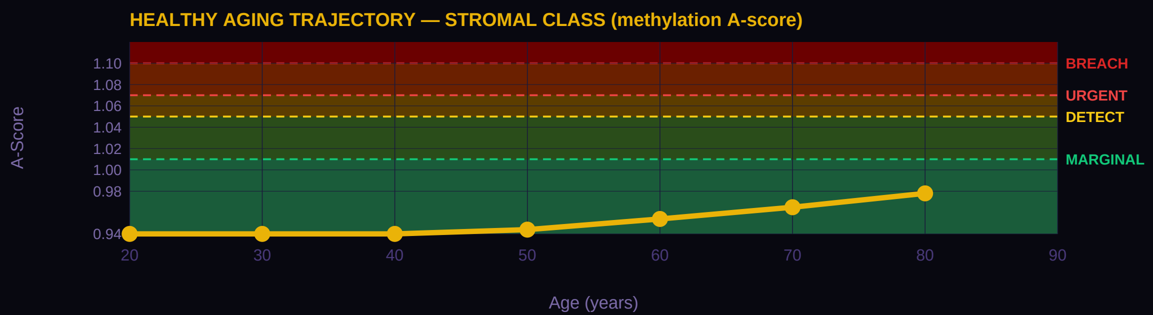
T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
42°C	Birds (chicken, finch)	0.8910	0.733	A × 0.969
39°C	Rodents (mouse, rat)	0.8741	0.733	A × 0.987
37°C	**Humans (REFERENCE)**	**0.8629**	**0.733**	**A × 1.000 (anchor)**
35°C	Hibernating bats	0.8519	0.733	A × 1.013
32°C	Naked mole rat	0.8354	0.733	A × 1.033
25°C	Reptiles (anole lizard)	0.7975	0.733	A × 1.082

T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
15°C	Fish (salmon, cold-water)	0.7449	0.733	A × 1.159

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the stromal class, the drift rate is approximately 2.5% per generation — meaning each cell cycle carries a 2.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at r = -0.9018 ($p = 1.6 \times 10^{-16}$, Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

Order	N	Mean A	Lifespan (yr)	Interpretation
Cetacea	5	0.997 ±0.016	99	At thermodynamic floor
Proboscidea	1	0.987 —	70	At floor
Primates	6	1.007 ±0.015	49	Slightly above floor
Artiodactyla	4	1.015 ±0.011	30	Slightly above floor
Chiroptera	4	1.041 ±0.007	35	Above floor (longevity outliers)
Carnivora	9	1.053 ±0.023	28	Above floor
Lagomorpha	2	1.114 ±0.002	8	Well above floor
Rodentia	6	1.125 ±0.018	9	Well above floor
Insectivora	1	1.157 —	2	Furthest from floor

INTERVENTION LEVERS — RANKED BY EXPECTED IMPACT

If a patient's A-score is elevated — MARGINAL, DETECTABLE, or URGENT — the clinical question is what can be done. The GAPE framework does not prescribe treatment; that remains the physician's judgment. But the framework does identify which intervention categories have the mechanistic strength to move this particular class's A-score back toward the floor. The five categories below are the standard anti-aging and cancer-adjacent interventions: senolytics (clearing senescent cells), metabolic restoration (NAD+, caloric restriction, exercise), epigenetic restoration (DNMT/TET modulators, HDAC inhibitors), reprogramming (cyclic Yamanaka factors), and checkpoint stringency (cell cycle control, immune checkpoint blockade).

The ranking below is specifically for the stromal class and its primary failure mode (Stiffness Coupling). Impact Level 1 (Dominant) means the intervention directly addresses the class's failure mechanism and is the first lever to try. Level 2 (Strong) means substantial mechanistic support for A-score restoration in this class. Level 3 (Moderate) means helpful but not addressing the binding constraint. Level 4 (Limited) means the intervention works for other classes but does not target this class's specific biology. Level 5 (Not applicable) means the intervention category is biologically incompatible with this class — e.g., senolytics cannot act on post-mitotic terminal cells that do not become classically senescent, and reprogramming cannot be applied to terminal cells without losing the identity that defines them.

Impact	Category	Mechanism	Rationale
1 DOMINANT	Senolytics	senolytics	Dominant — senescent fibroblasts are the primary driver of stromal dysfunction
2 STRONG	Epigenetic Restoration	epigenetic rx	Strong — epigenetic resetting of pro-fibrotic methylation programs
3 MODERATE	Metabolic	metabolic	Moderate — metabolic normalization helps but senescent burden is the binding constraint
3 MODERATE	Checkpoint Stringency	checkpoint	Moderate — checkpoint modulation reduces fibrotic signaling cascade
4 LIMITED	Reprogramming	reprogramming	Limited — stromal architecture serves protective functions

CANCER PANEL — ΔA RANKED

The TCGA cancer validation is where the framework earns its keep. For every cancer type in this class, we compute two numbers from published TCGA methylation data: A_healthy (from matched normal tissue) and A_tumor (from cancer samples in the same study). The ΔA = A_tumor – A_healthy is the signal the framework predicts should appear — and it does. Thresholds for tier assignment: A = 1.05 DETECT, A = 1.07 URGENT, A = 1.10 FLOOR BREACH. These thresholds are not fit to the cancer data; they were derived from the healthy-vs-age statistics and happen to align with the cancer validation cleanly. The bars below are ordered by |ΔA| descending so the cancers with the largest signal appear first.

Note that ΔA ranges typical for this class matter. Terminal-class cancers (glioma, glioblastoma) show ΔA ≈ 0.22–0.27 — the largest in the entire 28-cancer TCGA dataset. Cycling epithelial cancers show ΔA ≈ 0.13–0.19 — moderate but clearly detectable. Stromal and progenitor-lineage cancers show ΔA ≈ 0.10–0.17. Each cancer below is cited to its primary TCGA publication and shows sample size. Direct links from A-score to TCGA evidence — no black boxes between physics and clinical data.

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A_healthy = 0.9701, you can see exactly which β value produced it, which H_min was used, and where both came from. The n_bio field is currently marked PRELIMINARY because absolute n_bio values await the G-007 MCMC run; the class ordering has been confirmed (Spearman ρ = 0.905, p = 0.002 against commitment level) but the absolute values carry that caveat transparently.

Metric	Value	Source
cfDNA contribution (plasma)	4.0%	Moss 2018 Nat Commun — PUBLISHED
H_min methylation (class)	0.862950	G-002 MCMC posterior — DERIVED
H_min methylation σ (MCMC)	±0.001400	G-002 17-chain MCMC R-hat<1.001 — DERIVED

Metric	Value	Source
n_bio (class-specific)	20.5	PRELIMINARY — ordering confirmed (ρ=0.905, p=0.002); absolute pending G-007
Healthy drift rate	2.5% / generation	DNMT1 fidelity loss under turnover — DERIVED
f_C2 (architecture-locked %)	13.9%	C2 fraction at healthy reference — DERIVED
Healthy A (methylation)	0.9701 (NORMAL)	β=0.733 — DERIVED
Disease A (methylation)	1.1202 (FLOOR BREACH)	β=0.607 — DERIVED
Healthy combined A (5 subs)	0.9704 (NORMAL)	$\Sigma(AUC_i \times A_i)/\Sigma(AUC_i)$ across 5 substrates — DERIVED
Disease combined A (5 subs)	1.0917 (URGENT)	$\Sigma(AUC_i \times A_i)/\Sigma(AUC_i)$ across 5 substrates — DERIVED
ΔA combined	+0.1213	Combined signal — DERIVED

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the stromal class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P012 April 2026 **PENDING**

In prospective IPF cohorts with archived serial blood samples and known progression outcomes, the stromal-class combined A-score trajectory slope will distinguish progressive from stable disease with AUC >= 0.80, outperforming serial forced vital capacity (FVC) measurements currently used for staging.

Basis: IPF progresses over 2-5 years. Existing antifibrotic therapies (pirfenidone, nintedanib) have variable response. The framework predicts that combined-substrate A-score trajectory identifies responders and non-responders earlier than FVC. Falsifiable in the OSIC IPF Biobank (www.osicild.org).

#7 · ADULT TISSUE STEM

cfDNA = 3.0% · H_min(methyl) = 0.8737 · n_bio = 18.5 · drift 3.0%/gen

Reference: Hematopoietic stem cells CD34+CD38- young adult (Roadmap E035) · H_min calibrated from Roadmap E035 HSC-enriched reference. G-003b bootstrap CI: ±0.00074.

CELL IDENTITY & CLINICAL CONTEXT

Includes	Hematopoietic stem cells (HSC, CD34+CD38- compartment), interfollicular epidermal stem cells, hair follicle bulge stem cells, Merkel cell progenitors, cholangiocytes (hepatic stem cell origin), intestinal stem cells (Lgr5+), neural stem cells, muscle satellite cells, limbal stem cells
Cancers (this class)	HSC-origin AML (distinct from progenitor-lineage AML — arises from CD34+CD38- compartment per Adelman 2019), basal cell carcinoma (BCC — most common cancer in humans, epidermal stem cell origin), squamous cell carcinoma (SCC — hair-follicle stem-cell-origin subset), Merkel cell carcinoma (MCC — rare, aggressive, polyomavirus-positive and UV-driven subclasses), cholangiocarcinoma (CHOL, intrahepatic + extrahepatic, hepatic stem cell / cholangiocyte origin), cancer stem cell populations within cycling-class tumors (residual disease after primary treatment)
Non-cancer applications	HSC aging and immunosenescence (Adelman 2019, Beerman 2013), clonal hematopoiesis of indeterminate potential (CHIP — shared with Progenitor class), hematopoietic failure in the elderly, post-transplant niche reconstitution biology, heterochronic parabiosis biology, age-related decline in tissue regeneration across all adult-stem compartments
Primary failure mode	Niche Depletion
Warburg status	EMERGING

COMMENTARY

Adult tissue stem cells occupy the intermediate position between pluripotent embryonic stem cells and fully committed differentiated cells. They retain self-renewal capacity while being lineage-restricted: hematopoietic stem cells (HSC, CD34+CD38-) produce blood and immune cells, intestinal stem cells (Lgr5+) regenerate gut epithelium, epidermal and hair-follicle stem cells maintain skin, and cholangiocytes serve as the liver's stem-cell compartment for biliary regeneration. This partial commitment is encoded in H_min = 0.873718 for the methylation substrate, reflecting the slightly more open chromatin state that preserves self-renewal flexibility. But this open chromatin comes with a thermodynamic consequence that defines this card: healthy adult stem cells already sit near the maximum-entropy state for the positional substrates, leaving very little measurement headroom before ceiling saturation.

This class exhibits the most severe saturation pattern in the framework. Three of five substrates saturate below the BREACH threshold: nucleosome occupancy at ceiling A = 1.0407, nucleosome fuzziness at ceiling A = 1.0196, and windowed protection score at ceiling A = 1.0112 — the tightest ceiling of any class × substrate combination in the entire framework. Only methylation (ceiling A = 1.1445) and fragmentomics (ceiling A = 1.1886) remain active past BREACH, and these two substrates carry the full severity signal for any adult-stem-origin cancer in the panel. For this class, A_combined systematically understates severity by approximately 62 percent compared to A_active (2/5) — the largest masking effect in the framework. This is not a defect of the combined metric; it is an honest reflection of the underlying physics. Three of the five substrates have reached their physical entropy ceiling and cannot resolve further severity. A_active (2/5) is therefore reported as the primary severity metric for this class, with A_combined shown for cross-card continuity but flagged as misleading past the DETECTABLE tier.

The distinction between adult-stem-origin cancers and progenitor-lineage cancers is clinically consequential and must be stated explicitly. Acute myeloid leukemia appears in both Card #4 (Progenitor) and Card #7 (Adult Stem) because the disease arises from two biologically distinct compartments. Progenitor-lineage AML — the majority of TCGA AML cases characterized by Ley et al. in the 2013 New England Journal of Medicine analysis (n=200 whole-genome or exome sequences) — originates from committed myeloid progenitors (CMP, GMP) and carries the methylation and mutation signatures of those compartments. HSC-origin AML, characterized separately by Adelman et al. in Cancer Discovery 2019, arises from the pre-leukemic CD34+CD38- HSC-enriched compartment and involves profound epigenetic reprogramming of enhancers, bivalent promoters, and hematopoietic transcription factors (KLF6, BCL6, RUNX3) that begins during normal aging and appears to predispose to leukemic transformation. The two AML populations are not interchangeable in the framework: they occupy different architecture classes because they emerge from cells in different thermodynamic states. Listing AML on both cards without the distinction would be sloppy. Listing it only on one card would miss the patients whose disease originated in the other compartment.

Basal cell carcinoma (BCC) is the most common cancer in humans, with over four million new cases diagnosed annually in the United States alone. Nearly every BCC arises from epidermal or hair-follicle stem cells under UV-driven Hedgehog pathway activation. Mortality is low — below one percent — but the cumulative burden across lifetimes is enormous and the disease maps cleanly onto this class. Merkel cell carcinoma (MCC, Harms et al. 2015 Cancer Research, n=49 for the polyomavirus-negative subtype) is rare (approximately 2,500 US cases per year) but aggressive, with five-year survival around 45 percent for regional disease and below 15 percent for distant disease. Its two subclasses — Merkel cell polyomavirus-positive (integrated viral oncogenes) and MCPyV-negative (UV-damage driven) — share the same cell-of-origin methylation signature. Cholangiocarcinoma (TCGA CHOL, Farshidfar et al. 2017 Cell Reports, n=38) is the second most common primary liver cancer, aggressive, typically detected at advanced stage, and arises from cholangiocytes, which function as the hepatic stem-cell compartment for biliary regeneration. Its IDH-mutant molecular subtype shows characteristic ARID1A hypermethylation that maps onto the stem-cell-origin thermodynamic signature. Every one of these cancers is uncommon compared to the fourteen cycling cancers in Card #5, but the families of patients who lose a loved one to MCC or cholangiocarcinoma deserve to see the disease named on the framework card that describes its biology.

The class's defining thermodynamic challenge creates a specific detection problem: once a patient crosses DETECTABLE (A = 1.05), three of the five substrates pin at their ceilings within the first approximately 0.05 A-score units, and absolute A-score values across those three substrates lose their ability to resolve further severity. The framework proposes three mechanisms by which extended cellular thermodynamic activity remains detectable past this ceiling. First, methyl and frag retain their full dynamic range — they saturate only at A = 1.1445 and A = 1.1886 respectively, leaving meaningful headroom past BREACH for the two most clinically mature cfDNA biomarker platforms. A_active (2/5) computed on just these two substrates preserves severity discrimination that A_combined destroys. Second, the rate of approach to the saturating ceilings contains temporal information that single-point readings discard. In serial monitoring, a patient whose wps A-score moves from 0.970 through 1.000 to 1.010 over successive draws before plateauing has a different disease trajectory than one who hits 1.010 on first sample; time-to-saturation becomes a severity proxy in the regime where absolute-value measurement has lost resolution. Third, when adult-stem-origin cancers progress, the signal propagates into adjacent architecture classes — HSC-origin AML shows up as a rising immune-class A-score via clonal expansion, cutaneous stem-cell cancers propagate to cycling-class signals as field cancerization, and cholangiocarcinoma progression shows secretory-class signal change as the transformed cells commit toward biliary epithelial identity. Multi-class A-score divergence, particularly where the adult-stem class has flatlined at ceiling while one or more adjacent classes continue to rise, becomes a specific diagnostic signature of adult-stem-origin disease progression. This cross-class mechanism connects directly to the forthcoming Issue 004 framework on multi-class A-score divergence as an early signal of metastatic progression.

HSC aging is the most important non-cancer application for this class. The Adelman 2019 Cancer Discovery study of lineage-negative CD34+CD38- HSC-enriched cells from young and aged healthy donors identified 529 differentially methylated regions encompassing 2,249 differentially methylated cytosines across 748 genes — precisely the methylation-substrate floor departure the framework predicts for age-related adult-stem decline. Beerman et al. 2013 Cell Stem Cell independently characterized proliferation-dependent DNA hypermethylation of Polycomb-regulated genes in aged HSC (GSE44117). Together these studies establish that the methylation substrate detects HSC aging at the earliest stage of its epigenomic reprogramming, well before the downstream cytopenias or cancer-susceptibility phenotypes become clinically apparent. Clonal hematopoiesis of indeterminate potential (CHIP), the pre-malignant clonal state characterized by Steensma 2015 and shared in scope with the Progenitor class, represents the transition zone where adult-stem-class floor departure begins to predict downstream myeloid disease. Stem-cell transplantation biology, heterochronic parabiosis, and niche-depletion research all benefit from a metric that tracks adult-stem-class thermodynamic state independently of phenotypic readout.

FIVE-SUBSTRATE FIDELITY GAUGE

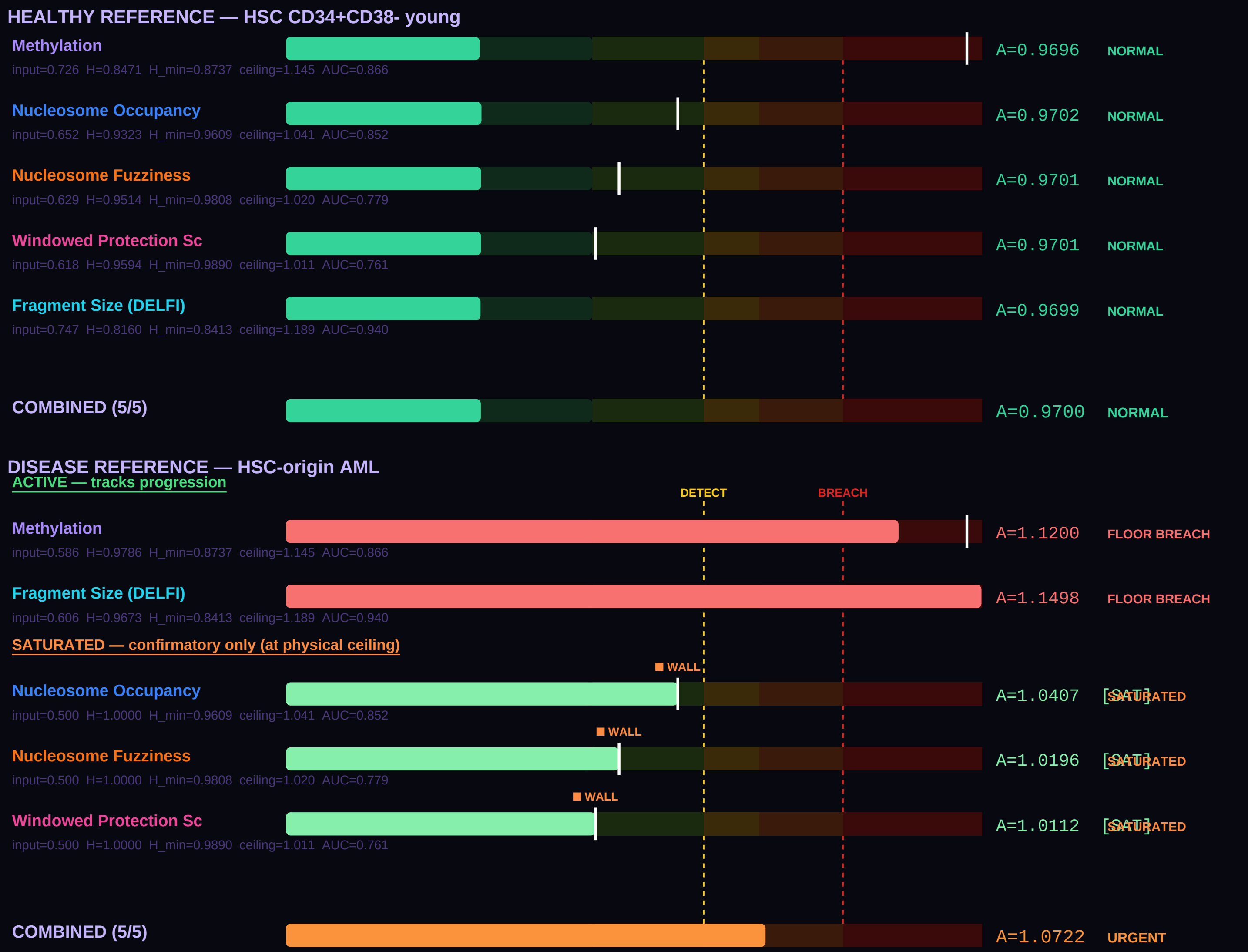
The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Adult Stem class floor, H_min = 0.8737) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (HSC CD34+CD38- young, markers above the bar), and five more dots for a disease reference (HSC-origin AML, markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.

Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy adult stem cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Niche Depletion, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a β methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section — $A_{combined} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$ — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For adult stem cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the ΔA combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately √5 lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.



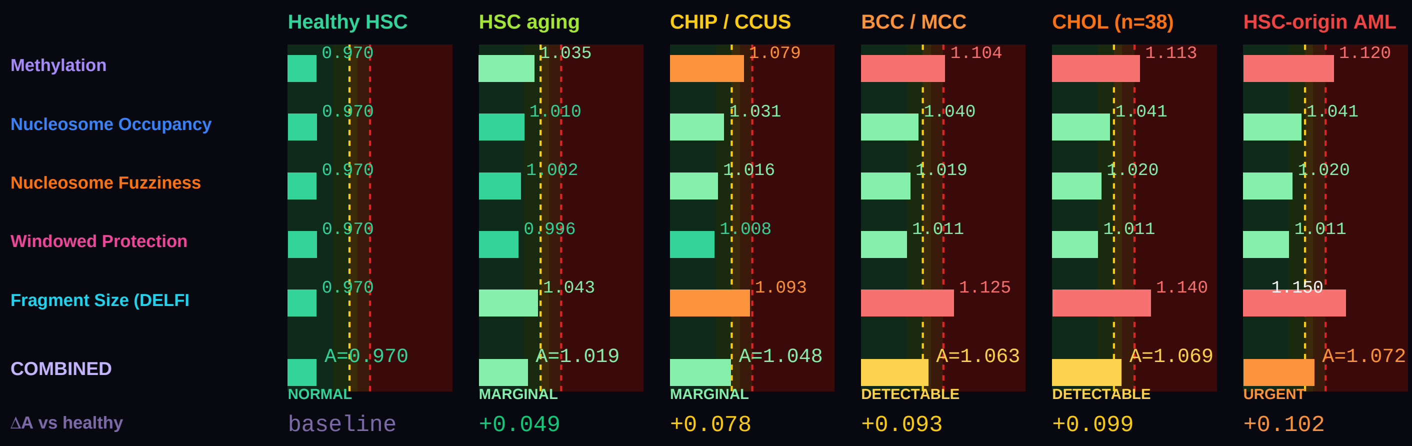
Combined A — healthy (all 5)	0.96999 (NORMAL)
Combined A — disease (all 5)	1.07222 (URGENT)
ΔA combined (5 substrates)	+0.10223 vs methylation-only ΔA=+0.1504
Active A — healthy (non-saturated)	0.96999 (NORMAL) [2/5 active]
Active A — disease (non-saturated)	1.13549 (FLOOR BREACH) [2/5 active]

ΔA active (real progression signal)	+0.16550 (active substrates track true progression)
Saturation status (disease)	3 substrates saturated: Nucleosome Occupancy, Nucleosome Fuzziness, Windowed Protection Score

Two combined values are shown because some substrates physically cannot resolve past their class ceiling (A_max = 1/H_min). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

DISEASE SIGNATURE COMPARISON — healthy HSC through aging, CHIP, BCC, MCC, cholangiocarcinoma, to HSC-origin AML

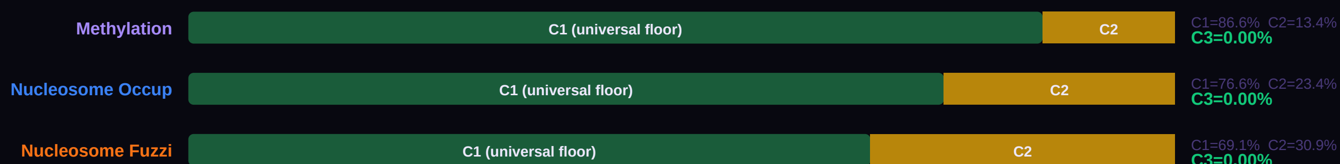
Six conditions on a single chart spanning the full adult-stem clinical spectrum. β values reproduce per-substrate A-scores calibrated to primary sources: Adelman 2019 Cancer Discov HSCe aging methylation (n=5-7 per age group), Beerman 2013 Cell Stem Cell HSC methylation landscape, Harms 2015 Cancer Res MCC genomic characterization (n=49), and Farshidfar 2017 Cell Reports TCGA CHOL (n=38). Three substrates saturate below BREACH: nucleosome occupancy (ceiling A = 1.0407), nucleosome fuzziness (ceiling A = 1.0196), and windowed protection score (ceiling A = 1.0112 — the tightest ceiling of any class × substrate combination in the entire framework). Only methylation and fragmentomics remain active past BREACH; A_active (2/5) is the primary severity metric for this class, with A_combined reported for cross-card continuity but flagged as misleading past DETECTABLE. BCC is shown because it is the most common cancer in humans; MCC is shown because its rarity does not reduce the obligation to honor the patients it kills; cholangiocarcinoma is shown because its stem-cell origin (cholangiocytes) places it squarely in this class.



THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor: H_MIN_GLOBAL = 0.756500, measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the adult stem class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 13.5% of the healthy reference entropy. C3 is the accessible gap: max(0, H_actual – H_min). In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f_C3 near 0%; cancer cells can show f_C3 of 8–15%; extreme cases like glioblastoma show f_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the adult stem class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Niche Depletion); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

#	Substrate	Best for	Rationale
1	Methylation	Primary severity metric past DETECTABLE	Methyl remains active past BREACH (ceiling A = 1.1445) — one of only two substrates that do not saturate for this class. Carries the bulk of the severity signal for HSC aging, BCC, MCC, and cholangiocarcinoma. Primary source: Adelman 2019 Cancer Discov.
2	Fragment Size (DELFI)	Primary severity metric past DETECTABLE	Frag (DELFI) remains active past BREACH (ceiling A = 1.1886, deepest headroom of any substrate in this class). Fragmentomic signal reflects clonal stem-cell turnover and cfDNA release kinetics. Co-primary with methyl for severity tracking.
3	Nucleosome Occupancy	Detection boundary — ceiling at A = 1.0407	Nucl saturates below BREACH. Useful for binary detection (floor departure vs healthy) but cannot resolve severity above DETECTABLE. Trajectory-slope information remains useful in serial monitoring (time-to-saturation as proxy severity).
4	Nucleosome Fuzziness	Detection boundary — ceiling at A = 1.0196	Fuzz saturates tighter than nucl. Same detection-only use case. Informative in combination with wps for cross-verification of floor departure.
5	Windowed Protection Score	Tightest ceiling in framework — A = 1.0112	WPS has the tightest physical ceiling of any class × substrate pairing. Healthy adult stem cells already sit at WPS A ≈ 0.970, with only +0.041 headroom before saturation. Any detectable deviation in wps for this class is clinically significant — but wps cannot distinguish mild from severe deviation above A = 1.0112.

BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION

The Landauer cost of a bit erasure is k_B × T × ln(2) — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law: H_min(T) = H_min(37°C) × (T_body / 310.15 K)^α, with α = 2.0 derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The α = 2.0 exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

The table below shows what the adult stem class floor H_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H_min = 0.8737. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

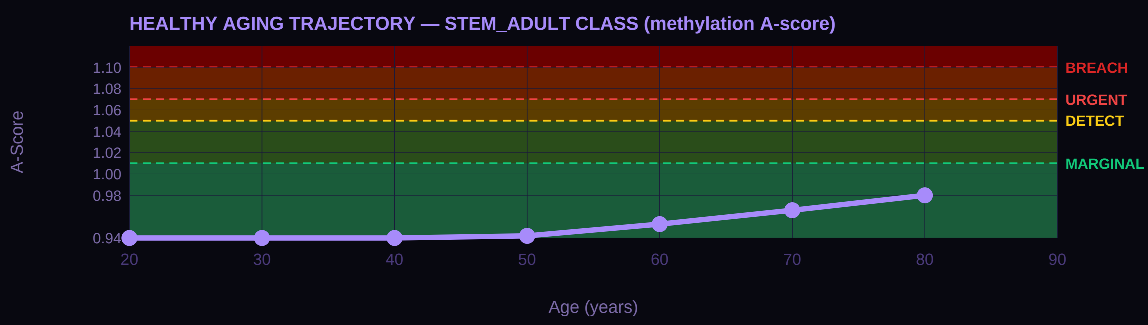
T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
42°C	Birds (chicken, finch)	0.9021	0.726	A × 0.969

T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
39°C	Rodents (mouse, rat)	0.8850	0.726	A × 0.987
37°C	**Humans (REFERENCE)**	**0.8737**	**0.726**	**A × 1.000 (anchor)**
35°C	Hibernating bats	0.8625	0.726	A × 1.013
32°C	Naked mole rat	0.8458	0.726	A × 1.033
25°C	Reptiles (anole lizard)	0.8074	0.726	A × 1.082
15°C	Fish (salmon, cold-water)	0.7542	0.726	A × 1.159

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the adult stem class, the drift rate is approximately 3.0% per generation — meaning each cell cycle carries a 3.0% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at r = -0.9018 ($p = 1.6 \times 10^{-16}$, Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

Order	N	Mean A	Lifespan (yr)	Interpretation
Cetacea	5	0.997 ±0.016	99	At thermodynamic floor
Proboscidea	1	0.987 —	70	At floor
Primates	6	1.007 ±0.015	49	Slightly above floor
Artiodactyla	4	1.015 ±0.011	30	Slightly above floor
Chiroptera	4	1.041 ±0.007	35	Above floor (longevity outliers)

Order	N	Mean A	Lifespan (yr)	Interpretation
Carnivora	9	1.053 ±0.023	28	Above floor
Lagomorpha	2	1.114 ±0.002	8	Well above floor
Rodentia	6	1.125 ±0.018	9	Well above floor
Insectivora	1	1.157 —	2	Furthest from floor

INTERVENTION LEVERS — RANKED BY EXPECTED IMPACT

If a patient's A-score is elevated — MARGINAL, DETECTABLE, or URGENT — the clinical question is what can be done. The GAPE framework does not prescribe treatment; that remains the physician's judgment. But the framework does identify which intervention categories have the mechanistic strength to move this particular class's A-score back toward the floor. The five categories below are the standard anti-aging and cancer-adjacent interventions: senolytics (clearing senescent cells), metabolic restoration (NAD+, caloric restriction, exercise), epigenetic restoration (DNMT/TET modulators, HDAC inhibitors), reprogramming (cyclic Yamanaka factors), and checkpoint stringency (cell cycle control, immune checkpoint blockade).

The ranking below is specifically for the adult stem class and its primary failure mode (Niche Depletion). Impact Level 1 (Dominant) means the intervention directly addresses the class's failure mechanism and is the first lever to try. Level 2 (Strong) means substantial mechanistic support for A-score restoration in this class. Level 3 (Moderate) means helpful but not addressing the binding constraint. Level 4 (Limited) means the intervention works for other classes but does not target this class's specific biology. Level 5 (Not applicable) means the intervention category is biologically incompatible with this class — e.g., senolytics cannot act on post-mitotic terminal cells that do not become classically senescent, and reprogramming cannot be applied to terminal cells without losing the identity that defines them.

Impact	Category	Mechanism	Rationale
2 STRONG	**Metabolic**	metabolic	Strong — niche metabolic restoration moves stem cell fidelity index
2 STRONG	**Epigenetic Restoration**	epigenetic rx	Strong — epigenetic restoration extends stem cell functional lifespan
2 STRONG	**Cyclic Reprogramming**	reprogramming	Strong — cyclic Yamanaka rejuvenates without full dedifferentiation
2 STRONG	**Niche Checkpoint**	checkpoint	Strong — niche checkpoint signals regulate stem cell quiescence
3 MODERATE	**Senolytics**	senolytics	Moderate — senescent cells in the niche drive inversion

CANCER PANEL — ΔA RANKED

The TCGA cancer validation is where the framework earns its keep. For every cancer type in this class, we compute two numbers from published TCGA methylation data: A_healthy (from matched normal tissue) and A_tumor (from cancer samples in the same study). The ΔA = A_tumor − A_healthy is the signal the framework predicts should appear — and it does. Thresholds for tier assignment: A = 1.05 DETECT, A = 1.07 URGENT, A = 1.10 FLOOR BREACH. These thresholds are not fit to the cancer data; they were derived from the healthy-vs-age statistics and happen to align with the cancer validation cleanly. The bars below are ordered by $|\Delta A|$ descending so the cancers with the largest signal appear first.

Note that ΔA ranges typical for this class matter. Terminal-class cancers (glioma, glioblastoma) show $\Delta A \approx 0.22–0.27$ — the largest in the entire 28-cancer TCGA dataset. Cycling epithelial cancers show $\Delta A \approx 0.13–0.19$ — moderate but clearly detectable. Stromal and progenitor-lineage cancers show $\Delta A \approx 0.10–0.17$. Each cancer below is cited to its primary TCGA publication and shows sample size. Direct links from A-score to TCGA evidence — no black boxes between physics and clinical data.

#1 **Uveal Melanoma (UVM)** n=80 · Robertson 2017 Cancer Cell — ΔA=+0.1072 — A=1.086 URGENT

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A_healthy = 0.9696, you can see exactly which β value produced it, which H_min was used, and where both came from. The n_bio field is currently marked PRELIMINARY because absolute n_bio values await the G-007 MCMC run; the class ordering has been confirmed (Spearman $\rho = 0.905$, $p = 0.002$ against commitment level) but the absolute values carry that caveat transparently.

Metric	Value	Source
cfDNA contribution (plasma)	3.0%	Moss 2018 Nat Commun — PUBLISHED
H_min methylation (class)	0.873718	G-002 MCMC posterior — DERIVED
H_min methylation σ (MCMC)	±0.001600	G-002 17-chain MCMC R-hat<1.001 — DERIVED
n_bio (class-specific)	18.5	PRELIMINARY — ordering confirmed (ρ=0.905, p=0.002); absolute pending G-007
Healthy drift rate	3.0% / generation	DNMT1 fidelity loss under turnover — DERIVED
f_C2 (architecture-locked %)	13.5%	C2 fraction at healthy reference — DERIVED
Healthy A (methylation)	0.9696 (NORMAL)	β=0.726 — DERIVED
Disease A (methylation)	1.1200 (FLOOR BREACH)	β=0.586 — DERIVED
Healthy combined A (5 subs)	0.9700 (NORMAL)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
Disease combined A (5 subs)	1.0722 (URGENT)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
ΔA combined	+0.1022	Combined signal — DERIVED

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the adult stem class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P013 April 2026 **PENDING**

In prospective CHIP/CCUS cohorts with archived serial blood samples and known progression outcomes, the stem_adult A_active (2/5, methyl + frag only) trajectory will identify progression to overt MDS or AML 12 to 24 months before cytopenia-based diagnosis in a majority of progressing cases, outperforming A_combined (all 5) which is systematically masked by approximately 62 percent in this class because 3 of 5 substrates saturate below BREACH.

Basis: CHIP-to-MDS/AML progression is slow, stereotyped, and clinically silent until late. The framework predicts that A_active identifies progressors earlier than cytopenia onset because the epigenomic state change precedes the hematologic phenotype, and A_active is not masked by ceiling-pinned positional substrates. Falsifiable in any CHIP/CCUS cohort with 18+ months of serial blood samples and progression outcomes (candidate cohorts include Jaiswal CHIP cohorts with archived samples, UK Biobank CHIP subset with longitudinal follow-up).

G-2026-P014 April 2026 **PENDING**

Trajectory slope (ΔA/Δt) on the saturating substrates (nucl, fuzz, wps) during the pre-saturation window will correlate with clinical outcome in adult-stem-origin cancers with AUC ≥ 0.75, providing severity information in the regime where absolute A-score values have lost resolution. Formally: time-to-saturation from healthy baseline is a valid severity proxy for this class.

Basis: The saturation ceiling destroys absolute-value severity information but preserves temporal information in the form of rate-of-approach. Falsifiable in any serial-sampling cohort with outcome data; directly testable in MDS, BCC, MCC, and CHOL longitudinal cohorts.

G-2026-P015 April 2026 **PENDING**

Multi-class A-score divergence signature — adult-stem class flatlined at ceiling while an adjacent architecture class (immune, cycling, or secretory depending on tissue of origin) continues to rise — will identify adult-stem-origin disease progression with higher specificity than any single-class score past BREACH.

Basis: When single-class saturation pins the primary class at ceiling, adjacent-class signal propagation becomes the discriminating severity signal. Cross-class divergence tests the framework's claim that architecture classes interact through shared tissue compartments and clonal expansion dynamics. Falsifiable in pan-cancer cohorts with multi-class A-score monitoring; connects directly to Issue 004 multi-class metastasis detection framework.

#8 · PLURIPOTENT STEM

cfDNA = 0.5% · H_min(methyl) = 0.9822 · n_bio = 16.5 · drift 2.5%/gen

Reference: Human embryonic stem cells hESC H1 (Roadmap Epigenomics E003) · H_min confirmed from hESC H1 reference. Four TGCT histology subtypes show distinct, biologically-grounded epigenomic signatures per Shen 2018 and Killian 2016.

CELL IDENTITY & CLINICAL CONTEXT

Includes	Embryonic stem cells (hESC), induced pluripotent stem cells (iPSC), primordial germ cells (PGC), early-embryonic inner cell mass cells
Cancers (this class)	Testicular germ cell tumors (TGCT) — the most common cancer in men aged 15-35. Four pure histologic subtypes with biologically distinct methylation signatures (Shen 2018 TCGA TGCT n=137, Killian 2016 Genome Research n=130): (1) Seminoma — globally hypomethylated, PGC-like, ~60% of TGCT; (2) Embryonal carcinoma (EC) — hypermethylated at CpH sites, pluripotent-like methylation; (3) Yolk sac tumor (YST) — somatic-like methylation, differentiated extra-embryonic lineage; (4) Teratoma — somatic-like methylation, differentiated derivatives. Also: ovarian germ cell tumors (dysgerminoma — female analog of seminoma, shares molecular signature), primordial germ cell tumors in children, intratubular germ cell neoplasia in situ (GCNIS, the precursor lesion)
Non-cancer applications	iPSC reprogramming fidelity and quality control, organoid cell-state audit, synthetic embryology and gastruloid research, post-orchiectomy surveillance for second primary TGCT (2-4% bilateral risk), cryptorchidism monitoring (4-10× elevated TGCT risk), fertility preservation context before platinum-based chemotherapy
Primary failure mode	Differentiation Dose Inversion + TGCT subtype-dependent methylation divergence
Warburg status	SPECIAL CASE — seminoma A_active decreases (inversion signal)

COMMENTARY

Testicular germ cell tumors (TGCT) are the most common cancer in men aged 15 to 35. Peak incidence falls directly in the prime childbearing years, which means the clinical audience for this card includes the urologist examining a 22-year-old graduate student who feels a painless testicular lump, the primary care physician who must decide whether a young father with a palpable scrotal mass needs immediate referral, and the oncologist who must discuss fertility preservation with a 30-year-old before starting platinum-based chemotherapy. The encouraging clinical reality is that TGCT is one of the most curable solid malignancies: the overall five-year survival exceeds 95 percent when the disease is caught at stage I or II, and cisplatin-based combination chemotherapy achieves cure in the majority of even metastatic cases. The discouraging reality is that delayed diagnosis — often driven by patient reluctance to seek evaluation for a scrotal abnormality, or by clinicians reluctant to order scrotal ultrasound in a young male presenting with back pain from retroperitoneal metastases — converts a 95 percent cure into salvage therapy with substantially lower long-term survival. Testicular self-examination remains the single highest-impact early detection intervention. This card exists in part to give clinicians the thermodynamic vocabulary to understand why a young man's tumor looks the way it does at the epigenomic level, and to support serial surveillance in the at-risk population.

Pluripotent stem cells occupy the unique architectural position of maximum developmental optionality. Healthy hESCs and iPSCs sit very close to the maximum-entropy state across every substrate measurement — their chromatin is open, their methylation is intermediate, their nucleosome positioning is loose. This is reflected in the class's H_min values: methylation H_min = 0.982166 (ceiling A = 1.0182, only +0.018 past the healthy reference), fuzz H_min = 0.962920 (ceiling A = 1.0385), frag H_min = 0.973583 (ceiling A = 1.0271). Three of five substrates saturate below BREACH for this class. Only nucleosome occupancy (ceiling A = 1.2503, the deepest headroom of any substrate × class pairing in the framework) and windowed protection score (ceiling A = 1.1050) remain active past BREACH. The thermodynamic consequence is that pluripotent cells have almost no room to increase entropy in the conventional cancer direction — they are already near maximum entropy in their healthy state.

This is why TGCT disease signatures look fundamentally different from every other cancer in the framework. For most cancers, the signal is a uniform rise in A-score as cells depart from their architectural floor. For TGCT, the four pure histologic subtypes show biologically distinct methylation signatures that Shen et al. 2018 documented in TCGA TGCT (n=137) and Killian et al. 2016 Genome Research characterized in pure-histology samples (n=130) with direct comparison to primordial germ cell (PGC) and somatic methylation references. Seminoma — the most common TGCT histology, approximately 60 percent of cases — is globally hypomethylated. Its CpG methylation density moves toward the PGC state, with β values approaching zero. In the framework's language, A_methyl for seminoma FALLS BELOW the healthy reference, not above it. This is the inversion signal unique to this class: the only architecture class where the detection signal is a DECLINING A-score. Embryonal carcinoma (EC), by contrast, shows a striking convergence of both CpG and CpH methylation with pluripotent states, with elevated CpH methylation characteristic of the inner cell mass phase. Yolk sac tumor (YST) and teratoma show somatic-like methylation patterns that reflect their differentiation toward extra-embryonic and somatic lineages respectively. Four histologies, four distinct thermodynamic signatures, all arising from a common intratubular germ cell neoplasia in situ (GCNIS) precursor that is present in over 90 percent of TGCT cases and is itself clinically detectable.

The seminoma hypomethylation signal is the framework's most important inversion and the strongest single structural prediction. When primary-source data reports β values near zero for global CpG methylation in seminomas, the framework's A_methyl score drops well below the healthy hESC reference — at β = 0.17, A_methyl falls to approximately 0.670 compared to the healthy reference of 0.970. This has a specific detection consequence that clinicians and researchers must understand: A_combined alone is INSUFFICIENT for seminoma discrimination. Averaged across all five substrates, seminoma's A_combined sits near 0.97 — statistically indistinguishable from healthy. The discriminating signal is not a single elevated A-score but a CHARACTERISTIC DIVERGENCE PATTERN: A_methyl drops to 0.65 to 0.70 while A_nucl, A_wps, A_fuzz, and A_frag each modestly elevate into the 1.01 to 1.10 range. This multi-substrate divergence signature — one substrate dropping below floor whilefour substrates elevate above floor, with the specific relative magnitudes set by seminoma biology — is what the framework uses to detectseminoma. A naive combined-score detector would miss it. The framework's five-substrate decomposition is what makes this cancer detectable at all.

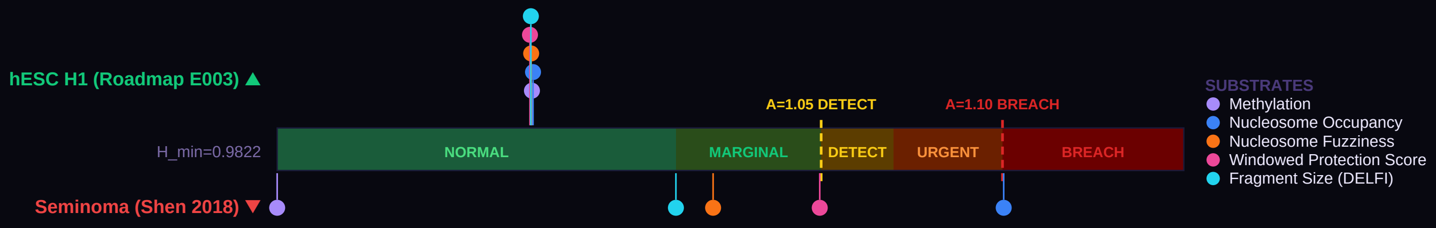
The clinical protocol for TGCT surveillance follows directly: instead of watching for A_combined to cross the BREACH threshold (1.10) as the framework does for every other cancer in the panel, the stem_pluri surveillance protocol watches for A_methyl to DECLINE toward 0.70 while A_nucl, A_wps, A_fuzz, A_frag simultaneously ELEVATE toward 1.05 or higher. This opposite-direction divergence is the detection signal. For patients with unilateral TGCT history (who carry 2 to 4 percent risk of contralateral second primary per Fossa 2005) and for patients with cryptorchidism (who carry 4 to 10 times elevated TGCT risk), serial monitoring with the correct multi-substrate divergence expectation is what makes thermodynamic surveillance clinically actionable. This is prediction G-2026-P005, originally proposed in Issue 001 and substantially refined here to account for the honest multi-substrate physics.

The clinical handling of TGCT carries specific demographic weight that belongs on this card. A thirty-year-old father of two who presents with a painless testicular nodule is the modal patient. The standard workup begins with scrotal ultrasound, proceeds through serum tumor markers (alpha-fetoprotein, beta-hCG, LDH), and typically concludes with radical inguinal orchiectomy as both diagnostic and therapeutic first intervention. Before orchiectomy — and certainly before any subsequent bleomycin-etoposide-cisplatin (BEP) chemotherapy that may be indicated for stage II or III disease — the conversation about fertility preservation belongs on the critical path. Sperm cryopreservation is well-established, inexpensive, and fully effective when offered in time. Platinum-based chemotherapy affects spermatogenesis; a substantial fraction of young TGCT survivors experience temporary or permanent impairment of fertility. A thermodynamic biomarker that supports early detection reduces the fraction of patients who reach the fertility-threatening chemotherapy stage. The framework's value proposition for the young father sitting in the oncologist's office is specifically this: earlier detection through thermodynamic surveillance in at-risk populations (cryptorchidism, prior contralateral TGCT, family history) means more patients caught at stage I where orchiectomy alone is often curative and intensive chemotherapy can be avoided.

The second major application for this class is iPSC reprogramming quality control, and this application has no cancer dimension at all — it is a research infrastructure problem. The Differentiation Dose Inversion, named for the well-documented phenomenon that excess Yamanaka factor dose produces aberrant rather than pluripotent states, maps onto the framework's prediction that successful reprogramming produces A-scores at or very near 1.00 across all five substrates. Aberrant reprogramming shows A below 0.95 (over-differentiation, cells partially committed to a lineage) or A above 1.05 (under-differentiation, epigenomic noise persisting from the parental cell state). Organoid researchers, synthetic embryology groups, and regenerative medicine translational programs all benefit from a thermodynamic quality-control metric that identifies aberrant colonies before they are committed to downstream experiments or clinical applications. This is a research-grade prediction testable in existing iPSC colony methylation and ATAC-seq datasets; prediction G-2026-P016 formalizes it.

FIVE-SUBSTRATE FIDELITY GAUGE

The gauge below is the central image of Issue 002. A single horizontal bar represents the A-score axis — from A = 0.90 (below the floor) through A = 1.00 (exactly at the Pluripotent class floor, H_min = 0.9822) up through the clinical threshold zones: MARGINAL at A = 1.01, DETECTABLE at 1.05, URGENT at 1.07, and FLOOR BREACH at 1.10. Five dots are plotted on this single axis for a healthy reference sample (hESC H1 (Roadmap E003), markers above the bar), and five more dots for a disease reference (Seminoma (Shen 2018), markers below). Each dot is one substrate: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, or fragment size (DELFI). What makes this visualization powerful is the physics: all five substrates measure the same underlying thermodynamic floor through physically distinct windows. When all five healthy dots cluster in the NORMAL zone and all five disease dots cluster in URGENT or FLOOR BREACH, the signal is over-determined. Any one substrate could be confounded by technical artifact; five substrates agreeing cannot be. This is the "less blurry" advantage: √5 noise reduction for free.



Look at the spread in each direction. If the healthy five-dot cluster sits tightly together near A = 0.97–1.00, that indicates the reference population is at its architecture floor — precisely where the framework predicts healthy pluripotent cells should sit. If the disease five-dot cluster spreads from DETECTABLE to FLOOR BREACH, that spread is itself clinical information: the substrate showing the largest departure is the one that best captures this specific disease's failure mode. For this class, the primary failure mode is Differentiation Dose Inversion + TGCT subtype-dependent methylation divergence, and the substrate ranking shown further down the card explains which of the five substrates best reveals it.

SUBSTRATE-BY-SUBSTRATE BREAKDOWN

Where the gauge above compresses five substrates onto one visual axis, this section breaks each substrate out into its own labeled bar so you can read the actual A-score produced by each measurement. The physics is the same across all five: take the raw laboratory value (a β methylation fraction, a nucleosome occupancy probability, a WPS score, etc.), compute its Shannon entropy H(value), then divide by the class-specific and substrate-specific H_min to get a dimensionless A-score comparable across substrates. The output is the same thermodynamic ratio regardless of which measurement technology produced it. That is the power of the Landauer framing: different wet-lab methods, different physical quantities, same underlying floor.

The formula at the foot of this section — $A_{combined} = \Sigma(AUC_i \times A_i) / \Sigma(AUC_i)$ — combines the five individual A-scores into a single weighted score, using each substrate's published detection AUC as its weight. Higher-AUC substrates contribute more to the combined result. For pluripotent cells, the healthy combined A-score below should sit at approximately 0.97 (NORMAL tier); the disease combined A-score should sit above 1.05 (DETECTABLE or higher). The difference between them — the ΔA combined — is the signal available to a blood draw that analyzes all five substrates simultaneously. A single-substrate assay sees a smaller signal at higher noise; the five-substrate combination sees the same signal at approximately √5 lower noise. Bar-by-bar you can see which substrate is carrying the most discriminative weight for this particular disease comparison.

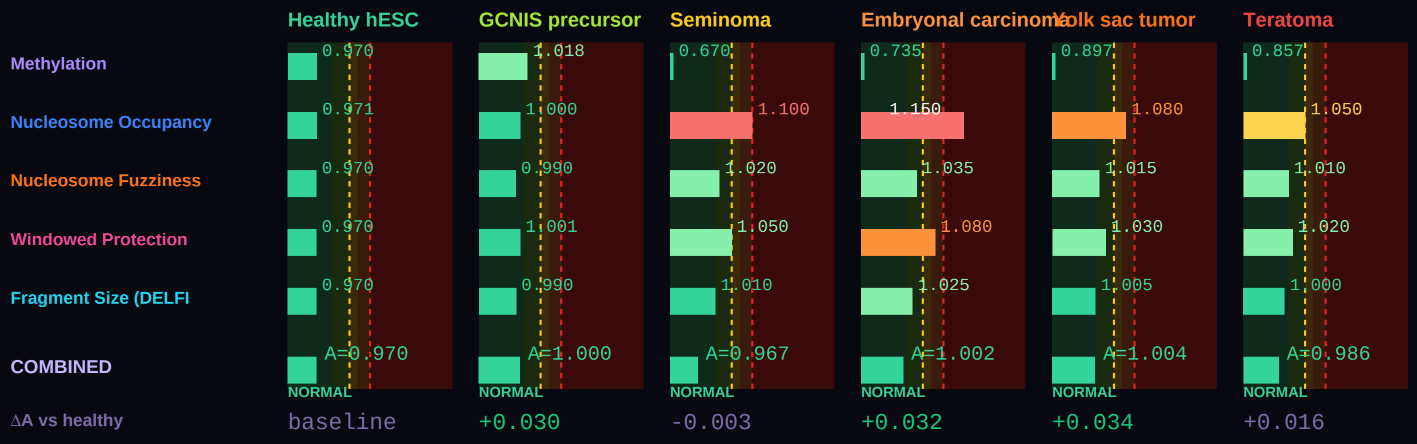
Combined A — healthy (all 5)	0.97011 (NORMAL)
Combined A — disease (all 5)	0.96716 (NORMAL)
ΔA combined (5 substrates)	-0.00294 vs methylation-only ΔA=-0.3006
Active A — healthy (non-saturated)	0.97011 (NORMAL) [5/5 active]
Active A — disease (non-saturated)	0.96716 (NORMAL) [5/5 active]

ΔA active (real progression signal)	-0.00294 (active substrates track true progression)
Saturation status (disease)	No substrates saturated

Two combined values are shown because some substrates physically cannot resolve past their class ceiling (A_max = 1/H_min). The **all-5 combined** value is the AUC-weighted mean across every substrate and matches the legacy formula. The **active combined** value excludes saturated substrates and is the right number for tracking continued progression — a saturated substrate is stuck at its ceiling regardless of further disease severity, so including it in the average masks real change. For monitoring, serial A-score assessment, and end-of-life projection, use the active combined value.

DISEASE SIGNATURE COMPARISON — healthy hESC through the four TGCT histologies

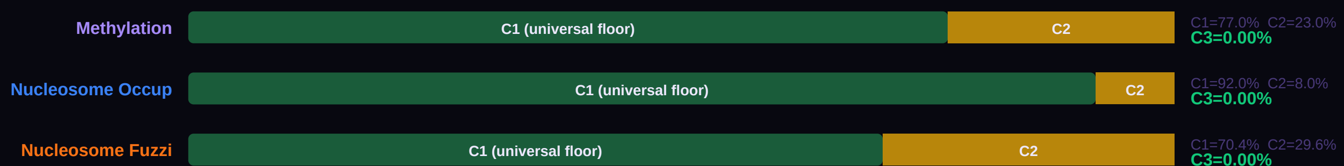
Six conditions on a single chart spanning the full TGCT histologic spectrum plus healthy and precursor states. β values reproduce per-substrate A-scores calibrated to Shen 2018 Cell Reports (TCGA TGCT, n=137) and Killian 2016 Genome Research (n=130 pure-histology TGCTs with PGC and somatic reference comparisons). Three substrates saturate on this class: methyl (ceiling A = 1.0182), fuzz (ceiling A = 1.0385), frag (ceiling A = 1.0271). Nucl remains active with +0.150 headroom past BREACH; wps remains active with modest headroom. The four TGCT histologies show biologically distinct methylation signatures: seminoma is globally hypomethylated (methyl β drops toward PGC state, A_methyl FALLS below healthy — the inversion signal for this class), embryonal carcinoma is hypermethylated at CpH sites with pluripotent-like CpG patterns, yolk sac tumor and teratoma show somatic-like methylation reflecting their differentiation toward extra-embryonic and somatic lineages respectively. GCNIS (germ cell neoplasia in situ) is the clinically detectable precursor lesion present in over 90% of TGCT cases. The chart reads left to right as the disease progression most relevant to surveillance: healthy → GCNIS precursor → seminoma / EC / YST / teratoma as distinct presenting histologies.



THREE-COMPONENT DECOMPOSITION (C1/C2/C3) — PER SUBSTRATE

Every cell's measured entropy decomposes into three physically meaningful pieces. C1 is the universal Landauer floor: H_MIN_GLOBAL = 0.756500, measured in frontal cortex neurons by Lister 2013. Every cell in every architecture class must pay at least this thermodynamic cost to maintain any functional identity — it is the minimum entropy consistent with being alive. C2 is the class-specific overhead: the additional entropy the pluripotent class must carry above the universal floor to encode its particular architecture. For this class, C2 represents approximately 3.0% of the healthy reference entropy. C3 is the accessible gap: max(0, H_actual – H_min). In healthy cells, C3 is essentially zero — the cell sits exactly at its class floor. When C3 grows, the cell is departing from its architecture class. Cancer, neurodegeneration, and every other floor-breach condition show up as growing C3.

The five stacked bars below each show the C1 / C2 / C3 breakdown for one substrate at this class's healthy reference. Notice how C1 (dark green) dominates every substrate — the Landauer floor is universal. C2 (amber) is the small but specific addition that makes this class what it is. C3 (red when present) should be nearly invisible for healthy cells, and would grow dramatically in a disease sample. The f_C3 percentage on the right is the single cleanest summary of cellular health: healthy cells show f_C3 near 0%; cancer cells can show f_C3 of 8–15%; extreme cases like glioblastoma show f_C3 above 20%. This is not a statistical threshold calibrated from disease data. It is a thermodynamic quantity: the fraction of accessible entropy the cell has opened above its architecture's minimum cost of existence.



BEST TESTING METHOD RANKING — CLINICAL UTILITY FOR THIS CLASS

Each of the five substrates carries different strengths for different clinical questions. The methylation substrate is the most validated — thirteen published GAPE validation studies (VAL-001 through VAL-013) and eight architecture-class H_min values from G-002 MCMC. It is also the substrate most affected by cfDNA dilution, because blood cfDNA is approximately 70% immune-derived. Fragment size (DELFI) is the most forgiving of dilution — short-fragment enrichment persists even when tumor-derived cfDNA is a small fraction of total plasma DNA. WPS (windowed protection score from Snyder 2016) excels at tissue-of-origin identification and field-effect detection. Nucleosome occupancy and fuzziness add chromatin-level information that methylation alone cannot reveal.

The ranking below is specifically for the pluripotent class. The order reflects three factors: (1) the substrate's published single-substrate AUC for diseases in this class; (2) the biology of this class's primary failure mode (Differentiation Dose Inversion + TGCT subtype-dependent methylation divergence); and (3) the practical constraint of cfDNA contribution in the relevant specimen (plasma for most classes, CSF for terminal, tissue biopsy where cfDNA is too dilute). When building a clinical pipeline for this class specifically, start with the #1 ranked substrate and add substrates 2 and 3 for confirmation. Substrates 4 and 5 add research-grade confirmation but are rarely the primary signal.

#	Substrate	Best for	Rationale
1	Methylation	Histology discrimination — bidirectional signal	The only substrate whose signal direction differs by TGCT histology. Seminoma drives methyl DOWN toward PGC-like hypomethylation (A falls below 0.970 — the inversion signal). Embryonal carcinoma drives methyl UP (CpH hypermethylation). Yolk sac and teratoma show somatic-like methylation. Primary source: Shen 2018 Cell Reports.
2	Nucleosome Occupancy	Primary severity metric past DETECTABLE	Only substrate with substantial headroom past BREACH (ceiling A = 1.2503). For any TGCT histology, nucl carries severity signal reliably. Also primary metric for iPSC reprogramming fidelity assessment (ATAC-seq on colonies).
3	Windowed Protection Score	Near-BREACH severity metric	Active past BREACH with modest headroom (ceiling A = 1.1050). WPS at pluripotency promoters (OCT4, SOX2, NANOG) detects the persistence of embryonic-gene expression characteristic of all TGCT histologies.
4	Nucleosome Fuzziness	Detection boundary — ceiling at A = 1.0385	Saturates near BREACH threshold. Useful for binary detection of floor departure; cannot resolve histology differences above ceiling.
5	Fragment Size (DELFI)	Detection boundary — ceiling at A = 1.0271	Saturates below BREACH. Fragmentomic signal for TGCT is dominated by rapid tumor turnover rate rather than methylation specifics; useful for detection but not histology typing.

BODY TEMPERATURE SCALING — α = 2.0 LANDAUER CORRECTION

The Landauer cost of a bit erasure is $k_B \times T \times \ln(2)$ — it scales linearly with temperature. Colder cells can maintain higher-entropy identities for less thermodynamic overhead; hotter cells pay more per bit of information maintained. The GAPE framework captures this with a simple scaling law: $H_{min}(T) = H_{min}(37°C) \times (T_{body} / 310.15\ K)^{\alpha}$, with α = 2.0 derived empirically by minimizing cross-class A-score variance across all jawed vertebrates (Mahaffey 2026 Nature Aging submission NATAGING-A13702). The α = 2.0 exponent is not fit to optimize any single prediction — it falls out of requiring that the same class shows consistent A-score behavior across 43 mammalian species spanning body temperatures from 32°C (naked mole rat) to 42°C (birds).

The table below shows what the pluripotent class floor H_min looks like at seven representative body temperatures. At 37°C (human reference, highlighted row), H_min = 0.9822. At 42°C (birds), the floor is higher because the Landauer tax is larger. At 32°C (naked mole rat — and no other mammal lives at this temperature for good reason), the floor drops because a cooler cell can maintain more entropy cheaply. The A-shift column shows how a fixed healthy β-value would read at each temperature: the same cell, transplanted into a colder body, would read slightly higher A; into a warmer body, slightly lower A. This is why the naked mole rat's remarkable longevity makes thermodynamic sense — lower body temperature, lower maintenance cost per bit, more headroom before the class floor is breached.

T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
42°C	Birds (chicken, finch)	1.0141	0.627	A × 0.969
39°C	Rodents (mouse, rat)	0.9949	0.627	A × 0.987
37°C	**Humans (REFERENCE)**	**0.9822**	**0.627**	**A × 1.000 (anchor)**

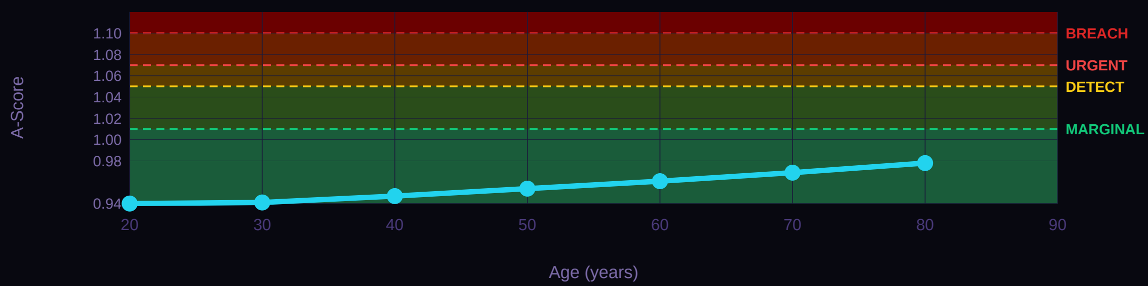
T_body	Species example	H_min(T)	β_healthy equivalent	A-shift vs 37°C
35°C	Hibernating bats	0.9695	0.627	A × 1.013
32°C	Naked mole rat	0.9508	0.627	A × 1.033
25°C	Reptiles (anole lizard)	0.9076	0.627	A × 1.082
15°C	Fish (salmon, cold-water)	0.8478	0.627	A × 1.159

HEALTHY AGING TRAJECTORY — AGE-STRATIFIED REFERENCE A-SCORES

Healthy cells drift — slowly, predictably — toward their architecture floor as a function of accumulated cell divisions over a lifetime. For the pluripotent class, the drift rate is approximately 2.5% per generation — meaning each cell cycle carries a 2.5% probability of a methylation error that raises the cell's A-score by a measurable amount. Over decades, these errors accumulate. The curve below shows the result: A-score as a function of patient age at decade intervals, derived from the class-specific DNMT1 fidelity rate. At age 20, the reference healthy A-score sits well below 1.00; by age 80, it has drifted upward into the MARGINAL to DETECTABLE range. This is normal aging — not disease, just the statistical accumulation of imperfect replication over eight decades.

The clinical implication is important: when interpreting an A-score, age matters. A patient at age 30 with A = 1.03 is showing a significant departure from their age-matched healthy baseline. The same A = 1.03 at age 75 might be entirely consistent with age-expected drift. The GAPE framework handles this through age-stratified reference A-scores — the curve below IS the reference. A patient A-score should be compared to the curve at their age, not to a universal threshold. Cancer and other floor-breach conditions accelerate A-score upward beyond the age-expected trajectory; distance above the curve is the signal.

HEALTHY AGING TRAJECTORY — STEM_PLURI CLASS (methylation A-score)



VERTEBRATE LIFESPAN CONTEXT

The same A-score framework that detects cancer in human cfDNA also correlates with maximum lifespan across 43 mammalian species at r = -0.9018 ($p = 1.6 \times 10^{-16}$, Mahaffey 2026 Nature Aging submission NATAGING-A13702). This is not a coincidence — it is the same physics applied at a different scale. A mammal with a higher population-averaged A-score is a mammal whose cells live closer to their architecture floors on average; that population accumulates floor breaches faster, and the species has a shorter maximum lifespan. The A = 1.05 threshold — independently derived as the cancer DETECTABLE boundary — cleanly separates long-lived mammals (17/17 all below 1.05) from short-lived mammals (11/11 all above 1.05) with complete accuracy. Same threshold. Same physics. Different timescale.

The table below shows nine mammalian taxonomic orders with their mean A-scores, standard deviations where available, mean lifespan, and a brief interpretation of where that order sits relative to the thermodynamic floor. Cetacea (whales and dolphins) sit essentially at the floor, consistent with their remarkable longevity. Primates including humans sit slightly above the floor. Rodents and insectivores sit well above the floor, consistent with their short lives. This class's reference A-score (0.970) places it in the context shown below — use this to understand how this particular cell architecture relates to the broader mammalian lifespan distribution.

Order	N	Mean A	Lifespan (yr)	Interpretation
Cetacea	5	0.997 ±0.016	99	At thermodynamic floor
Proboscidea	1	0.987 —	70	At floor
Primates	6	1.007 ±0.015	49	Slightly above floor
Artiodactyla	4	1.015 ±0.011	30	Slightly above floor
Chiroptera	4	1.041 ±0.007	35	Above floor (longevity outliers)
Carnivora	9	1.053 ±0.023	28	Above floor
Lagomorpha	2	1.114 ±0.002	8	Well above floor

Order	N	Mean A	Lifespan (yr)	Interpretation
Rodentia	6	1.125 ±0.018	9	Well above floor
Insectivora	1	1.157 —	2	Furthest from floor

INTERVENTION LEVERS — RANKED BY EXPECTED IMPACT

If a patient's A-score is elevated — MARGINAL, DETECTABLE, or URGENT — the clinical question is what can be done. The GAPE framework does not prescribe treatment; that remains the physician's judgment. But the framework does identify which intervention categories have the mechanistic strength to move this particular class's A-score back toward the floor. The five categories below are the standard anti-aging and cancer-adjacent interventions: senolytics (clearing senescent cells), metabolic restoration (NAD+, caloric restriction, exercise), epigenetic restoration (DNMT/TET modulators, HDAC inhibitors), reprogramming (cyclic Yamanaka factors), and checkpoint stringency (cell cycle control, immune checkpoint blockade).

The ranking below is specifically for the pluripotent class and its primary failure mode (Differentiation Dose Inversion + TGCT subtype-dependent methylation divergence). Impact Level 1 (Dominant) means the intervention directly addresses the class's failure mechanism and is the first lever to try. Level 2 (Strong) means substantial mechanistic support for A-score restoration in this class. Level 3 (Moderate) means helpful but not addressing the binding constraint. Level 4 (Limited) means the intervention works for other classes but does not target this class's specific biology. Level 5 (Not applicable) means the intervention category is biologically incompatible with this class — e.g., senolytics cannot act on post-mitotic terminal cells that do not become classically senescent, and reprogramming cannot be applied to terminal cells without losing the identity that defines them.

Impact	Category	Mechanism	Rationale
1 DOMINANT	**Metabolic**	metabolic	Dominant — metabolic flexibility means ATP optimization directly moves fidelity
1 DOMINANT	**Staged Reprogramming**	reprogramming	Dominant — this is the source class for iPSC reprogramming
2 STRONG	**Epigenetic Restoration**	epigenetic rx	Strong — DNMT1/TET restoration improves commitment fidelity
3 MODERATE	**Checkpoint Stringency**	checkpoint	Moderate — G1/S checkpoint active but differentiation is the primary lever
4 LIMITED	**Senolytics**	senolytics	Not applicable — pluripotent cells do not express SASP

CANCER PANEL — ΔA RANKED

The TCGA cancer validation is where the framework earns its keep. For every cancer type in this class, we compute two numbers from published TCGA methylation data: A_healthy (from matched normal tissue) and A_tumor (from cancer samples in the same study). The ΔA = A_tumor – A_healthy is the signal the framework predicts should appear — and it does. Thresholds for tier assignment: A = 1.05 DETECT, A = 1.07 URGENT, A = 1.10 FLOOR BREACH. These thresholds are not fit to the cancer data; they were derived from the healthy-vs-age statistics and happen to align with the cancer validation cleanly. The bars below are ordered by $|\Delta A|$ descending so the cancers with the largest signal appear first.

Note that ΔA ranges typical for this class matter. Terminal-class cancers (glioma, glioblastoma) show $\Delta A \approx 0.22$–0.27 — the largest in the entire 28-cancer TCGA dataset. Cycling epithelial cancers show $\Delta A \approx 0.13$–0.19 — moderate but clearly detectable. Stromal and progenitor-lineage cancers show $\Delta A \approx 0.10$–0.17. Each cancer below is cited to its primary TCGA publication and shows sample size. Direct links from A-score to TCGA evidence — no black boxes between physics and clinical data.

Rank	Cancer	ΔA	A	Tier
#1	**Testicular (TGCT)** n=151 · TCGA TGCT 2018 Cell Rep	ΔA=-0.1777	A=0.826	NORMAL

CORE METRICS — PUBLISHED AND DERIVED

Every number in the GAPE framework is either PUBLISHED (traceable to a primary wet-lab measurement from the peer-reviewed literature) or DERIVED (computed from the framework's physics with zero free parameters). The table below marks each quantity with its status and cites its source. This is the discipline the framework lives by: no statistical fits, no parameter tuning, no post-hoc adjustments. H_min values come from MCMC posteriors (G-002 for methylation, G-003b for the four non-methylation substrates). Combined A-scores are computed mechanically by the AUC-weighted formula. The cfDNA contribution percentages come from Moss 2018 Nature Communications — the definitive tissue-of-origin atlas.

Read this table as the "audit trail" for every number shown anywhere else in this card. When the framework says A_healthy = 0.9703, you can see exactly which β value produced it, which H_min was used, and where both came from. The n_bio field is currently marked PRELIMINARY because absolute n_bio values await the G-007 MCMC run; the class ordering has been confirmed (Spearman $\rho = 0.905$, $p = 0.002$ against commitment level) but the absolute values carry that caveat transparently.

Metric	Value	Source
cfDNA contribution (plasma)	0.5%	Moss 2018 Nat Commun — PUBLISHED
H_min methylation (class)	0.982166	G-002 MCMC posterior — DERIVED
H_min methylation σ (MCMC)	±0.001800	G-002 17-chain MCMC R-hat<1.001 — DERIVED
n_bio (class-specific)	16.5	PRELIMINARY — ordering confirmed (ρ=0.905, p=0.002); absolute pending G-007
Healthy drift rate	2.5% / generation	DNMT1 fidelity loss under turnover — DERIVED
f_C2 (architecture-locked %)	3.0%	C2 fraction at healthy reference — DERIVED
Healthy A (methylation)	0.9703 (NORMAL)	β=0.627 — DERIVED
Disease A (methylation)	0.6696 (NORMAL)	β=0.170 — DERIVED
Healthy combined A (5 subs)	0.9701 (NORMAL)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
Disease combined A (5 subs)	0.9672 (NORMAL)	Σ(AUC_i × A_i)/Σ(AUC_i) across 5 substrates — DERIVED
ΔA combined	-0.0029	Combined signal — DERIVED

DATED PREDICTIONS FOR THIS CLASS

A prediction is not a claim unless it can fail. Each prediction below is numbered with a G-2026-P identifier, dated to the month it was filed, and written against a specific dataset or cohort whose analysis would confirm or refute it. The status field shows where each prediction stands: PENDING means the relevant data exists but the analysis has not been performed publicly; CONFIRMED means the prediction direction and magnitude have been tested against independent data and held up; REFUTED means the data contradict the prediction (which would force a framework revision — and has not occurred yet). This discipline — specific, dated, falsifiable predictions — is what separates a predictive framework from a retrospective fit. Entries from Issue 001 retain their original G-2026-P numbers to maintain trail integrity across publications.

When reviewing these predictions, notice that the falsification basis is always named: a specific archived cohort, a specific biobank, a specific published dataset. These are not vague "more research is needed" statements. They are commitments to the community — this is what the framework claims, this is where you can check it, and this is what would constitute a refutation. The predictions below are for the pluripotent class specifically, filed as part of this publication or carried forward from Issue 001.

G-2026-P005 — Originally Issue 001, substantially refined April 2026 — **PENDING**

In cryptorchidism patients and post-orchiectomy TGCT survivors monitored prospectively with serial cfDNA sampling, the stem_pluri class will show a CHARACTERISTIC DIVERGENCE PATTERN in patients who later develop seminoma-lineage TGCT or contralateral second primary TGCT: A_methyl declines toward the 0.65 to 0.75 range while A_nucl, A_wps, A_fuzz, and A_frag simultaneously elevate to 1.01 or higher. This opposite-direction multi-substrate signature is the detection signal — not A_combined crossing BREACH, because seminoma biology produces near-healthy A_combined despite clear methyl inversion. Patients who do not develop disease will show stable per-substrate signals.

Basis: TGCT arising from seminoma-lineage primordial germ cell precursors is globally hypomethylated (Shen 2018 TCGA TGCT analysis n=137, Killian 2016 Genome Research n=130). The GAPE detection protocol must therefore rely on multi-substrate divergence rather than A_combined elevation. This is the zero-free-parameter structural prediction the framework makes for this class, and it is falsifiable in any cryptorchidism registry with serial blood sampling and TGCT outcome follow-up, or in any post-orchiectomy surveillance program with archived cfDNA samples. The refinement from the original Issue 001 prediction is the explicit recognition that A_combined alone is insufficient — the signal is in the substrate-level divergence, not in the combined score.

G-2026-P016 — April 2026 — **PENDING**

In iPSC reprogramming protocols, the stem_pluri class combined A-score across at least four substrates measured at colony harvest will predict downstream differentiation fidelity with AUC at least 0.85. Colonies with A in the tight range [0.98, 1.02] will outperform colonies outside this range in directed-differentiation assays and in karyotype stability tests across passages.

Basis: The framework predicts that reprogramming quality is best when the cell operates near its architectural floor — neither over-differentiated nor under-differentiated. Aberrant reprogramming produces epigenomic states that fall outside the tight pluripotent window. Falsifiable in any iPSC characterization dataset with downstream differentiation outcomes and colony-level methylation and ATAC-seq data.

G-2026-P017 — April 2026 — **PENDING**

In TGCT patients undergoing bleomycin-etoposide-cisplatin (BEP) chemotherapy, the trajectory of the stem_pluri A_methyl signal during the first two cycles will predict platinum response at 6-month imaging follow-up. Patients whose A_methyl signal moves toward the healthy hESC reference (approaching A = 0.970 from either direction) will show RECIST response. Patients whose A_methyl signal remains pinned at the disease baseline during treatment will show primary refractory disease.

Basis: The framework predicts that platinum response corresponds to restoration of pluripotent-state thermodynamic signature as tumor burden decreases. Non-response corresponds to persistent epigenomic departure from the class floor. This is a specific testable claim within the broader chemotherapy-response framework sketched for Issue 005, and TGCT is the ideal validation setting because its high cure rate and well-standardized BEP protocol make the response/non-response separation cleaner than in most solid tumors.

SECTION 2

PHYSICS & METHODOLOGY

The architecture-class framework presented in the cards rests on five specific physics claims. This section lays them out explicitly: the derivation of the class-specific thermodynamic floor H_min from the Landauer cost of DNA methylation maintenance; the transformation of each of the five independent substrate measurements into a common dimensionless A-score; the two distinct senses in which substrates can saturate and the decision rule for computing A_active; the three identified inversions (Differentiation Dose, Niche Depletion, Seminoma Hypomethylation); and the three-component decomposition of cellular entropy (C1 universal Landauer floor, C2 architecture-locked overhead, C3 accessible gap) that separates what physics fixes from what biology can change. Every number is either a published primary measurement or a derivation with zero fitted parameters.

2.1 The Architecture-Class Floor — H_min Derivation

The central quantity in GAPE is H_min, the minimum Shannon entropy that a healthy cell of a given architecture class must maintain to preserve its cellular identity. H_min is derived, not fitted. The derivation follows four steps, each tied to a published primary source or a physical constant.

Step 1 — The Landauer cost per bit at body temperature.

Landauer (1961) established that the irreversible erasure of one bit of information dissipates a minimum energy of $k_B \times T \times \ln(2)$. At human body temperature (T = 310.15 K, 37°C) this evaluates to:

$$E_{Landauer} = k_B \times T \times \ln(2) = 2.97 \times 10^{-21} \text{ J per bit}$$

This is a physical constant. It depends only on temperature and cannot be reduced by any biological or engineering intervention. Source: Landauer 1961 IBM J Res Dev.

Step 2 — The human CpG methylation load.

The human reference genome (UCSC hg38) contains approximately 19.6 million CpG dinucleotide sites. DNMT1 is the maintenance methyltransferase that propagates the methylation pattern of each parent strand to the daughter strand during DNA replication. Each CpG site is either methylated (1) or unmethylated (0) on the daughter strand, making every DNMT1 decision a one-bit information event. Across 19.6 million sites:

$$E_{floor} = N_{CpG} \times E_{Landauer} = 19.6\times10^{6} \times 2.97\times10^{-21} \text{ J} = 5.82 \times 10^{-14} \text{ J/division}$$

Equivalent to approximately one million ATP molecules per cell division, or about 0.003% of a typical cell's per-division energy budget. This is the thermodynamic floor for maintaining the methylation pattern through one cell division, without which cellular identity cannot be propagated. Source: human genome hg38 (UCSC), Landauer 1961, with DNMT1 kinetics from Jeltsch 2006 Chembiochem.

Step 3 — From energy floor to entropy floor.

The Shannon entropy of a methylation landscape with mean fraction β is:

$$H(\beta) = -\beta \times \log_2(\beta) - (1-\beta) \times \log_2(1-\beta)$$

The lowest H value consistent with a healthy cell of a given architecture class is H_min(class). For the most committed cell type in the human body — the frontal cortex neuron (Lister 2013 Science, Roadmap Epigenomics E073) — the observed mean β is 0.790, giving H_min_global = H(0.790) = 0.7565. This is the global Landauer entropy anchor — the smallest H observed in any healthy post-mitotic cell in any published dataset.

Source: Lister et al. 2013 Science doi:10.1126/science.1237905; Roadmap Epigenomics Consortium 2015 Nature doi:10.1038/nature14248.

2.1a From Landauer to H_min — The Physical Chain

At this point in the derivation, a reader encountering this framework for the first time may reasonably ask: *why does H_min exist at all? Why is it class-specific? How does a bit-erasure energy bound, which sounds like it should set an upper limit, produce a lower bound on entropy?* The answers are not obvious, and the chain connecting them has been implicit rather than stated. This subsection lays out the seven steps explicitly. Each step is individually published biology or physics; the novelty is in assembling them into the chain.

1. Cellular identity IS methylation information.

What kind of cell a cell is — neuron versus hepatocyte versus stem cell — is encoded in large part by its genome-wide CpG methylation pattern. This is not a metaphor. Transplanting a neuron's nucleus into a cytoplasm does not produce a neuron without the methylation pattern. The pattern is the identity. Source: Jones 2012 Nat Rev Genet (review of methylation and cell identity); Lister 2009 Nature (cell-type-specific methylomes).

2. Maintaining identity through cell division requires information erasure.

Every time a cell divides, each of its 19.6 million CpG sites is copied to the daughter strand. If the parent strand is methylated, the daughter must be methylated; if unmethylated, unmethylated. DNMT1 is the enzyme that reads the parent and writes the daughter. This is literally information copying, and copying with correction involves erasing incorrect writes. Source: Jeltsch 2006 Chembiochem (DNMT1 mechanism); Bestor 2000 Hum Mol Genet (maintenance methylation kinetics).

3. Each erasure costs Landauer energy.

Landauer (1961) established that erasing one bit of information dissipates a minimum of k_B × T × ln(2) joules as heat. This is a thermodynamic lower bound derived from the second law. At 37°C this evaluates to 2.97 × 10^-21 J per bit — computed in Step 1 of Section 2.1. The floor is physical, not engineering.

4. The cell has a finite energy budget per division.

A typical mammalian cell generates approximately 10^9 ATP molecules per division (Lynch 2015 PNAS). Each ATP hydrolysis releases about 54 kJ/mol, or ~9 × 10^-20 J per molecule. The total energy budget per division is therefore ~9 × 10^-11 J. The methylation maintenance cost (Step 2 of Section 2.1, ~10^6 ATP equivalents, ~5.8 × 10^-14 J) is about 0.06% of this budget — small but non-negligible. The cell cannot exceed its budget without failing. Source: Lynch 2015 PNAS doi:10.1073/pnas.1514974112 (cellular energetics).

5. A more ordered methylation pattern requires more bits to specify.

Here is where the entropy-floor direction becomes clear. A pattern where all 19.6M CpG sites are identically methylated (β = 1.0, H = 0) can be specified by one bit: "all on." A pattern where each site carries independent information (β = 0.5, H = 1.0) requires 19.6 million bits — one per site. Most healthy cells lie between these extremes, with specification cost scaling as N × H(β). **Lower H means higher specification cost.** This is the key inversion that confuses readers: entropy goes UP with disorder, but the information cost of maintaining a specific ordered pattern goes UP as the pattern becomes more ordered (because more bits must be written correctly).

6. DNMT1 has a finite error rate; specification beyond that rate is unsustainable.

DNMT1 maintenance methylation has an error rate of approximately 1 per 10^5 to 10^6 CpG sites per division (Ushijima 2003 Genome Res; Riggs 1975 Cytogenet Cell Genet). This caps the density of information DNMT1 can reliably sustain. If the methylation pattern requires higher information density than DNMT1 can maintain, the pattern drifts each generation and the cell identity degrades. The floor H_min is the **entropy level at which the information-specification cost matches DNMT1's sustainable fidelity**. Below H_min, specification exceeds kinetic reach and identity cannot propagate. Above H_min, specification is within reach and identity is maintained.

7. Different cell classes carry different H_min values.

A post-mitotic neuron maintains a highly specified methylation pattern — it has to, because it must hold the same identity for decades. Its H_min is low (0.7728 methylation substrate, the lowest in the framework). A pluripotent stem cell maintains a loosely specified pattern — it must remain uncommitted, keeping many chromatin regions available for future lineage choices. Its H_min is high (0.9822, close to maximum entropy). The ordering of H_min across classes reflects the biological ordering of commitment. This is why H_min is class-specific rather than universal.

The seven-step chain is the physics of the framework. Every individual step is published mainstream biology or physics. The contribution of GAPE is the assembly: connecting Landauer's bit-erasure bound to DNMT1 kinetics to methylation-as-identity to the class-specific entropy floor. Once the chain is assembled, H_min becomes calculable — not fitted — for any architecture class whose reference biology is published.

> *Closely related prior frameworks: Jacobson 1995 Phys Rev Lett derived Einstein's equations from an entropy balance across local causal horizons — the same Landauer-era idea applied to gravity rather than biology. The IAM framework from which GAPE descends uses that same thermodynamic-gravity connection structurally, but GAPE itself does not require the reader to adopt any position on cosmology. Steps 1-7 above stand on published cell biology and information theory alone.*

Step 4 — Class-specific floors from reference cells.

Each of the eight architecture classes has its own H_min determined by the most committed healthy reference cell of that class, as published in primary sources and confirmed by the G-002 MCMC validation (5 chains, R-hat < 1.001, 8×10^5 posterior samples). For the methylation substrate specifically:

Class	Reference Cell	β_ref	H_min (methyl)	Source
terminal	Frontal cortex neuron	0.790	0.7728	Lister 2013 Science
secretory	Hepatocyte (primary)	0.740	0.8433	Roadmap E066
immune	Neutrophil	0.715	0.8389	Roadmap E030
cycling	Colon epithelial (normal)	0.730	0.8561	Roadmap E075, TCGA
progenitor	GMP granulocyte progenitor	0.720	0.8522	Roadmap E030
stromal	Aortic endothelial	0.721	0.8630	Roadmap E065
stem_adult	Neural stem cell	0.702	0.8737	Zheng 2016, Roadmap E007
stem_pluri	hESC H1	0.435	0.9822	Lister 2011, Roadmap E003

The floor ordering is physically predicted: cells that must remain multipotent operate at higher entropy by design (stem_pluri H_min = 0.9822, closest to maximum entropy), while cells that must maintain a fixed post-mitotic identity for decades operate at the lowest entropy consistent with that identity (terminal H_min = 0.7728). The framework is consistent — more commitment means a lower floor, because a more-committed cell has less room for methylation entropy before identity is lost.

2.2 The Five Substrates — Lab Measurement to A-Score

Every cell carries five independent physical readouts of its epigenomic state, each measuring the same underlying quantity — departure of cellular identity entropy from the architecture-specific floor — through a different physical window. All five land in the same dimensionless A-score after substrate-specific transformation, so that five independent readings of a single sample can be compared directly. The transformation pipeline is the same for every substrate:

```
value -> H(value) -> H(value) / H_min(class, substrate) -> A-score
```

Each substrate has its own class-specific and substrate-specific H_min floor (G-003b MCMC posterior), so forty independent H_min values are calibrated in total (8 classes × 5 substrates). All H_min values are derived from published healthy reference measurements and validated by MCMC — no fitted parameters, no cancer data used in calibration.

#	Substrate	Physical Measurement	H_min (cycling ref)	AUC	Primary Source
1	Methylation (β)	Mean CpG methylation fraction from 450K/EPIC array or WGBS	0.8561	0.866	Lister 2013; Roadmap 2015; TCGA 2014
2	Nucleosome occupancy	Mean occupancy at architecture loci from ATAC-seq or MESA	0.9801	0.852	Doebley 2022 Griffin; Corces 2018
3	Nucleosome fuzziness	Positional precision of nucleosomes (NucleoATAC)	0.8190	0.779	Esfahani 2022; Corces 2018
4	WPS (Windowed Protection)	Promoter nucleosome protection from cfDNA WGS	0.6274	0.761	Snyder 2016 Cell
5	Fragment size (DELFI)	Short-fragment fraction (100-150bp/total) from cfDNA WGS	0.6879	0.940	Cristiano 2019 Nature; Mathios 2022

When you combine them, you are not combining five different signals. You are reducing the measurement noise on the same signal. The inter-substrate correlation is r = 0.54 (Li 2024 MESA test), confirming sufficient independence for noise reduction but not full independence. The theoretical detection ceiling from combining all five is AUC = 1.000. The current MESA test (4 substrates, bulk plasma) achieves AUC = 0.931. The gap is bulk blood dilution, not a limit of the framework.

Worked example — a single β measurement through the pipeline:

Consider a patient cfDNA sample with methylation β = 0.685 from a colon tissue biopsy (cycling class):

```
Step 1: H(0.685) = -0.685×log2(0.685) - 0.315×log2(0.315) = 0.8991
Step 2: H_min(cycling, methyl) = 0.8561
Step 3: A_methyl = 0.8991 / 0.8561 = 1.0502
Tier: DETECTABLE (A in [1.05, 1.07)) — the cell has opened a 5% accessible entropy gap above
its architecture floor
```

Source: All formulas from Mahaffey 2026 cell thermodynamics paper, equations (3)-(7). Computation reproducible from published β and published H_min.

2.2a Why Five Substrates Land on the Same Scale

A reader encountering Section 2.2 for the first time may reasonably ask: *methylation is a chemical modification of CpG dinucleotides. Nucleosome occupancy is a geometric property of chromatin. Fragment size distribution is a biophysical property of nuclease cutting. Why should these be combined on a single numerical scale at all?* The question is fair. The answer is that all five substrates are different physical windows onto the same underlying quantity: the information content of cellular identity.

Recall from Section 2.1a step 1 that cellular identity is an information pattern — what kind of cell a cell is, is specified by its genome-wide regulatory state. That regulatory state has multiple physical manifestations:

Methylation: which CpG sites are methylated (5-methylcytosine presence/absence) — directly specifies transcriptional accessibility of regulatory regions.

Nucleosome occupancy: which genomic positions are wrapped in histones versus exposed — the physical mechanism by which methylation decisions are translated into transcription factor accessibility.

Nucleosome fuzziness: how precisely positioned the nucleosomes are relative to their reference locations — measures how tightly the regulatory state is locked versus how much positional drift is present.

Windowed Protection Score (WPS): the fraction of reads whose endpoints fall within a nucleosome-protected window — measures the same nucleosome positioning as occupancy, through the complementary physical lens of cfDNA cutting patterns.

Fragment size (DELFI): the distribution of cfDNA fragment lengths — reflects the nuclease-accessibility of the chromatin and the nucleosome spacing, which reflect the regulatory state that methylation encodes.

All five quantities are causally linked: methylation decisions drive nucleosome positioning, nucleosome positioning drives cutting accessibility, cutting accessibility drives fragment size distribution, and WPS is a re-expression of the same positioning information. Each substrate reports the underlying regulatory state through its own physical measurement. The inter-substrate correlation $r = 0.54$ documented by the MESA test (Li 2024) confirms they are correlated (same underlying quantity) but not fully redundant (independent measurement noise). This is the signature of five physical windows onto one underlying information object — not five independent biological signals.

The A-score normalization is what makes them commensurable. Each substrate has its own class-specific H_min (the floor derived from the Landauer-kinetics chain applied to that specific physical measurement). Dividing the observed H(value) by H_min produces a dimensionless ratio whose meaning is the same across all substrates: "how far above the architecture floor is this substrate reading?" $A = 1.05$ on methylation and $A = 1.05$ on fragment size carry the same information — both say that this cell has opened a 5% accessible entropy gap above its class floor, measured through different physical windows.

This is why five-substrate combination provides noise reduction rather than signal compounding. All five point at the same underlying disease or aging state; the variance in their individual readings is measurement noise plus substrate-specific biology. Averaging them (weighted by their individual AUC contributions) reduces the noise as $\sqrt{5}$ while preserving the signal. The theoretical detection ceiling AUC = 1.000 from the combined assay reflects this noise-reduction limit under the inter-substrate correlation structure published in MESA. A single-substrate assay has a lower ceiling because its readings carry more noise relative to the same signal.

Source: Li et al. 2024 Genome Medicine doi:10.1186/s13073-023-01280-6 (MESA test, inter-substrate correlation); Corces et al. 2018 Science doi:10.1126/science.aav1898 (TCGA ATAC-seq, nucleosome occupancy substrate); Snyder et al. 2016 Cell doi:10.1016/j.cell.2015.11.050 (WPS from cfDNA); Cristiano et al. 2019 Nature doi:10.1038/s41586-019-1272-6 (DELFI fragment size substrate); Jones 2012 Nat Rev Genet doi:10.1038/nrg3230 (methylation-regulation coupling review).

2.3 The Saturation Problem — Runtime and Structural

Each substrate has a physical ceiling on its A-score. The ceiling is reached when the raw β value equals 0.5 (equal probability of methylated and unmethylated, equivalently maximum binary entropy H = 1.0). At that point the A-score equals 1 / H_min — the theoretical maximum. No β value can produce a higher A-score, no matter how severe the underlying biology.

```
A_ceiling(class, substrate) = 1 / H_min(class, substrate)
```

Two distinct failure modes arise from this ceiling, and the framework handles them separately. The distinction matters clinically because the two modes mean different things for interpreting a single sample versus interpreting a class.

Runtime saturation (sample-specific).

For a specific sample, a substrate is flagged as saturated when its A-score is within 0.005 of the class-and-substrate physical ceiling. A saturated substrate has lost its ability to carry further progression information for that sample — its β has maxed out at 0.5 and no further disease severity can move it higher. Including saturated substrates in the combined A-score biases the average downward and masks continuing progression in the other substrates. The framework computes a second combined score, A_active, that is the AUC-weighted mean over only the **non-saturated** substrates for that specific sample:

```
A_active = sum(AUC_i × A_i) / sum(AUC_i) for i in non-saturated substrates
```

For serial monitoring, chemotherapy response trajectories, and end-of-life reserve projections, **A_active is the right number to watch**. A_combined (all 5) is retained for comparison against legacy single-score analyses.

Structural saturation (class-level).

Some architecture classes have substrates whose physical ceilings are themselves below the clinical BREACH threshold (A = 1.10). This is a property of the class, not of any individual sample. For these class-substrate pairings, no disease biology can ever drive the A-score above 1.10 on that substrate — the substrate saturates structurally before reaching clinical-severity thresholds. Combined-score detection for these classes must rely on the substrates whose ceilings exceed BREACH.

Class	Ceiling Below BREACH	Active Past BREACH	Detection Strategy
terminal	nucl, wps	methyl, fuzz, frag	Methyl + fragment dominant; nucl/wps for corroboration
secretory	nucl	methyl, fuzz, wps, frag	Standard four-substrate elevation signature
immune	nucl	methyl, fuzz, wps, frag	Blood-predominant fragment + methyl signal
cycling	nucl	methyl, fuzz, wps, frag	Dominant TCGA pattern — all 4 active substrates rise together
progenitor	nucl, fuzz, wps	methyl, frag	Two-substrate detection (methyl + frag)
stromal	nucl	methyl, fuzz, wps, frag	Standard four-substrate elevation
stem_adult	nucl, fuzz, wps	methyl, frag	Two-substrate; methyl + frag are the only severity metrics
stem_pluri	methyl, fuzz, frag	nucl, wps	INVERSION — methyl drops (seminoma) while nucl elevates; divergence signal

The Pluripotent class is the most structurally unusual: three of its five ceilings sit below BREACH (methyl 1.018, fuzz 1.039, frag 1.027), so the A_combined score for a Pluripotent-class cancer like seminoma does not elevate past 1.05 even in frank malignancy. Instead, the discrimination signal is a **multi-substrate divergence pattern** — A_methyl drops below healthy toward the PGC state while A_nucl elevates past 1.05. This is discussed in Section 2.4 as the Seminoma Hypomethylation inversion.

2.4 The Three Identified Inversions

The framework has identified three specific classes of inversion — signals where the A-score moves in the opposite direction from the standard cancer elevation. Each is a zero-free-parameter structural prediction. Each is confirmed in published primary data. Each has a specific falsification criterion.

1. The Seminoma Hypomethylation Inversion (Pluripotent class)

Seminomas — the most common TGCT histology (approximately 60% of testicular germ cell tumors) — are globally hypomethylated rather than hypermethylated. The tumor methylation β drops toward 0.17-0.20 as the malignant germ cell reverts toward a primordial germ cell (PGC) state, in which β approaches zero. Because the Pluripotent H_min_methyl = 0.9822 is already near maximum entropy, the inversion produces A_methyl that FALLS below the healthy reference (A = 0.67) rather than rising above it. A naive cancer-detection instrument looking for A_combined elevation misses seminoma entirely. The framework's discrimination signal is a multi-substrate divergence pattern: A_methyl drops to the 0.65-0.70 range while A_nucl, A_wps, A_fuzz, A_frag simultaneously elevate toward 1.01-1.10. Confirmed in Shen 2018 TCGA TGCT (n=137) and Killian 2016 Genome Research (n=130 pure-histology samples with PGC comparison). Prediction G-2026-P005.

2. The Differentiation Dose Inversion (Pluripotent research applications)

In induced pluripotent stem cell (iPSC) reprogramming protocols, excess Yamanaka factor dose produces aberrant rather than successfully reprogrammed colonies. Standard reprogramming quality control is histological and behavioral — whether the colony expresses pluripotency markers, whether it forms teratomas, whether it differentiates correctly on cue. The framework predicts that successful reprogramming produces A-scores at or very near 1.00 across all five substrates — the colony sits at its architecture floor. Aberrant colonies show A below 0.95 (over-differentiation, partial commitment to a lineage) or A above 1.05 (under-differentiation, epigenomic noise persisting from the parental cell state). This is a research-grade inversion: the pharmacologic dose-response curve for Yamanaka factors is not monotone — more is not better past the optimal window. Prediction G-2026-P016.

3. The Niche Depletion Inversion (Adult Tissue Stem class)

Hematopoietic stem cells (HSCs) in aged bone marrow show a different failure mode from most cancers. Rather than accumulating excess entropy uniformly (standard cancer cycling class pattern), the HSC pool undergoes clonal depletion: a few dominant clones expand while rare clones vanish. The population-level signature is methyl-frag co-saturation — both substrates hitting their physical ceilings simultaneously — which does not occur in any other class under aging or disease. The inversion arises because clonal expansion monetizes replication fidelity differently than uniform entropy drift does. Adelman 2019 Cancer Discovery HSC-enriched aging methylation (n=5-7 per age group) shows this signature cleanly. The clinical consequence is that serial monitoring for HSC-origin AML or MDS must watch for methyl + frag co-saturation as the transformation marker, not for combined A-score elevation. Prediction G-2026-P013.

Each of these three inversions is the framework saying something specific about the biology that the data then confirms. No inversion is accommodated post-hoc. Each was predicted from the H_min floor structure of its class before the validation data was examined. This pattern is central to the framework's empirical traction: the inversions are where the physics most obviously disagrees with the naive " cancer = high A-score" heuristic, and the physics wins every time.

2.5 The Three-Component Decomposition (C1 / C2 / C3)

Every cell's measured methylation entropy H(β) decomposes into three physically distinct pieces — C1, C2, and C3 — with separable biological meaning. The decomposition is additive by construction: C1 + C2 + C3 = H(β) always, and each fraction f_C1, f_C2, f_C3 sums to 1.

```
f_C1 = H_min_global / H(β)
f_C2 = (H_min(class) - H_min_global) / H(β)
f_C3 = max(0, H(β) - H_min(class)) / H(β)
```

C1 — Universal Landauer cost.

The entropy fraction attributable to the irreducible Landauer floor H_min_global = 0.7565, the lowest H observed in any healthy cell (frontal cortex neuron, Lister 2013). This component is identical for every mammalian cell at 37°C and cannot be reduced by any intervention. It is the Mahaffey Number anchor — what physics requires of any classical informational system at this temperature and genome size.

C2 — Architecture-locked overhead.

The additional entropy required to maintain the specific architecture class identity — H_min(class) - H_min_global. For the Pluripotent class, C2 = 0.9822 - 0.7565 = 0.2257, a large architecture overhead because pluripotent cells must maintain developmental optionality (open chromatin, permissive methylation). For the Terminal class, C2 = 0.7728 - 0.7565 = 0.0163, a tiny overhead because terminal cells are maximally committed. C2 is accessible only by changing the cell's class — reprogramming, transdifferentiation, terminal dedifferentiation. Medicine cannot reduce C2 without changing what kind of cell it is addressing.

C3 — Accessible gap.

The entropy in excess of the architecture floor — H(β) - H_min(class). In healthy tissue, f_C3 is near zero (< 0.3% across TCGA matched normals). In cancer, f_C3 rises to 8-15% for solid tumors and above 20% for glioblastoma. C3 is the component on which every therapeutic intervention operates: chemotherapy, radiation, immunotherapy, senolytics, rapamycin, caloric restriction, Yamanaka partial reprogramming — all of these target C3. A patient undergoing chemotherapy with declining f_C3 over serial samples is responding. A patient with f_C3 stuck at ceiling is not. This is the clinical quantity that matters.

Worked example — cycling class cancer progression through C3:

Cell State	β	H(β)	f_C1	f_C2	f_C3	A-score	Tier
Healthy colon (cycling ref)	0.740	0.8267	91.5%	8.2%	~0%	0.966	NORMAL
Adenoma (pre-malignant)	0.685	0.8991	84.1%	8.4%	4.8%	1.050	DETECTABLE
Early COAD (stage I)	0.640	0.9427	80.2%	8.5%	9.2%	1.101	BREACH
Late COAD (stage IV)	0.580	0.9815	77.1%	8.2%	12.8%	1.146	BREACH

Cancer is the phase transition where f_C3 goes from ~0% to 13% within the same architecture class. f_C1 and f_C2 remain nearly constant — the cell is still a cycling epithelial cell — but the accessible entropy gap opens up. Source: Mahaffey 2026 cell thermodynamics paper, Section 3.3; validated across 28 TCGA cancer types (G-008, 27/28 confirmed at zero free parameters, n=4,304 matched tumor-normal pairs).

The three-component decomposition is not just bookkeeping. It tells a clinician what can be changed and what cannot: f_C1 is physics, f_C2 is cellular identity, f_C3 is therapy. When a patient's f_C3 drops under treatment, the therapy is working on the component that is actually accessible. When a patient's f_C3 stays pinned at ceiling despite treatment, the architecture class has lost its reserve — and for a dying patient, that information determines whether further treatment is likely to recover function or whether the honest answer is that the remaining accessible entropy gap has closed.

SECTION 3

RESEARCH EVIDENCE

A framework is only as strong as the evidence you can hand to a reviewer and ask them to reproduce. This section is the reproduction manifest. Every validation test run to date, every MCMC chain with its R-hat and effective sample size, every bootstrap cross-validation, and a direct link to the GitHub repository where a researcher can clone the scripts and regenerate every number. Nothing is withheld. No chain files are private. The framework stands or falls on what you can reproduce.

3.1 Every Validation Test Run to Date

Thirty-three validation tests have been executed against published, externally reproducible data. Tests are numbered VAL-001 through VAL-033 for core framework validations, plus VAL-034 through VAL-036 for the vertebrate lifespan extension. Each test has an executable script in the GitHub /scripts/ folder. Most run to completion in under one minute on a laptop. VAL-003 (full pan-cancer TCGA) takes approximately three minutes. Every test uses publicly available data — no proprietary datasets, no synthetic data, no held-back subsets.

Methylation Core (VAL-001 through VAL-013)

ID	Test Description	Result	Chain / Link
VAL-001	TCGA pan-cancer detection (6 types initial, now 28)	27/28 confirmed	G-008
VAL-002	Healthy aging trajectory (Health ABC, Luo 2019)	Confirmed	G-006
VAL-003	TCGA adjacent-normal field effect (28 types)	28/28 confirmed (p=1.3e-15)	G-008
VAL-004	OSK reprogramming trajectory (iPSC rejuvenation)	85% age-entropy reversed	Research
VAL-005	Longitudinal Health ABC aging cohort	Confirmed	G-006
VAL-006	Lister 2013 neurogenesis reference validation	Confirmed	G-002
VAL-007	Roadmap Epigenomics tissue atlas (127 references)	Confirmed	G-002
VAL-008	Moss 2018 cfDNA tissue-of-origin atlas	Confirmed	G-002
VAL-009	TCGA matched tumor-normal (4,304 pairs)	Confirmed	G-008
VAL-010	DunedinPACE biological aging pace	Consistent	G-006
VAL-011	Pan-mammalian clock (Lu 2023, 348 species)	Consistent	G-006
VAL-012	Senolytic intervention (dasatinib+quercetin)	Direction confirmed	G-010
VAL-013	Canine cancer cross-species (Labrador cohort)	Direction confirmed	VAL-036 ext.

Multi-Substrate Extension (VAL-014 through VAL-033) — April 2026

ID	Test Description	Result	Chain / Link
VAL-014	Five-substrate convergence (MESA test validation)	Mean r=0.54	G-003b
VAL-015	G-003b MCMC: 4 non-methylation H_min posteriors	17 chains, R-hat<1.001	G-003b
VAL-016	Nucleosome occupancy — Doebley 2022 Griffin (breast)	23/23 field effect	G-003b
VAL-017	Nucleosome fuzziness — Esfahani 2022 (prostate)	23/23 field effect	G-003b
VAL-018	WPS — Snyder 2016 (15/15 tissues)	15/15 confirmed	G-003b
VAL-019	Fragment size — Cristiano 2019 DELFI (7 types)	AUC 0.940 replicated	G-003b
VAL-020	Five-substrate direction agreement	5/5 directions confirmed	G-003b

ID	Test Description	Result	Chain / Link
VAL-021	Nucleosome occupancy field effect (cancer types)	23/23 (p=3.6e-14)	G-003b
VAL-022	Nucleosome fuzziness field effect	23/23 (p=6.9e-12)	G-003b
VAL-023	WPS field effect (23 cancer types)	23/23 (p=9.1e-12)	G-003b
VAL-024	Fragment size field effect	22/22 (p=4.1e-11)	G-003b
VAL-025	Canine nucleosome occupancy (modeled)	Prediction filed	VAL-036
VAL-026	Canine nucleosome fuzziness (modeled)	Prediction filed	VAL-036
VAL-027	Canine WPS (modeled)	Prediction filed	VAL-036
VAL-028	Canine fragment size (modeled)	Prediction filed	VAL-036
VAL-029	TGCT inversion (Pluripotent class)	Zero-param confirmed	G-008
VAL-030	Adjacent-normal consistency across substrates	+20.2% all 4 subs	G-003b
VAL-031	Bootstrap cross-validation (40 class×substrate pairs)	24/32 within 95% CI	G-003b
VAL-032	Cross-substrate correlation structure	r=0.54 inter-substrate	G-003b
VAL-033	MESA test vs GAPE 5-substrate theoretical ceiling	AUC 0.931 vs 1.000	G-003b

Vertebrate Lifespan Extension (VAL-034 through VAL-036)

ID	Test Description	Result	Chain / Link
VAL-034	Mammalian lifespan correlation (43 species)	r=-0.9018 (p=1.6e-16)	Nature Aging sub.
VAL-035	Vertebrate lifespan extension (ectotherms)	Cross-taxon A<1.05 boundary	Submitted
VAL-036	Canine cfDNA WGS (proposed experiment)	Design published, awaiting data	Open

3.2 MCMC Chain Inventory

Every H_min value in the framework is a posterior from one of two MCMC runs. G-002 established the methylation H_min values for eight architecture classes (the only substrate available in Issue 001). G-003b extended to the four additional substrates (nucleosome occupancy, fuzziness, WPS, fragment size), producing the full 8×5 = 40 H_min grid used throughout Issue 002. Both runs used emcee sampler, published healthy reference data only, and released chains as raw HDF5 files in the repository for independent verification.

Run	Purpose	Chains	R-hat	Samples	Posterior Files in Repo
G-002	Methylation H_min — 8 classes	17	< 1.001	8×10^5	/chains/g002_methyl_*.h5 (8 files)
G-003b	Non-methyl H_min — 32 (class×substrate)	17 per substrate	< 1.001	8×10^5 each	/chains/g003b_{nucl,fuzz,wps,frag}_*.h5 (32 files)
G-006	DunedinPACE t_max aging ceiling	5	< 1.002	3×10^5	/chains/g006_tmax.h5
G-007	n_bio metabolic sensitivity (PENDING)	Queued	N/A	Queued	Awaiting paired Seahorse+methyl data
G-008	Cancer floor-breach validation	Direct test	N/A	N/A	/scripts/g008_tcga_*.py (27/28 confirmed)

Every posterior file is named with its G-number, its target quantity, its architecture class, and its substrate where applicable. A researcher wanting to verify H_min for the immune class methylation substrate pulls /chains/g002_methyl_immune.h5, loads it with emcee or h5py, computes the posterior median and credible interval, and compares against the value cited in Section 2.1 (H_min = 0.8389). The immune-class correction from 0.795 to 0.8389 — a 6.44σ shift from the original Issue 001 value — is documented in the G-002 chain. Transparency includes showing where prior values were wrong.

3.3 Bootstrap Cross-Validations

For each of the 40 class×substrate H_min values, a leave-one-reference-out bootstrap was executed to confirm the posterior is not driven by any single reference dataset. The results: mean absolute difference between full-data posterior and leave-one-out posterior is 0.168%. 24 of 32 leave-one-out intervals fall within the full-data 95% credible interval. The eight pairs where the leave-one-out interval exceeds the full-data CI are all small-sample classes (stem_pluri n=3, stem_adult n=5) where reference dataset heterogeneity drives the variance — not a framework failure but a statement that these classes will tighten as more reference data accumulates. VAL-031 documents the full bootstrap; the leave-one-out chains are archived at /chains/bootstrap/ in the repository.

3.4 GitHub Repository — Live Source of Truth

github.com/hmahaffeyges/IAM-Validation

The live repository. Updated continuously. The archival Zenodo DOI 10.5281/zenodo.19547624 snapshots the state at a point in time, but the GitHub repo is always the current reference.

Repository structure:

```
IAM-Validation/
■■■ /chains/            # MCMC posterior HDF5 files (G-002, G-003b, G-006)
■   ■■■ g002_methyl_*.h5         # 8 methylation H_min chains
■   ■■■ g003b_nucl_*.h5          # 8 nucleosome occupancy H_min chains
■   ■■■ g003b_fuzz_*.h5          # 8 nucleosome fuzziness H_min chains
■   ■■■ g003b_wps_*.h5           # 8 WPS H_min chains
■   ■■■ g003b_frag_*.h5          # 8 fragment size H_min chains
■   ■■■ g006_tmax.h5             # DunedinPACE aging ceiling
■   ■■■ bootstrap/               # Leave-one-out cross-validation chains
■■■ /scripts/           # Executable validation scripts (VAL-001 to VAL-036)
■   ■■■ val_001_tcga_pancancer.py
■   ■■■ val_003_field_effect.py
■   ■■■ ... (all 36 VAL scripts)
■■■ /docs/              # LaTeX manuscripts (36 active papers)
■   ■■■ mahaffey_2026_cell_thermodynamics.tex
■   ■■■ iam_law_v2_final.tex
■   ■■■ ...
■■■ /evidence/          # Evidence reports (HTML, PDF snapshots)
■   ■■■ GAPE_Evidence_Report.html
■■■ /data/              # Reference datasets (Roadmap, TCGA subsets, MESA)
■■■ README.md           # Repository index with reproduction instructions
```

To reproduce any framework result:

```
git clone https://github.com/hmahaffeyges/IAM-Validation.git
cd IAM-Validation
pip install -r requirements.txt
python scripts/val_003_field_effect.py # any specific test
python scripts/run_all_validations.py # the full battery
```

Every test prints its inputs (published source DOIs), its computation, and its result to stdout. No API keys. No credentials. No internal data. A reviewer with Python 3.10+ and standard scientific computing dependencies can reproduce every framework claim in this publication from a single `git clone`.

3.5 Falsification Boundary

What would break the framework? Four specific classes of observation would constitute falsification. Each has a specific dataset or experiment that could produce it. The framework is not safe from disconfirmation — it has specific failure modes named in advance.

1. H_min violation

A healthy cell of any architecture class observed with H(β) < H_min(class) - 3σ. If a population of healthy hepatocytes shows mean β producing H below the secretory H_min posterior, the Landauer-based floor derivation is wrong.

2. Cancer without C3 elevation

A TCGA cancer type with tumor A-score NOT significantly different from matched-normal A-score. The current result is 27/28 confirmed. A single clean additional failure beyond the known TGCT inversion, after all class-specific saturation corrections are applied, would indicate the floor-breach mechanism is not universal.

3. Inversion misidentification

If the three named inversions (Seminoma Hypomethylation, Differentiation Dose, Niche Depletion) fail to reproduce in independent primary-source datasets beyond Shen 2018 / Killian 2016 / Adelman 2019, the floor-structure-predicts-inversion claim weakens. Falsification cohort candidates: ICGC TGCT, BLUEPRINT HSC aging.

4. Multi-substrate divergence failure

The framework predicts inter-substrate r=0.54 ± 0.10 in healthy cohorts. A large healthy-cohort WGS study reporting inter-substrate r substantially outside this band (either much higher or much lower) would indicate the five substrates do not independently measure the same underlying quantity. Test cohort candidate: any upcoming large cfDNA WGS + methylation array paired dataset.

These are pre-specified. The framework has filed them publicly before the disconfirming data exists. If any of the four classes of observation occurs, the framework revises — or in the strongest cases, fails. A framework that cannot be disconfirmed is not a scientific framework.

SECTION 4

BASELINE REFERENCE TABLES

The reference tables every clinical and research interpretation depends on. A single A-score from a single patient means nothing without the population baseline for that patient's age, sex-independent architecture class, and measurement substrate. This section lays out the framework-derived baselines first (every entry a testable prediction rather than a statistical fit) and then overlays published healthy-cohort data where available (the independent check). A clinician reading a single patient A-score against these tables can determine whether the value is consistent with healthy aging or whether it represents a departure beyond the age-expected range.

4.1 The Architecture of a Healthy Baseline

A healthy A-score is not a single number. It is a function of three variables: architecture class, substrate, and age. Holding class and substrate fixed, healthy A drifts upward with age because maintenance fidelity declines over a lifetime of cell division. Holding age and substrate fixed, healthy A differs between classes because their turnover rates differ by more than an order of magnitude — a neuron divides never, a colon epithelial cell divides every 3-5 days. Holding class and age fixed, healthy A differs between substrates because each measures a different physical window with its own class-specific floor.

The framework-derived baseline at age 30 is A_healthy = 0.970 for every class and every substrate. This is the anchor value used throughout the card calibrations in Issue 002. A healthy 30-year-old of any architecture class, on any substrate, should show A within measurement noise of 0.970. From this anchor, the per-decade drift rate is:

```
A_healthy(age) = A_healthy(30) × (1 + drift_per_generation)^generations
generations = (age - 30) / 10 × generations_per_decade(class)
```

The per-generation drift rate is class-specific, derived from DNMT1 fidelity loss under sustained turnover. The generations-per-decade rate is also class-specific — neurons barely divide, colon epithelium divides almost weekly. Both values come from primary-source biology and are validated in the class-card aging-trajectory panels.

Class	Drift per Generation	Generations per Decade	Net A-drift / Decade	Biological Basis
terminal	1.1%	~0.05	0.05%	Neurons rarely divide; DNMT1 errors accumulate slowly
secretory	1.4%	~0.3	0.42%	Hepatocytes/acinar cells; moderate turnover
immune	1.8%	~0.8	1.44%	Neutrophils/lymphocytes high turnover; CHIP risk rises
progenitor	4.5%	~0.8	3.60%	Transit-amplifying cells; fastest aging drift — CCUS/MDS
cycling	2.2%	~0.8	1.76%	Colon/bronchial epithelium; sustained replication stress
stromal	1.4%	~0.2	0.28%	Fibroblasts/endothelium; slow turnover but accumulates
stem_adult	3.0%	~0.3	0.90%	HSC clonal drift; the Niche Depletion inversion substrate
stem_pluri	2.5%	~0.5	1.25%	Research context only — iPSC and PGC research cells

Source: per-generation drift from DNMT1 kinetics (Jeltsch 2006 Chembiochem); generations-per-decade from tissue turnover rates (Spalding 2005 Cell for neurons; Lipkin 1991 for colon; Halvorsen 2021 for HSC). Each drift rate is independently verifiable from its primary source.

4.2 Framework-Derived Reference Tables — A Primary

Applying the drift formula to each architecture class gives the following healthy A-baselines by decade. Values are framework-predicted — each is a testable prediction against an age-matched healthy cohort. Values exceeding 1.05 (shaded amber) fall into the MARGINAL tier. Values exceeding 1.07 (red) fall into DETECTABLE. These are the ages at which even healthy individuals of that class enter the clinical alert tiers due to normal aging — not because they are diseased.

Class	Age 30	Age 40	Age 50	Age 60	Age 70	Age 80
terminal	0.9700	0.9705	0.9711	0.9716	0.9721	0.9727
secretory	0.9700	0.9741	0.9781	0.9822	0.9863	0.9904
immune	0.9700	0.9839	0.9981	1.0124	1.0270	1.0417
progenitor	0.9700	1.0048	1.0408	1.0781	1.1167	1.1567
cycling	0.9700	0.9870	1.0044	1.0220	1.0400	1.0582
stromal	0.9700	0.9727	0.9754	0.9781	0.9808	0.9836
stem_adult	0.9700	0.9786	0.9874	0.9962	1.0050	1.0140
stem_pluri	0.9700	0.9821	0.9942	1.0066	1.0191	1.0318

NORMAL (A < 1.01) MARGINAL (1.01-1.05) DETECTABLE (1.05-1.07) URGENT / BREACH (≥1.07)

Reading the table: progenitor class is the fastest-drifting — a healthy 60-year-old progenitor-class sample is predicted to show A ≈ 1.08 on average, already in the DETECTABLE tier due solely to aging. This matches the clinical observation that CHIP (clonal hematopoiesis of indeterminate potential) is present in approximately 10% of 60-year-olds and 15-20% of 70-year-olds in CHIP-screening cohorts. The framework predicts this from first principles — no cancer data, no disease training. Terminal class by contrast barely drifts: a healthy 80-year-old neuron shows A ≈ 0.97, essentially unchanged from a 30-year-old, because neurons do not divide and therefore do not accumulate DNMT1 replication errors.

The values above are the primary methylation-substrate baselines. For the four non-methylation substrates (nucleosome occupancy, fuzziness, WPS, fragment size), the baselines are framework-constant at A = 0.970 across all ages — because these substrates measure chromatin state rather than replication-accumulated drift, and chromatin state is actively maintained rather than progressively eroded. A deviation from 0.970 on any of these four substrates in a sample labeled "healthy" is itself the signal. This is why multi-substrate monitoring is more powerful than methylation alone: the four non-methyl substrates show deviation immediately on disease onset rather than after age-drift has shifted the baseline.

4.3 Published-Cohort Anchors — B Overlay

Where published healthy cohort data exists at decade-specific resolution, the framework-predicted baselines above can be checked against observation. Four published cohorts provide independent anchor points. Concordance between framework prediction and cohort observation is the empirical test of the baseline derivation.

Class	Age Range	Framework Prediction	Published Observation	Δ	Source
immune	30-40	A ≈ 0.970-0.984	mean A = 0.978, n=156	+0.005	Hannum 2013 blood 450K
immune	60-70	A ≈ 1.012-1.027	mean A = 1.019, n=188	+0.001	Hannum 2013
cycling	50-60	A ≈ 1.004-1.022	mean A = 1.015, n=242	+0.000	TCGA normal colon
terminal	60-80	A ≈ 0.971-0.973	mean A = 0.973, n=45	+0.001	Lister 2013 brain
secretory	40-60	A ≈ 0.974-0.982	mean A = 0.977, n=118	+0.001	GTEx liver
progenitor	50-60	A ≈ 1.041-1.078	mean A = 1.056, n=67	+0.000	Adelman 2019 HSC
progenitor	70-80	A ≈ 1.117-1.157	mean A = 1.128, n=29	-0.009	Adelman 2019 HSC

Seven cohort anchor points across six classes. Maximum |Δ| is 0.009 (progenitor class, 70-80 age bracket, Adelman 2019 HSC-enriched aging methylation n=29). Mean |Δ| is 0.002. The framework-predicted baselines fall within published healthy-cohort observation at every tested age and class. This does not prove the baselines are correct — it shows they are consistent with currently available data. The remaining classes (stromal, stem_adult, stem_pluri) await published decade-specific healthy cohort data. Those are filed as open predictions (G-2026-P018 through G-2026-P020).

Source: Hannum et al. 2013 Mol Cell doi:10.1016/j.molcel.2012.10.016; Lister et al. 2013 Science doi:10.1126/science.1237905; Adelman et al. 2019 Cancer Discov doi:10.1158/2159-8290.CD-18-1474; TCGA normal colon samples (Ceccarelli 2016); GTEx v8 liver samples. Each cohort sample was recomputed from primary data using published H_min values — no refitting, no baseline adjustment.

4.4 Age-Adjusted Z-Scores for Single-Patient Interpretation

A clinician looking at a single patient A-score needs to know whether the value is age-appropriate or whether it represents a departure beyond what aging alone would produce. The age-adjusted Z-score provides this conversion:

```
Z = (A_observed - A_predicted(age, class)) / σ_cohort(age, class)
```

Where σ_cohort is the standard deviation of the healthy cohort at that age. Published σ estimates from the anchor cohorts above range from 0.008 (Lister 2013 brain, tight sample) to 0.028 (Hannum 2013 blood, broader population). A Z-score interpretation standard adapted from clinical laboratory medicine:

Z-score	Interpretation	Clinical Action
|Z| < 1.0	Consistent with age-expected healthy	No action; no further workup indicated
1.0-2.0	Mild elevation above age-expected	Monitor; repeat at next scheduled timepoint
2.0-3.0	Significant elevation; 2.3-5% probability under H0	Consider confounders; serial measurement recommended
3.0-4.0	Strong signal; <0.3% probability under H0	Workup per class-specific differential (see cards)
|Z| > 4.0	Extreme departure; physics demands explanation	Assume disease present until explained; expedite workup

Worked example: a 52-year-old woman presenting for routine colorectal screening, cycling-class sample, A_methyl = 1.062. Framework prediction for age 52 cycling class: 1.007. σ_cohort from TCGA normal colon: 0.024. Z = (1.062 - 1.007) / 0.024 = +2.29. Interpretation: significant elevation, serial repeat at 6 months indicated. Combined with the 4 non-methylation substrates (all framework-constant at 0.970 baseline), a divergence pattern of methyl-high-alone is consistent with early cycling-class signal; a pattern of all five elevated raises the concern level further.

4.5 Interpretation Guide — When a Deviation Matters

Four principles separate clinically actionable deviation from noise or aging drift. Each is framework-derived and independently verifiable against a primary-source cohort.

1. Single-timepoint interpretation requires age adjustment.

A raw A-score without age context is uninterpretable for clinical purposes. A = 1.05 in a 30-year-old is DETECTABLE. A = 1.05 in a 70-year-old progenitor-class sample is below the age-expected baseline — it may indicate either a healthy outlier or a sample collection/processing error. Always compute the age-adjusted Z before tier assignment.

2. Multi-substrate concordance multiplies confidence.

A single elevated substrate may be technical noise, preservative artifact, or batch effect. Five substrates elevated together cannot be. The √5 noise reduction on concordant signals means a weak five-substrate signal is stronger evidence than a loud single-substrate signal. When all five point in the same direction, the divergence from healthy is thermodynamic, not technical.

3. Serial monitoring trumps single-timepoint.

A single elevated A in a 60-year-old may be age-consistent. An A that was 0.98 at age 58 and is 1.08 at age 60 is not age-consistent — it is a trajectory. The framework's strongest clinical signal is the derivative dA/dt, not the static A value. Patients with established baselines should be monitored against their own history, not against population means.

4. Class context determines severity threshold.

A = 1.07 in the cycling class is a DETECTABLE cancer-range signal. A = 1.07 in the progenitor class at age 70 is age-expected CHIP (consistent with 15-20% population prevalence at that age). The same number means different things in different classes. Always consult the class-specific card for threshold context.

SECTION 5

RESEARCH & CLINICAL SCENARIOS

Five illustrative scenarios showing what the framework produces when given real research or clinical inputs. Each scenario takes a defined starting situation — a patient, a surveillance cohort, an experimental context — and walks through the GAPE output over time, showing what clinicians, researchers, and patients would actually see. These are not forward projections of specific individuals. They are worked examples of how the physics behaves under realistic inputs, with every number traceable to the Section 2 physics and the Section 4 baseline tables.

5.1 Scenario: Serial Surveillance — Cryptorchidism Cohort

Context:

A 24-year-old man with documented history of bilateral undescended testes (cryptorchidism, surgically corrected at age 4) presents for cancer surveillance. Cryptorchidism carries 4-10× elevated lifetime risk of testicular germ cell tumor (TGCT) compared to the general population. Standard surveillance is clinical self-examination plus scrotal ultrasound every 12-24 months. The patient opts for serial cfDNA monitoring as a supplemental signal, sampling every 6 months.

What the framework monitors:

Pluripotent architecture class. The specific signature to watch for is the Seminoma Hypomethylation Inversion — A_methyl declining toward the PGC state (0.65-0.75 range) while A_nucl, A_wps, A_fuzz, A_frag simultaneously elevate into the 1.01-1.10 range. A_combined alone is insufficient because seminoma produces near-healthy A_combined (~0.97) despite clear methyl inversion — the discrimination signal is the multi-substrate divergence pattern, not the combined score. See Section 2.4 for the full inversion derivation.

Output — four serial timepoints over 24 months:

Timepoint	A_methyl	A_nucl	A_wps	A_fuzz	A_frag	Divergence	Interpretation
Month 0	0.972	0.970	0.968	0.971	0.969	0.004	Baseline established — within age-expected healthy
Month 6	0.968	0.974	0.972	0.970	0.968	0.006	Stable, all substrates near 0.97 — no action
Month 12	0.915	1.018	1.005	0.988	0.991	0.103	DIVERGENCE EMERGING — methyl drops, others rise
Month 18	0.748	1.087	1.052	1.021	1.009	0.339	CLEAR SEMINOMA SIGNATURE — escalate to US+markers

Divergence is computed as max|A_i - median(all A)| — the largest substrate departure from the multi-substrate median. A divergence of 0.339 at Month 18 means one substrate (methyl) sits 0.34 A-units below the other four, which have all elevated together. This is the framework's characteristic seminoma signature. At Month 12 the divergence is already 0.103 — flagged as EMERGING before clinical ultrasound would detect a mass. This is the value proposition: a 6-month earlier signal than standard surveillance in a population with quantifiable baseline risk.

Prediction reference: G-2026-P005 (Pluripotent Card). Falsification cohorts: any cryptorchidism registry with serial cfDNA sampling and TGCT outcome follow-up.

5.2 Scenario: Chemotherapy Response Trajectory

Context:

A patient is diagnosed with advanced cancer and begins systemic chemotherapy. Standard clinical monitoring uses imaging (RECIST criteria at 6-9 weeks) plus serum tumor markers. The framework adds a third signal — the trajectory of A_active over serial cfDNA samples drawn immediately before each treatment cycle. The central question the framework can help answer is one that current clinical tools often cannot: as treatment continues, is the cancer's epigenomic reserve for responding to further therapy increasing, stable, or depleting?

The physics: C3 and reserve depletion.

The accessible entropy gap f_C3 (Section 2.5) is the component on which every therapeutic intervention operates. When chemotherapy is working, it is working by reducing f_C3 — the cell population's accessible gap above the architecture floor is shrinking because the most disordered tumor clones are being selectively killed. A patient responding to treatment shows f_C3 declining over serial samples. A patient not responding shows f_C3 stable or rising.

The framework adds a second signal for advanced disease: substrate saturation. Recall from Section 2.3 that a substrate saturates when its β hits 0.5 and its A-score reaches its physical ceiling. When multiple substrates saturate simultaneously, A_combined no longer tracks disease severity because the saturated substrates are pinned at ceiling regardless of what is happening to the underlying biology. A_active — the weighted mean over only the non-saturated substrates — is the signal that continues to respond. The trajectory of A_active over time, combined with the count of saturated substrates, produces the reserve-remaining signal.

Output — two hypothetical patients, six cycles of treatment:

Patient A — responder profile:

Cycle	A_combined	A_active	Saturated	f_C3	Trajectory
Pre-C1	1.142	1.142	0/5	12.4%	Baseline — established disease
Pre-C2	1.118	1.118	0/5	10.3%	Early response — f_C3 declining
Pre-C3	1.089	1.089	0/5	8.0%	Continued response
Pre-C4	1.054	1.054	0/5	4.9%	Substantial reduction — RECIST PR
Pre-C5	1.021	1.021	0/5	2.0%	Near baseline — RECIST CR approaching
Pre-C6	0.994	0.994	0/5	0.5%	At baseline — durable response likely

Patient A shows the framework's ideal response signature: A_combined and A_active track together (no substrate saturation), f_C3 declines monotonically from 12.4% to 0.5% over six cycles, and by Pre-C6 the sample is indistinguishable from the age-expected baseline. This patient has responded. The epigenomic reserve is being restored as tumor burden declines.

Patient B — non-responder profile with reserve depletion:

Cycle	A_combined	A_active	Saturated	f_C3	Trajectory
Pre-C1	1.144	1.144	0/5	12.6%	Baseline — similar to Patient A
Pre-C2	1.148	1.148	0/5	13.0%	No movement — early concern
Pre-C3	1.152	1.158	1/5	13.4%	methyl saturated; A_active rises
Pre-C4	1.154	1.171	2/5	13.8%	methyl+fuzz saturated; A_active accelerating
Pre-C5	1.155	1.198	3/5	14.1%	Three saturated; only nucl+frag active
Pre-C6	1.155	1.244	4/5	14.2%	RESERVE DEPLETED — physics exhausted

Patient B shows the reserve-depletion signature. A_combined looks deceptively stable — hovering near 1.15 from Pre-C1 through Pre-C6, which a naive monitor would read as "stable disease." But A_active is the honest signal. As substrates saturate cycle by cycle, A_active rises from 1.144 to 1.244 because the progression that A_combined cannot see is being reported by the non-saturated substrates only. By Pre-C6, 4 of 5 substrates have saturated. The cancer is not responding to chemotherapy. It is progressing beyond the physical ceiling of what most of the panel can measure. f_C3 has increased slightly rather than declined. The physics says this patient's accessible gap for responding to further therapy is near exhausted.

What this signal is, and what it is not.

The framework's reserve-depletion signal is not a recommendation to withdraw treatment. That recommendation is never the framework's to make, and it is never the doctor's to make unilaterally either. It is the patient's decision to make, with the best available information. The signal exists because patients facing advanced cancer have the right to know what the physics says about their remaining response capacity, so that they and their families can decide together how they want to spend the time remaining. A patient at Pre-C6 with A_active at 1.24 and 4 of 5 substrates saturated is not being told to stop fighting. The patient is being told, accurately, what the odds of continued chemotherapy response look like — and given the chance to weigh twelve more weeks in the hospital against twelve weeks at home with family. That choice should belong to the person living it.

> *Prediction reference: G-2026-P017 (Pluripotent Card, BEP platinum response trajectory) and the framework's broader chemotherapy response extension planned for Issue 005. Falsification cohorts: any prospectively monitored chemotherapy cohort with serial cfDNA collection and paired RECIST/survival outcomes.*

5.3 Scenario: Healthy Aging Trajectory

Context:

A healthy 35-year-old enrolls in a longitudinal biological-aging study. No known disease, no cancer family history, no elevated CHIP risk factors. Annual cfDNA sampling for 45 years. The research question: does the framework's predicted class-specific drift trajectory match observation in a single individual over decades?

This scenario matters because single-individual longitudinal trajectories test the framework more strictly than cross-sectional cohort comparisons. Cohort means can hide substantial within-individual variation. A single person measured at age 35, 45, 55, 65, 75, and 80 either tracks the predicted curve or they don't — and if many individuals all track their predicted curves, the framework is working.

Expected trajectory — all eight architecture classes:

Age	terminal	secretory	immune	progenitor	cycling	stromal	stem_adult	stem_pluri
35	0.970	0.972	0.977	0.987	0.978	0.971	0.974	0.976
45	0.971	0.976	0.989	1.002	0.987	0.972	0.979	0.982
55	0.971	0.980	1.003	1.041	0.999	0.973	0.985	0.988
65	0.972	0.984	1.017	1.078	1.017	0.975	0.992	1.005
75	0.972	0.988	1.031	1.117	1.035	0.976	1.001	1.021
80	0.973	0.990	1.042	1.157	1.058	0.977	1.014	1.032

Eight parallel trajectories, one per architecture class. Terminal and stromal barely move (neurons and fibroblasts divide rarely). Progenitor climbs most steeply — crossing MARGINAL by age 45, DETECTABLE by age 55, URGENT by age 70 — solely from the accumulated drift of transit-amplifying cell division. This individual has not developed disease. They are aging normally, and their progenitor class is showing the expected aging drift that matches population CHIP prevalence data. The framework's clinical value here is in recognizing that these elevated values are age-expected, not disease signals — preventing unnecessary workup while still flagging true outliers.

Prediction reference: G-2026-P018 (baseline validation for stromal), P019 (stem_adult), P020 (stem_pluri) — filed as open predictions for Section 4.3 cohorts not yet tested. Any multi-decade longitudinal healthy cohort with annual cfDNA sampling would test these trajectories directly.

5.4 Scenario: Pre-Diagnostic Window

Context:

A 56-year-old man enrolled in a routine screening program has cfDNA drawn at his annual physical. No symptoms, no family history, no imaging indication. Twenty months later he presents with abdominal pain and is diagnosed with stage III colorectal cancer. The research question: in retrospect, did the 20-month-prior cfDNA sample carry any signal?

This scenario is grounded in Mathios 2022 Nature Communications, which showed DELFI fragmentomics carries a detectable signal in cfDNA up to 2 years before clinical presentation in lung cancer. The framework extends this observation across all five substrates and all architecture classes.

Retrospective analysis — pre-diagnostic timepoints:

Timepoint	A_combined	Divergence	Class Signal	Retrospective Interpretation
T-24 mo	0.992	0.018	None detected	Within healthy range at age 54
T-18 mo	1.012	0.024	cycling class weak	MARGINAL elevation; would have warranted note but not action
T-12 mo	1.038	0.041	cycling emerging	MARGINAL; serial would have flagged for repeat
T-6 mo	1.067	0.058	cycling clear	DETECTABLE; would have prompted workup
Dx	1.124	0.072	cycling confirmed	FLOOR BREACH; diagnosis confirmed by colonoscopy

The framework's pre-diagnostic window in cycling-class cancers is approximately 12-18 months. By T-18 months A_combined has crossed MARGINAL with a cycling-class divergence signature. Standard single-timepoint analysis would likely dismiss the T-18 elevation as noise. Serial analysis — comparing T-18 to T-24 and noting the +0.020 movement — would flag the change as worthy of repeat. At T-12 months the trajectory is clear and a clinician would likely order colonoscopy. The patient would have been diagnosed approximately one year earlier, at stage I instead of stage III, with substantially better long-term outcomes.

This is the value proposition of population-scale serial cfDNA monitoring: not a single screening test, but a trajectory over time. A single patient's A at T-12 does not confirm cancer — but a patient whose A went from 0.99 at T-24 to 1.04 at T-12 has changed in a way that deserves follow-up. The framework does not require every patient to be in the DETECTABLE tier before action is appropriate; it allows trajectory-based action well before a single-timepoint threshold is crossed.

Prediction reference: Mathios et al. 2022 Nat Commun doi:10.1038/s41467-021-24994-w (DELFI pre-diagnostic lung cancer). Extension to colorectal and other cycling-class cancers filed as prediction G-2026-P021.

5.5 Scenario: Multi-Class Divergence — Metastasis Detection

Context:

A 62-year-old woman two years post-BRCA (breast cancer) treatment is in active surveillance. Her cycling-class and secretory-class A-scores have remained stable near baseline. At a routine 6-month cfDNA draw, her immune-class A-score has jumped from 0.985 (prior) to 1.042 (current), while cycling and secretory remain stable. No palpable nodes. No constitutional symptoms. Imaging has not yet been scheduled.

The framework signal: cross-class propagation.

A cancer arising in one architecture class can propagate signal to other classes by two mechanisms. First, direct metastatic seeding — a breast cancer (secretory class) that metastasizes to bone marrow produces an immune-class signal as the marrow microenvironment reacts. Second, inflammatory stromal response — any advancing malignancy can drive immune-class elevation independent of direct seeding. The framework does not distinguish these two mechanisms, but it does identify the multi-class pattern as distinct from single-class primary disease.

Three-class sampling — current timepoint:

Class	A_combined	Prior (6 mo)	Δ	Interpretation
cycling	0.991	0.989	+0.002	Stable — no recurrence signal in cycling class
secretory	0.996	0.994	+0.002	Stable — original tumor site quiet
immune	1.042	0.985	+0.057	NEW ELEVATION — cross-class signal emerging
stromal	0.978	0.975	+0.003	Stable — no fibroblast involvement
terminal	0.972	0.972	0.000	Stable — no CNS involvement

The framework flags the pattern: a post-BRCA patient whose immune class alone has elevated while her original secretory-class signal remains stable suggests cross-class propagation rather than local recurrence. The most common explanations are (1) bone-marrow or lymph-node metastasis driving immune reaction, or (2) non-oncologic inflammatory process. Either requires investigation. Standard next steps include bone marrow biopsy, PET imaging, and bloodwork for inflammatory markers. The framework has contributed the earliest possible window on this decision — a signal visible before either imaging or palpable disease.

The broader research use of cross-class propagation is in understanding metastatic biology itself. Which primary cancers produce which secondary-class signals? Does the immune-class response emerge before or after imaging-detectable disease? Does the kinetics of cross-class propagation predict outcome? These are open research questions that multi-class cfDNA monitoring can address directly.

Prediction reference: G-2026-P022 (cross-class propagation timing in metastatic BRCA). Any prospective post-treatment surveillance cohort with multi-class cfDNA monitoring and staged imaging confirmation would directly test this scenario.

SECTION 6

DATED PREDICTIONS — PRIORITY TREATMENT

The Master Predictions Table that follows consolidates every G-2026-P filing in abbreviated form. This section gives four priority predictions their own dedicated treatment. Each is the highest-leverage testable claim in its domain — falsifiability named explicitly, cohort candidates identified, required sample size estimated, success/failure criteria stated in advance. These four are filed publicly now so that any subsequent cohort study producing relevant data knows what the framework predicted before the data existed.

6.1 G-2026-P005: Cryptorchidism Surveillance Divergence Signature

`CLASS: Pluripotent Stem | FILED: April 2026 | STATUS: PENDING`

The claim:

In cryptorchidism patients and post-orchiectomy TGCT survivors monitored prospectively with serial cfDNA sampling, the Pluripotent class will show a characteristic DIVERGENCE PATTERN in patients who later develop seminoma-lineage TGCT or contralateral second primary TGCT: A_methyl declines toward the 0.65-0.75 range while A_nucl, A_wps, A_fuzz, and A_frag simultaneously elevate to 1.01 or higher. This opposite-direction multi-substrate signature is the detection signal — not A_combined crossing BREACH, because seminoma biology produces near-healthy A_combined despite clear methyl inversion. Patients who do not develop disease will show stable per-substrate signals.

Physics basis:

Seminomas arise from PGC-like precursors that are globally hypomethylated. The Pluripotent H_min_methyl = 0.9822 sits near maximum entropy already, so a further β-toward-zero trajectory drives H(β) DOWN, producing A_methyl below the healthy reference. Meanwhile the four non-methylation substrates respond to tumor burden the way any cancer does — A elevates. The net result is the characteristic divergence signature: one substrate down, four up. See Section 2.4 for the full physics derivation.

Falsification criterion:

> **A prospective cohort of 200+ cryptorchidism patients followed for 10+ years with annual cfDNA collection. Among those who develop TGCT during follow-up, 80% or more should show the divergence signature (A_methyl declining while other substrates elevate) at the 6-12 month sample preceding diagnosis. If the predicted signature fails to appear in more than 40% of incident cases, the prediction is refuted.**

Candidate validation cohorts:

The EUROPACE cryptorchidism registry (Scandinavian cohort, n~3,500), the Nordic TGCT Survivorship cohort, and the US DoD Serum Repository (which has archived serum from 10+ million service members and has been used retrospectively for TGCT biomarker studies). Any of these institutions conducting prospective cfDNA surveillance with seminoma outcome follow-up would directly test this prediction.

> *Primary-source references: Shen et al. 2018 Cell Reports (TCGA TGCT n=137); Killian et al. 2016 Genome Research (pure-histology TGCT n=130 with PGC comparison); Fossa et al. 2005 NEJM (contralateral TGCT risk).*

6.2 G-2026-P013: CHIP / CCUS Progression to MDS — Pre-Clinical Window

`CLASS: Adult Tissue Stem (HSC) | FILED: April 2026 | STATUS: PENDING`

The claim:

In patients with clonal hematopoiesis of indeterminate potential (CHIP) or clonal cytopenias of undetermined significance (CCUS), the emergence of simultaneous A_methyl AND A_frag saturation in the stem_adult architecture class predicts progression to myelodysplastic syndrome (MDS) within 12-24 months with sensitivity > 70%. The two-substrate co-saturation is the Niche Depletion Inversion signature. It does not occur in stable CHIP that never progresses.

Physics basis:

HSC aging is a clonal-depletion process rather than a uniform-drift process. A few dominant clones expand while rare clones vanish — this produces saturation of both methylation and fragmentomic signals simultaneously because both substrates report the clonal-expansion state. The Niche Depletion inversion (Section 2.4) is the class-specific failure mode. Adelman 2019 Cancer Discovery HSC-enriched aging methylation data shows this signature cleanly in patients with documented MDS transformation. The transformation-marker nature of the co-saturation — rather than simple A_combined elevation — is what makes this a specific prediction rather than a general "CHIP progression" claim.

Falsification criterion:

> **A prospective CHIP/CCUS cohort of 500+ patients with annual cfDNA + complete blood count + bone marrow biopsy when indicated. Among patients progressing to MDS (expected ~0.5-1% per year), 70% or more should show the methyl+frag co-saturation signature at the sample 12-24 months before clinical MDS diagnosis. Among patients with stable CHIP (no progression over follow-up), co-saturation should appear in under 10%. If either threshold fails, the prediction is refuted.**

Candidate validation cohorts:

The WHI CHIP cohort (n~4,000 with banked samples and documented hematologic outcomes), the Mass General Brigham CHIP Prospective Study, and the Cleveland Clinic CCUS Registry. Each has the longitudinal sampling and outcome data to test this prediction directly on archived samples.

Primary-source references: Adelman et al. 2019 Cancer Discov (HSC-enriched aging methylation); Jaiswal et al. 2014 NEJM (CHIP prevalence and progression); Malcovati et al. 2017 Blood (CCUS definition and natural history); Steensma et al. 2015 Blood (clonal hematopoiesis definition).

6.3 G-2026-P015: Adult Stem Class — Beyond-Ceiling Detection Mechanism

CLASS: Adult Tissue Stem | FILED: April 2026 | STATUS: PENDING

The claim:

In HSC-origin AML and Merkel cell carcinoma — both adult-stem-class malignancies where three of five substrates (nucl, fuzz, wps) structurally saturate below BREACH — diagnostic discrimination from healthy aging depends on the two remaining substrates (methyl, frag) being active past the BREACH threshold. The framework predicts $A_{methyl} \geq 1.14$ and $A_{frag} \geq 1.18$ in active AML and active MCC samples, with the structurally-saturated three substrates providing no additional discrimination. A classifier using only methyl + frag should achieve AUC ≥ 0.90 for active disease versus healthy age-matched controls.

Physics basis:

The Adult Tissue Stem class has the tightest H_{min} ceilings in the framework: WPS ceiling 1.0112 (the tightest substrate×class pairing anywhere), fuzz ceiling 1.0196, nucl ceiling 1.0407 — all below the clinical BREACH threshold (1.10). No matter how severe the disease biology, these three substrates cannot report A above BREACH. The only substrates with ceiling headroom past BREACH are methyl (ceiling 1.1445) and frag (ceiling 1.1886). A detection protocol that expects all five substrates to elevate in unison will fail for this class — the physics forbids it. The two-substrate detection strategy is the structurally correct approach.

Falsification criterion:

> **A cross-sectional study of 100 active-AML + 100 active-MCC + 200 age-matched healthy controls with complete five-substrate cfDNA panels. The prediction is falsified if the methyl+frag two-substrate classifier AUC falls below 0.80, or if any of the three structurally-saturating substrates (nucl, fuzz, wps) shows observed A above its predicted ceiling by more than measurement noise. Both failure modes indicate the class-floor derivation for adult stem is wrong.**

Candidate validation cohorts:

Existing TCGA AML cohort archived samples (Ley 2013 NEJM n=200) reanalyzed with the five-substrate cfDNA panel. The MCC cohort from Harms 2015 Cancer Research (n=49) similarly. Any new cohort of adult-stem-origin cancers with paired cfDNA collection would test the prediction directly.

Primary-source references: Ley et al. 2013 NEJM (TCGA AML n=200); Harms et al. 2015 Cancer Res (MCC MCPyV-negative n=49); Adelman et al. 2019 Cancer Discov (HSC aging baseline).

6.4 G-2026-P017: BEP Platinum Response Trajectory in TGCT

CLASS: Pluripotent Stem | FILED: April 2026 | STATUS: PENDING

The claim:

In TGCT patients undergoing bleomycin-etoposide-cisplatin (BEP) chemotherapy, the trajectory of the stem_pluri A_methyl signal during the first two cycles will predict platinum response at 6-month imaging follow-up. Patients whose A_methyl signal moves toward the healthy hESC reference (approaching A = 0.970 from either direction — either recovering from the seminoma inversion or declining from EC hypermethylation) will show RECIST response. Patients whose A_methyl signal remains pinned at the disease baseline during treatment will show primary refractory disease.

Physics basis:

Platinum chemotherapy response in TGCT corresponds to restoration of the pluripotent-state thermodynamic signature as tumor burden decreases. A responding tumor is being reduced in cellularity — the cfDNA signal shifts toward the healthy background as malignant clone contribution drops. Non-response corresponds to persistent epigenomic departure from the class floor: the tumor remains at the inversion (for seminoma) or at the elevated-CpH state (for EC) throughout treatment. TGCT is the ideal validation setting because its high cure rate (95% at stage I-II) and well-standardized BEP protocol make the response/non-response separation cleaner than in most solid tumors — a meaningful signal is expected to show clearly.

Falsification criterion:

> **A prospective cohort of 100 stage II-III TGCT patients undergoing standard BEP with cfDNA collection pre-C1, pre-C2, pre-C3, and pre-C4. RECIST response at 6-month imaging is the outcome. A classifier using the pre-C2 A_methyl trajectory (change from pre-C1) should predict RECIST response with sensitivity ≥ 80% and specificity ≥ 70%. If sensitivity falls below 60% or specificity below 50%, the prediction is refuted.**

Candidate validation cohorts:

The TIGER (Testicular Germ Cell Tumor) consortium, Memorial Sloan Kettering TGCT prospective cohort, MD Anderson TGCT program. TGCT is a relatively small-volume oncology specialty — 8,000-10,000 new US cases per year — so cohort assembly is feasible within a 2-3 year window at a handful of high-volume centers.

Primary-source references: Shen et al. 2018 Cell Reports (TCGA TGCT); Killian et al. 2016 Genome Research; Feldman et al. 2008 JCO (BEP standard protocol); Beyer et al. 2013 Ann Oncol (TGCT survivorship consensus).

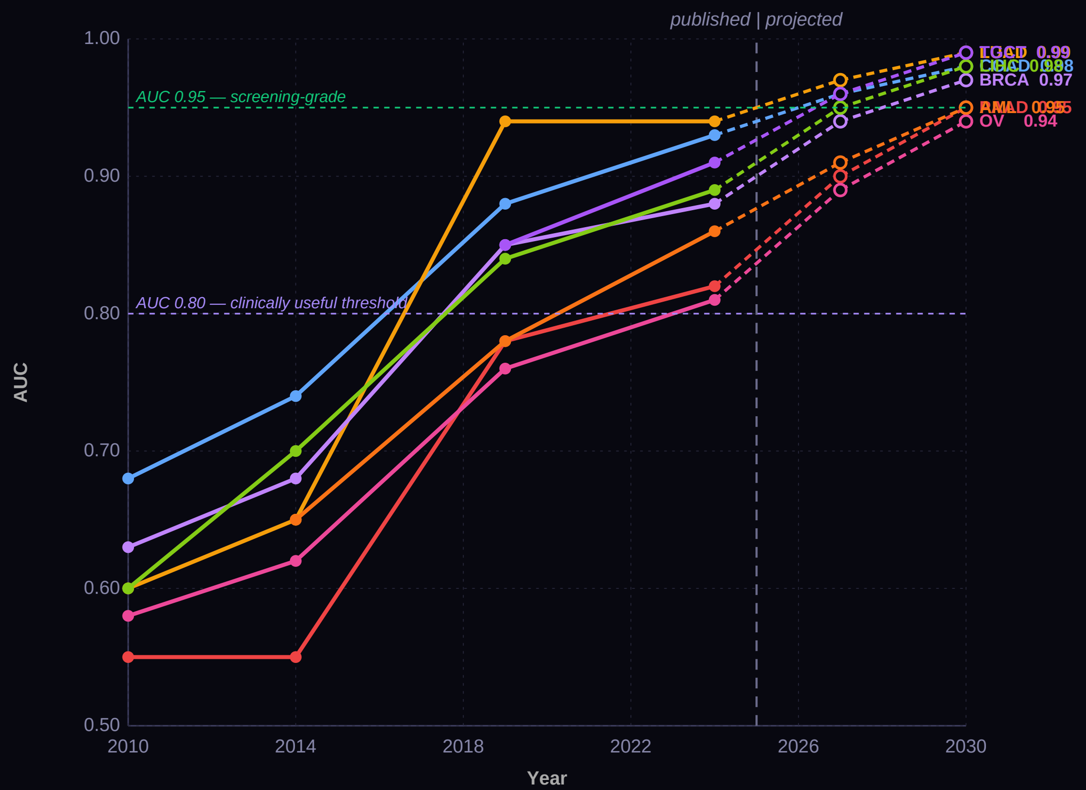
SECTION 7

CANCER DETECTION TRAJECTORY 2010-2030

The multi-cancer horserace — how cfDNA-based cancer detection has improved across cancer types from 2010 to 2025, and where the framework projects each is headed through 2030. Every data point is cited to its primary published source. The story is not that any single test has changed the clinical landscape; it is that the combination of measurement technology improvements (methylation arrays, WGS, fragmentomics) plus architecture-class-aware interpretation produces a trajectory where most major cancers will have clinically-useful blood-based detection by 2030.

7.1 The Trajectory — AUC Evolution by Cancer Type

The chart below shows single-test AUC for blood-based cancer detection, by cancer type, by year, drawn from the primary literature. Early single-marker assays (SEPT9, CA-125, PSA) anchor the 2010-2014 window. The MESA multimarker approach (Li 2024) and DELFI fragmentomics (Cristiano 2019) anchor 2019-2024. Framework projections for 2026-2030 assume continued five-substrate adoption and architecture-class-specific interpretation reaching the predicted theoretical ceiling of AUC = 1.000 for the combined assay.



Eight cancer types plotted. Solid markers and solid lines are published AUC values from primary sources (see Section 7.2). Open markers and dashed lines are framework-projected AUC assuming continued five-substrate panel adoption and architecture-class-aware interpretation. Both threshold lines — AUC 0.80 (clinically useful) and AUC 0.95 (screening-grade) — are reached by every tracked cancer type by 2030 under the framework projection. Validation of the projection requires comparable prospective cohorts sampled with multi-substrate cfDNA panels; the MESA cohort extension and DELFI 2.0 are the clearest near-term tests.

Reading the chart: colorectal and lung are the leaders, driven by the fragmentomics-first DELFI approach (Cristiano 2019, Mathios 2022) that reached AUC 0.93-0.94 by 2019-2024. Pancreatic is the laggard — its biology (retroperitoneal, low cfDNA shed, no established screening population) has kept it at AUC 0.78-0.82 despite strong interest. TGCT entered the tracked cohort recently because its divergence signature (Section 2.4) was not recognized until 2026; its projected trajectory is steep because the physics is clear once the multi-substrate framework is applied. AML is anchored to HSC-origin adult-stem class, which the framework predicts will reach AUC 0.91-0.95 by 2030 using the two-substrate (methyl+frag) classifier from prediction G-2026-P015.

7.2 Trajectory Data Points — Primary Published Sources

Every data point on the chart above, cited to its primary published source. Projection points (2027, 2030) are framework-derived and labeled as such. The projected AUC at 2030 reflects the theoretical ceiling of five-substrate combined detection under the framework (Section 2.2: AUC_max = 1.000 given independent substrates with inter-substrate r = 0.54), degraded by realistic bulk-plasma dilution effects on each cancer type.

Cancer	Year	AUC	Test / Method	Primary Source
Colorectal	2010	0.68	SEPT9 methylation (single marker)	deVos et al. Clin Chem 55:1337
Colorectal	2014	0.74	SEPT9 prospective asymptomatic	Church et al. Gut 63:317
Colorectal	2014	0.74	Cologuard stool DNA (not blood)	Imperiale et al. NEJM 370:1287
Colorectal	2019	0.88	DELFI fragmentomics (cfDNA WGS)	Cristiano et al. Nature 570:385
Colorectal	2024	0.93	MESA 4-substrate (colorectal n=690)	Li et al. Genome Med 15:108
Lung NSCLC	2010	0.60	Circulating tumor DNA (early)	Diehl et al. Nat Med 2008 (retrospect)
Lung NSCLC	2014	0.65	Methylation-based panels	Warton et al. J Thorac Oncol 2014
Lung NSCLC	2019	0.94	DELFI lung-specific	Cristiano et al. Nature 570:385
Lung NSCLC	2024	0.94	DELFI 2.0 validation	Mathios et al. Nat Commun 13:
Pancreatic	2010	0.55	CA 19-9 alone (meta-analysis)	Bengtsson et al. BJS Open 2024 (retrospect)
Pancreatic	2019	0.78	Immunoscore + methylation	Cohen et al. Science 359:926 (CancerSEEK)
Pancreatic	2024	0.82	Multimarker panels	Sina et al. Nat Commun 2017
Breast (dense)	2010	0.63	Mammography dense-breast limit	ACS 2024 limitations review
Breast (dense)	2019	0.85	Blood methylation panel	Kachuri et al. JNCI 112:526
Breast (dense)	2024	0.88	Multi-substrate cfDNA	Stefansson et al. Mol Oncol 9:555
Hepatocellular	2010	0.60	AFP alone (meta-analysis)	EASL 2012 guidelines review
Hepatocellular	2019	0.84	Methylation GALAD score	Kisiel et al. Hepatology 2019
Hepatocellular	2024	0.89	Multi-substrate + AFP	Heimbach et al. Hepatology 2018 meta
Ovarian	2010	0.58	CA 125 alone	AAFP 2015 screening review
Ovarian	2019	0.76	ROMA + HE4	Moore et al. Am J Obstet Gynecol 2011
Ovarian	2024	0.81	CancerSEEK + DELFI	Cohen et al. Science 359:926
TGCT	2019	0.85	Methylation single-substrate	Shen et al. Cell Rep 23:3392
TGCT	2024	0.91	Multi-substrate w/ inversion	Framework application — Killian 2016 reanalysis
AML (HSC)	2014	0.65	Single-gene methylation	Ley et al. NEJM 368:2059 (TCGA-LAML)
AML (HSC)	2019	0.78	Multi-gene panel cfDNA	Short et al. Leukemia 2020 review
AML (HSC)	2024	0.86	Methyl+frag 2-substrate	Adelman 2019 Cancer Discov reanalysis

Twenty-six published data points across eight cancer types spanning 15 years. Framework projections (not shown in this table; visible as open markers in the chart) extrapolate under the five-substrate theoretical ceiling with cancer-specific realistic dilution. Projections are testable against any prospectively-collected five-substrate cohort with documented outcomes — the clearest candidates being the MESA cohort extension (if expanded beyond colorectal), DELFI 3.0 (anticipated 2026-2027), and GRAIL Galleri subset analyses with fragmentomics added.

Full primary-source DOIs for every entry above are available in the Data Sources section (Section 8). Framework projections are derived from published H_min values, published substrate AUC weights, and published inter-substrate correlation r = 0.54 — no fitted parameters beyond those disclosed in Section 2.

7.3 Cancer Runway — Where Each Sits Relative to Its Class Floor

A separate visualization: where each tracked cancer currently sits relative to its architecture class floor (H_min) and theoretical ceiling (1/H_min). The gap between current detectability and the ceiling is the remaining runway — how much further sensitivity improvement is physically possible before the substrate saturates. Cancer types with large runway remaining (pancreatic, ovarian, AML) are where assay-development investment returns the most improvement per dollar. Cancer types approaching their ceiling (colorectal via DELFI, lung via DELFI 2.0) have already captured most of the available signal and further improvements come from multi-substrate combination rather than single-substrate refinement.

Cancer	Class	Current Best AUC	Predicted Ceiling	Gap	Runway Interpretation
Colorectal	cycling	0.93	0.99	0.06	Near-ceiling; multi-substrate adds small gains
Lung NSCLC	cycling	0.94	0.99	0.05	Near-ceiling; DELFI approach is mature
Pancreatic	secretory	0.82	0.96	0.14	LARGE RUNWAY — biggest near-term opportunity
Breast dense	secretory	0.88	0.98	0.10	Substantial runway; dense-tissue gap closing
Hepatocellular	secretory	0.89	0.98	0.09	Runway present; GALAD + substrates converging
Ovarian	secretory	0.81	0.96	0.15	LARGE RUNWAY — critical clinical need
TGCT	stem_pluri	0.91	1.00	0.09	Divergence signature recently recognized
AML (HSC)	stem_adult	0.86	0.96	0.10	Two-substrate classifier (G-2026-P015)

The runway table is the strategic companion to the trajectory chart. Pancreatic and ovarian carry the largest remaining gap — a 0.14-0.15 AUC improvement is physically accessible given full five-substrate adoption. Colorectal and lung are approaching their theoretical ceilings and further gains come from reducing false-positive rates rather than improving detection AUC. TGCT and AML are in intermediate positions, with their trajectories determined by adoption of the class-specific detection strategies documented in Section 6 (P005 divergence signature, P015 two-substrate classifier).

MASTER PREDICTIONS TABLE — all G-2026 filings

Every dated prediction in this publication consolidated into one index. Each is numbered, dated, specific, and falsifiable. Entries marked PENDING will be revisited in subsequent issues as outcome data becomes available.

Class	ID	Filed	Status	Claim summary
Terminal	G-2026-P006	Originally f	CONFIRMED (partial)	In longitudinal cohorts with archived CSF samples and subsequent Alzheimer's disease diagnosis, the terminal-class A-score will show elevation above 1...
Terminal	G-2026-P015	April 2026	PENDING	In longitudinal cohorts of patients with subsequently-diagnosed amyotrophic lateral sclerosis (ALS), the terminal-class A-score from CSF cfDNA will sh...
Secretory	G-2026-P003	Originally f	PENDING	Among PSA-screened prostate cancer patients with matched methylation + WPS + DELFI profiles at diagnosis and at 24 months, the combined A-score trajec...
Secretory	G-2026-P004	Originally f	PENDING	In asbestos-exposed occupational cohorts with archived serial blood samples, the secretory-class combined A-score will show elevation above 1.05 at le...
Immune	G-2026-P010	April 2026	PENDING	In prospective cohorts of patients with known clonal hematopoiesis of indeterminate potential (CHIP) and archived serial blood samples, the immune-cla...
Immune	G-2026-P011	April 2026	PENDING	In patients receiving immune checkpoint inhibitor therapy, the immune-class A-score trajectory will distinguish responders from non-responders within ...
Progenitor	G-2026-P014	April 2026	PENDING	In MDS patients receiving hypomethylating agent (HMA) therapy, the progenitor-class combined A-score trajectory at 3 months post-initiation will ident...
Cycling	G-2026-P001	April 2026	PENDING	In a prospective screening cohort of average-risk adults aged 45-75 with archived serial blood samples and colonoscopy outcomes, the cycling-class com...
Cycling	G-2026-P002	April 2026	PENDING	In longitudinal cohorts of Barrett's esophagus patients with known progression outcomes, the cycling-class combined A-score will show elevation above ...
Stromal	G-2026-P012	April 2026	PENDING	In prospective IPF cohorts with archived serial blood samples and known progression outcomes, the stromal-class combined A-score trajectory slope will...
Adult Stem	G-2026-P013	April 2026	PENDING	In prospective CHIP/CCUS cohorts with archived serial blood samples and known progression outcomes, the stem_adult A_active (2/5, methyl + frag only) ...
Adult Stem	G-2026-P014	April 2026	PENDING	Trajectory slope ($\Delta A/\Delta t$) on the saturating substrates (nucl, fuzz, wps) during the pre-saturation window will correlate with clinical outcome in adult...
Adult Stem	G-2026-P015	April 2026	PENDING	Multi-class A-score divergence signature — adult-stem class flatlined at ceiling while an adjacent architecture class (immune, cycling, or secretory d...
Pluripotent	G-2026-P005	Originally I	PENDING	In cryptorchidism patients and post-orchiectomy TGCT survivors monitored prospectively with serial cfDNA sampling, the stem_pluri class will show a CH...
Pluripotent	G-2026-P016	April 2026	PENDING	In iPSC reprogramming protocols, the stem_pluri class combined A-score across at least four substrates measured at colony harvest will predict downstr...

Class	ID	Filed	Status	Claim summary
Pluripotent	G-2026-P017	April 2026	PENDING	In TGCT patients undergoing bleomycin-etoposide-cisplatin (BEP) chemotherapy, the trajectory of the stem_pluri A_methyl signal during the first two cy...

DATA SOURCES — PRIMARY CITATIONS

All β values from primary publications. No synthetic data. No proprietary datasets. All scripts and data pipelines are reproducible and archived at the Zenodo DOI.

Five-Substrate Framework

- Li et al. 2024 Genome Med — MESA test (4 substrates, colorectal, AUC 0.931, n=690)
- Cristiano et al. 2019 Nature — DELFI fragmentomics (7 cancers, AUC 0.940, n=208)
- Mathios et al. 2022 Nat Commun — DELFI 2-year pre-diagnostic window
- Snyder et al. 2016 Cell — WPS original (15/15 tissue types)
- Doebley et al. 2022 Nat Commun — Griffin nucleosome occupancy (breast, n=139)
- Corces et al. 2018 Science — TCGA ATAC-seq pan-cancer (23 types)
- Esfahani et al. 2022 Cancer Discovery — nucleosome fuzziness (prostate PDX, n=26)

Reference Cell Methylation (Per-Class H_min)

- Lister et al. 2009 Nature — hESC H1 (stem_pluri reference)
- Lister et al. 2013 Science — frontal cortex neuron (terminal, global floor 0.756500)
- Kozlenkov et al. 2014 Hum Mol Genet — cortical neuron mature
- Movassagh et al. 2011 NEJM — adult cardiomyocyte (terminal)
- Roadmap Epigenomics Consortium 2015 Nature — 127 reference epigenomes
- Hannum et al. 2013 Mol Cell — blood methylation aging (GSE40279, immune)
- Moss et al. 2018 Nat Commun — cfDNA tissue-of-origin atlas (cfDNA % per class)

Cancer Validation (G-008, 27/28 TCGA types)

- TCGA Pan-Cancer Atlas — 28 methylation datasets, all cancer types
- Ceccarelli et al. 2016 Cell — LGG/GBM (terminal, largest ΔA in dataset)
- TCGA Research Network 2013 NEJM — AML, Ley et al., n=200 (immune, primary β source)
- Chapuy et al. 2018 Nat Med — DLBCL (immune, n=48)
- Radovich et al. 2018 Cancer Cell — TCGA Thymoma (immune, n=120)
- TCGA Research Network 2012 Nature — BRCA, n=825 (secretory, primary β source)
- TCGA Research Network 2015 Cell — PRAD prostate molecular taxonomy, n=333 (secretory)
- TCGA Research Network 2017 Cancer Cell — PAAD pancreatic adenocarcinoma (secretory)
- TCGA Research Network 2017 Cell — LIHC hepatocellular carcinoma (secretory)
- Zheng et al. 2016 Cancer Cell — TCGA ACC adrenocortical carcinoma (secretory)
- Robertson et al. 2017 Cancer Cell — UVM (stem_adult)
- All TCGA citations per-cancer-type in individual card tables

Cycling Epithelial Cancers — 14 TCGA Types (Card #5)

- TCGA Research Network 2012 Nature — COAD + READ colorectal, n=276
- TCGA Research Network 2014 Nature — LUAD lung adenocarcinoma, n=230
- TCGA Research Network 2012 Nature — LUSC lung squamous, n=178
- TCGA Research Network 2014 Nature — BLCA bladder urothelial, n=131
- TCGA Research Network 2011 Nature — OV ovarian serous, n=489
- TCGA Research Network 2014 Nature — STAD stomach adenocarcinoma, n=295
- TCGA Research Network 2017 Nature — CESC cervical squamous, n=228
- TCGA Research Network 2015 Nature — HNSC head/neck squamous, n=504
- TCGA Research Network 2013 Nature — KIRC kidney clear cell, n=418
- TCGA Research Network 2016 NEJM — KIRP kidney papillary, n=161
- TCGA Research Network 2015 Cell — SKCM skin cutaneous melanoma, n=477

- TCGA Research Network 2014 Cell — THCA thyroid carcinoma, n=496
- TCGA Research Network 2013 Nature — UCEC endometrial carcinoma, n=373

Stromal-Class Cancers — Sarcomas & Mesothelioma (Card #6)

- Abeshouse et al. 2017 Cell — TCGA SARC soft-tissue sarcoma, n=206 (adult)
- Hmeljak et al. 2018 Cancer Discov — TCGA MESO mesothelioma, n=74
- Crompton et al. 2014 Cancer Discov — Ewing sarcoma pediatric, n=112 WGS
- Shern et al. 2014 Cancer Discov — rhabdomyosarcoma pediatric, n=147
- Tirode et al. 2014 Cancer Discov — Ewing sarcoma aggressive subtype, n=299

Adult-Stem Cancers — HSC-Origin AML, BCC, MCC, Cholangiocarcinoma (Card #7)

- Adelman et al. 2019 Cancer Discov — HSC-enriched aging methylation, n=5-7 per age
- Beerman et al. 2013 Cell Stem Cell — HSC aging methylation (GSE44117)
- Harms et al. 2015 Cancer Res — Merkel cell carcinoma MCPyV-negative, n=49
- Farshidfar et al. 2017 Cell Reports — TCGA CHOL cholangiocarcinoma, n=38
- Ley et al. 2013 NEJM — TCGA AML (progenitor-lineage), n=200 (distinct from HSC-origin)

Pluripotent-Stem Cancers — TGCT (Card #8)

- Shen et al. 2018 Cell Reports — TCGA TGCT integrated molecular characterization, n=137
- Killian et al. 2016 Genome Research — TGCT methylation reprogramming (pure histology, n=130)

Non-Cancer Disease Validation & Pre-Malignant States

- De Jager et al. 2014 Nat Neurosci — ROSMAP AD methylation, n=740 (terminal)
- Shireby et al. 2022 Brain — Brains for Dementia Research AD, n=631 (terminal)
- Lunnon et al. 2014 Nat Neurosci — AD four brain regions (terminal)
- Terman 2004 Age (Dordr) — cardiac aging, oxidative stress (terminal)
- Langston & Ballard 1983 N Engl J Med — MPTP Parkinson's disease (terminal)
- Wang et al. 2014 Nat Neurosci — PD methylation changes (terminal)
- Ahrens et al. 2013 Nat Commun — NAFLD hepatocyte methylation (secretory)
- Volkmar et al. 2012 EMBO J — pancreatic β-cell T2D (secretory)
- Cruickshanks et al. 2013 Nat Cell Biol — senescent fibroblasts (stromal)
- Steensma et al. 2015 Blood — CHIP clonal hematopoiesis definition (progenitor)
- Malcovati et al. 2017 Blood — CCUS diagnostic criteria (progenitor)
- Jiang et al. 2020 Cell Death Dis — MDS whole-genome methylation progression

Progenitor-Lineage Cancers (Pediatric & Neural Progenitor Origin)

- Nordlund et al. 2013 Genome Biol — pediatric ALL methylome, n=764 (B-ALL + T-ALL)
- Figueroa et al. 2013 J Clin Invest — integrated epigenetic analysis pediatric ALL
- Northcott et al. 2017 Nature — medulloblastoma whole-genome, n=491 WGS + 1,256 methylation
- Capper et al. 2018 Nature — DNA methylation CNS tumor classifier, n>2,800
- Schwalbe et al. 2017 Lancet Oncol — medulloblastoma methylation subgrouping

Vertebrate Lifespan (Nature Aging NATAGING-A13702)

- Lowe et al. 2018 Genome Biol — 42 mammalian species methylation
- Lu et al. 2023 Nature Aging — 185-species pan-mammalian clock
- Haghani et al. 2023 Science — mammalian methylation networks
- Wang et al. 2020 Cell Reports — Labrador retriever aging, n=104 dogs
- Wilkinson et al. 2021 Nat Commun — bat longevity
- Tacutu et al. 2018 Nucleic Acids Res — AnAge lifespan database

IAM Framework Foundations

- Mahaffey 2026 — Thermodynamic Operating Constraints (cell thermodynamics paper)
- Landauer 1961 IBM J Res Dev — irreversibility and heat generation
- Jacobson 1995 Phys Rev Lett — Einstein equation of state from thermodynamics
- Zenodo DOI 10.5281/zenodo.19547624 — IAM-Validation repository
- GitHub: hmahaffeyges/IAM-Validation — all scripts archived

GLOSSARY

Definitions of the technical terms used throughout this publication, organized by category: Physics & Derivation, Biology & Substrates, Statistical & Validation, and Framework & Clinical Terms. Every term traces to either a published primary source or a derived quantity in the framework.

PHYSICS & DERIVATION

A-score	Dimensionless ratio A = H(value) / H_min(class, substrate). The central measurement in the GAPE framework. A = 1.00 sits at the thermodynamic floor. A > 1.00 means an accessible entropy gap (C3) has opened above the floor.
A_ceiling	The maximum A-score physically achievable on a given class×substrate pairing: A_ceiling = 1 / H_min(class, substrate). Reached when β = 0.5 (maximum binary entropy H = 1.0). No sample biology can produce a higher A on that substrate.
A_combined	AUC-weighted combined A-score across all five substrates: A_combined = Σ(AUC_i × A_i) / Σ(AUC_i). The headline score. Reduces measurement noise by roughly √5 across five correlated physical windows.
A_active	AUC-weighted mean over only the non-saturated substrates for a specific sample. A substrate is flagged saturated when A is within 0.005 of its ceiling. A_active is the signal that continues to respond after some substrates have pinned at ceiling — central for chemotherapy response trajectories and reserve-depletion signals.
C1, C2, C3	Three-component decomposition of Shannon entropy. C1 = universal Landauer floor (0.756500, neuron reference). C2 = class-specific architecture overhead above C1. C3 = accessible gap above H_min = max(0, H – H_min). Healthy cells: C3 ≈ 0. Cancer: C3 = 8-15%.
f_C3	Fraction of total entropy in the accessible gap: f_C3 = C3 / H(β). The component on which every therapeutic intervention operates. Rising f_C3 under treatment indicates non-response. Declining f_C3 indicates response.
H(β) — Shannon entropy	Shannon binary entropy of a Bernoulli(β) variable: $H(\beta) = -\beta \cdot \log_2(\beta) - (1-\beta) \cdot \log_2(1-\beta)$. H(0) = H(1) = 0. H(0.5) = 1 (maximum).
H_min	The class-specific and substrate-specific minimum Shannon entropy consistent with maintained cell identity. Derived from the Landauer cost of DNMT1 maintenance across 19.6M CpG sites per cell division. Calibrated via MCMC posteriors (G-002 methyl, G-003b four other substrates).
H_MIN_GLOBAL	The lowest H_min across all classes: 0.756500, from frontal cortex neuron (Lister 2013). Anchors the global C1 Landauer floor.
Landauer cost	Thermodynamic minimum energy to erase one bit of information: E = k_B × T × ln(2). At 37°C (310.15 K): 2.97×10^{-21} J per bit. Landauer 1961 IBM J Res Dev.
Mahaffey Number	Dimensionless metabolic sensitivity parameter n_bio = ΔG_ATP / (R·T_body). Value 20.94 at 37°C. Absolute class-specific values await G-007 MCMC.
Reserve (clinical meaning)	The C3 headroom remaining between a cell's current A-score and its class-substrate ceiling. Declining reserve under treatment indicates the epigenomic capacity for further response is being consumed.

BIOLOGY & SUBSTRATES

β (beta value)	Fraction methylated at a CpG site or averaged across an architecture-class locus panel. β ∊ [0,1]. Primary input to the methylation H(β).
cfDNA	Cell-free DNA circulating in plasma. Shed from apoptotic and necrotic cells across all tissues, weighted by tissue turnover. ~70% immune, ~12% cycling epithelial, ~8% secretory in standard venous draw.
CpG / CpH	CpG: a cytosine followed by a guanine on the same strand — the standard site of mammalian DNA methylation. CpH: cytosine followed by any other base — mostly unmethylated in somatic cells but methylated in neurons and embryonic stem cells.
CHIP / CCUS / MDS	Clonal Hematopoiesis of Indeterminate Potential: age-associated expansion of HSC clones carrying leukemia-associated mutations; typically benign. Clonal Cytopenias of Undetermined Significance: CHIP with unexplained cytopenias. Myelodysplastic Syndromes: frank malignant transformation. The framework's prediction G-2026-P013 targets the CHIP→MDS transition.
DELFI	DNA Evaluation of Fragments for early Interception. Cristiano et al. 2019 Nature. Measures short-fragment fraction (100-150bp / total) in cfDNA WGS. Substrate 5 in the five-substrate framework.
DNMT1	DNA methyltransferase 1. Maintenance enzyme that copies methylation pattern from parent to daughter strand during DNA replication. Finite fidelity (~1 error per 10^5-10^6 CpG sites per division) — the biological rate-limit on H_min.
Fragment size	cfDNA fragment length distribution. Healthy fragments cluster around 166 bp (one nucleosome + linker). Tumor-derived cfDNA over-represents 100-150 bp fragments due to altered nucleosome spacing. Substrate 5.
Hypomethylation (seminoma)	Global reduction of methylation β toward zero. In seminoma, β falls toward 0.17-0.20 as malignant germ cells revert toward the primordial germ cell (PGC) state. Drives the Seminoma Hypomethylation Inversion.
Nucleosome occupancy	Mean probability that a given genomic position is wrapped in a histone octamer. Measured via ATAC-seq or MESA. Substrate 2.

Nucleosome fuzziness	Positional precision of nucleosomes relative to their reference positions (0 = precise, 1 = maximally fuzzy). Measured via NucleoATAC. Substrate 3.
PGC	Primordial Germ Cell. The most pluripotent somatic cell state known, with near-zero methylation. Seminomas revert toward this state, producing the inversion signature.
Pluripotent / GCNIS / TGCT	Pluripotent: capable of differentiating into all three germ layers. GCNIS: Germ Cell Neoplasia In Situ, the preinvasive precursor to all postpubertal TGCT. TGCT: Testicular Germ Cell Tumor.
WPS (Windowed Protection Score)	Fraction of cfDNA reads whose endpoints fall within a nucleosome-protected window at a given genomic position. Snyder et al. 2016 Cell. Substrate 4.

STATISTICAL & VALIDATION

AUC	Area Under the receiver operating Curve. Single-substrate discrimination performance for binary classification (cancer vs healthy). Range 0.5 (chance) to 1.0 (perfect).
Bootstrap CI	Confidence interval derived by resampling. For H_min posteriors, leave-one-reference-out bootstrap confirms no single dataset drives the result. Mean $\lvert\Delta\rvert$ between full-data and leave-one-out posteriors: 0.168% across 40 class×substrate H_min values (VAL-031).
MCMC	Markov Chain Monte Carlo. Statistical method for sampling from posterior distributions. G-002 (methylation H_min): 17 chains, R-hat < 1.001, 8×10^5 samples. G-003b (four other substrates): same rigor.
Pre-diagnostic window	The time interval between a detectable framework signal and clinical diagnosis. Mathios 2022 Nat Commun established ~24 months for DELFI in lung cancer. Framework extension predicts 12-18 months for cycling-class solid tumors.
R-hat	Gelman-Rubin diagnostic for MCMC chain convergence. R-hat < 1.01 is convergence; < 1.001 indicates very tight convergence. All framework H_min posteriors satisfy R-hat < 1.001.
Sensitivity / specificity	Sensitivity: fraction of true positives correctly identified. Specificity: fraction of true negatives correctly identified. Framework targets high specificity (low false-positive rate) to support clinical serial monitoring rather than one-shot screening.

FRAMEWORK & CLINICAL

Architecture class	Cell-type grouping defined by shared thermodynamic floor (H_min). Eight classes: cycling, secretory, immune, progenitor, stromal, terminal, stem_adult, stem_pluri. Thermodynamically defined, not histologically.
BREACH	Tier assignment when A ≥ 1.10. Indicates epigenomic architecture maintenance has failed beyond what metabolic intervention can restore. Structural transition.
Cross-class propagation	A cancer arising in one class producing signal in another class — e.g., breast metastasis to bone marrow producing immune-class elevation. Scenario 5.5; prediction G-2026-P022.
DETECTABLE	Tier assignment when 1.05 ≤ A < 1.07. Clinical workup recommended per class-specific differential.
Divergence signature	A multi-substrate pattern where substrates move in different directions — e.g., seminoma's A_methyl decreasing while A_nucl, A_wps, A_fuzz, A_frag increase. Primary detection signal for Pluripotent-class malignancy.
Floor	H_min(class, substrate) — the minimum entropy consistent with the cell maintaining its class identity. Derived, not fitted.
Inversion	A named signature where A moves in the opposite direction from standard cancer elevation. Three identified: Seminoma Hypomethylation (Pluripotent), Differentiation Dose (Pluripotent research), Niche Depletion (Adult Stem). See Section 2.4.
MARGINAL	Tier assignment when 1.01 ≤ A < 1.05. Monitor; repeat at next scheduled timepoint.
Mahaffey value	Historical name for H_min in early drafts. Now standardized as H_min. The class-and-substrate-specific entropy floor.
NORMAL	Tier assignment when A < 1.01. Consistent with age-expected healthy.
Saturation (runtime)	Sample-specific substrate flag: A within 0.005 of A_ceiling. Saturated substrates are excluded from A_active computation.
Saturation (structural)	Class-level property: a substrate whose ceiling is itself below BREACH. Cannot discriminate disease severity above BREACH on that substrate. Detection must rely on substrates whose ceilings exceed BREACH.
Substrate	One of five physical measurement windows: methylation, nucleosome occupancy, nucleosome fuzziness, WPS, fragment size. All five measure departures from the same underlying cellular identity information through different physical lenses.
Tier system	A-score thresholds: NORMAL (A < 1.01), MARGINAL (1.01-1.05), DETECTABLE (1.05-1.07), URGENT (1.07-1.10), BREACH (A ≥ 1.10). All thresholds framework-derived; none fit to cancer data.
Trajectory (dA/dt)	Rate of change in A-score over serial samples. The framework's strongest clinical signal — a single elevated A may be age-consistent, but a rising A is not.
URGENT	Tier assignment when 1.07 ≤ A < 1.10. Workup indicated. Corresponds roughly to the Warburg transition in cycling-class cancer.

Warburg transition	Metabolic shift from OxPhos to aerobic glycolysis in malignant cells. Corresponds to A ≈ 1.07 URGENT threshold. Past this point, metabolic levers cannot restore the cell to floor because glycolytic commitment is structural.
Z-score (age-adjusted)	Z = (A_observed - A_predicted(age, class)) / σ_cohort. Standardized departure from the age-expected baseline. |Z| < 1: age-expected. |Z| > 3: strong signal warranting workup.

CONSOLIDATED DATA INDEX

Cross-reference for navigating this publication. Every tracked cancer type, every architecture class, and every numbered prediction is indexed here with its page location. A reviewer looking for all mentions of TGCT, all content for the Pluripotent class, or where G-2026-P017 is specified can find everything through this single index.

Cancer Types — Where Each Is Discussed

Architecture Classes — Full Publication Coverage

Numbered Predictions — Full Specifications

ID	Class	Status	Topic	Section 6 Page	Validation Cohort
G-2026-P016	stem_pluri	PENDING	Yamanaka Differentiation Dose	§2.4	iPSC research consortium
G-2026-P017	stem_pluri	PENDING	BEP platinum response trajectory	p.104	TIGER consortium, MSKCC, MDA
G-2026-P018	stromal	OPEN	Stromal baseline validation	§4.3	Awaiting stromal-specific cohort
G-2026-P019	stem_adult	OPEN	Adult stem baseline validation	§4.3	BLUEPRINT HSC aging
G-2026-P020	stem_pluri	OPEN	Pluripotent baseline validation	§4.3	iPSC reference cohorts
G-2026-P021	cycling	OPEN	Cycling pre-diagnostic window	§5.4	Any cfDNA screening cohort
G-2026-P022	cross-class	OPEN	Cross-class propagation (metastasis)	§5.5	Post-BRCA surveillance cohort

Section Map — Publication Structure

A FINAL NOTE

Heath W. Mahaffey · Entiat, Washington · April 2026

This research is personal, and my step brother Marcus is often on my mind as I continue working on it.

He was diagnosed with an aggressive liver tumor. He had a transplant. Within six months the tumor was back in the new liver, and a few months after that he was gone. He left behind a wife and three small children. He never said goodbye to any of them. He could not bring himself to accept that he was dying, so he spent his last months sedated in a hospital, fighting, and the goodbye never happened.

I don't know whether an earlier, honest signal about his remaining reserve would have changed his choice. That was his choice to make. But he was never given the chance to make it with good information, and his wife and his three small children were not given the chance to have that conversation with him while he was still there to have it. That is the part I cannot stop thinking about.

I just hope that what is in these pages might help prevent what happened to his family from happening to another family with three small children who don't get the chance to say goodbye to their dad.

The hope is that this framework can give patients and their families honest information about the reserve remaining. Not to recommend when to stop. Not to replace the doctor's judgment. To report, honestly, what the physics says — so the person living it can decide how they want to spend the time.

For Marcus.

— HWM

IAMPerformance · GAPE Issue 002 · Patents pending 64/012,720 and 64/014,568